

MODULE -1 INTRODUCTION

Managerial Economics: Meaning, Nature, Scope, & Significance, Uses of Managerial Economics, Role and Responsibilities of Managerial Economist. Theory of the Firm: Firm and Industry, Objectives of the firm, alternate objectives of firm.

Managerial theories: Baumol's Model, Marris's Hypothesis, Williamson's Model.

Economics is a social science concerned with the production, distribution, and consumption of goods and services.

Economics studies how individuals, businesses, governments, and nations make choices about how to allocate resources. Economics focuses on the actions of human beings, based on assumptions that humans act with rational behavior, seeking the most optimal level of benefit or utility.

Economics can generally be broken down into macroeconomics, which concentrates on the behavior of the economy as a whole (analyzes the decisions made by countries and governments), and microeconomics, which focuses on individual people and businesses (allocation of resources, and prices of goods and services, taxes, regulations and government legislation.).

MANAGERIAL ECONOMICS: MEANING & DEFINITION

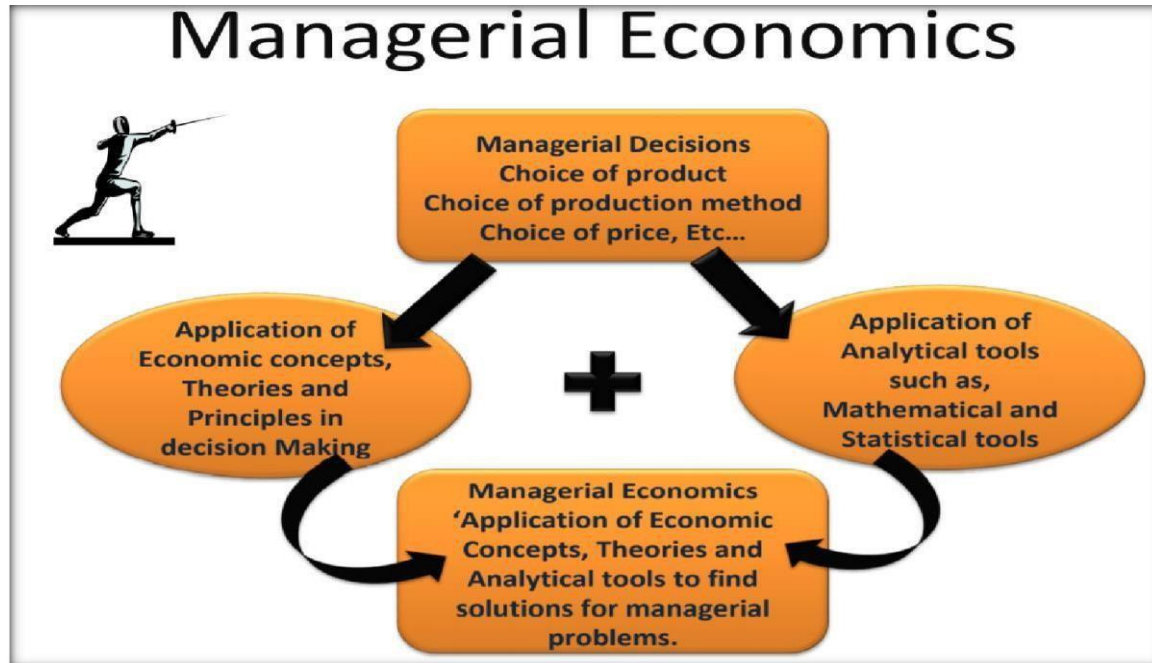
- “Managerial economics is concerned with the application of economic concepts and economic analysis to the problems of formulating rational managerial decisions.
- Branch of Economics: ‘Managerial Economics is the study of Economic Theories, Principles and Concepts which is used in Managerial Decision Making.’
- ‘Managerial Economics is the Application of various Theories, Concepts and Principles of Economics in the Business Decisions.’
- It also includes ‘The Application of Mathematical and Statistical tools in Management decisions.’

Definition:

1. “Managerial economics is the study of allocation of resources available to a firm among the activities of that unit” - **Hynes**.
2. “The integration of economic theory and business practice for the purpose of facilitating decision making and forward planning by management. – **Spencer and Seligman**.
3. “Managerial economics is the application of economic principles and methodologies to the decision-

making process within the firm or organization.”-**Douglas**.

4. “Managerial economics applies economic theory and methods to business and administrative decision-making.”- **Pappas & Hirschey**.



MACROECONOMICS:

- The study of economic activity by looking at the economy as a whole.
- Macroeconomics analyzes overall economic issues such as employment, inflation, productivity, interest rates, the foreign trade deficit, and the federal budget deficit. ...
- An example of macroeconomics is the study of U.S. employment.

MICROECONOMICS:

- Microeconomics is the social science that studies the implications of incentives and decisions, specifically about how those affect the utilization and distribution of resources.
- Microeconomics shows how and why different goods have different values, how individuals and businesses conduct and benefit from efficient production and exchange, and how individual's best coordinate and cooperate with one another.
- Generally speaking, microeconomics provides a more complete and detailed understanding than macroeconomics.

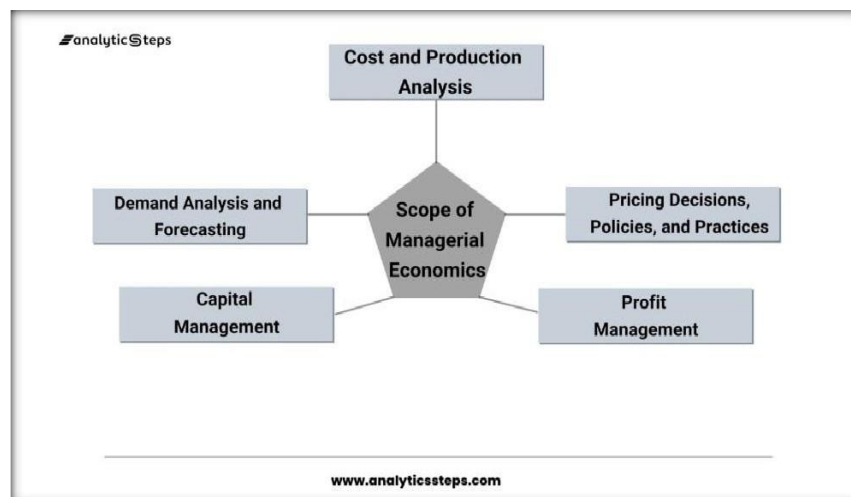
NATURE OF MANAGERIAL ECONOMICS:

1. **Art and Science:** Management theory requires a lot of critical and logical thinking and analytical skills to make decisions or solve problems. Many economists also find it a source of research, saying it includes applying different economic concepts, techniques and methods to solve business problems.
2. **Micro Economics:** In managerial economics, managers typically deal with the problems relevant to a single entity rather than the economy as a whole. It is therefore considered an integral part of microeconomics.
3. **Uses Macro Economics:** A corporation works in an external world, i.e. it serves the consumer, which is an important part of the economy. For this purpose, it is important that managers evaluate the various macroeconomic factors such as market dynamics, economic changes, government policies, etc., and their effect on the company.

Multidisciplinary: It uses many tools and principles that belong to different disciplines, such as accounting, finance, statistics, mathematics, production, operational research, human resources, marketing, etc.

4. **Prescriptive/Normative Discipline:** By introducing corrective steps it aims at achieving the objective and solves specific issues or problems.
5. **Management Oriented:** This serves as an instrument in managers' hands to deal effectively with business-related problems and uncertainties. This also allows for setting priorities, formulating policies, and taking successful decision-making.
6. **Pragmatic:** The solution to day-to-day business challenges is realistic and rational.
7. It is concerned with the **application of theories and principles of economics**.

SCOPE OF MANAGERIAL ECONOMICS:



Managerial economics is concerned with the application of economic concepts and analysis to the problem of formulating rational managerial decisions. There are four groups of problem in both decision making and forward planning.

1. **Demand Analysis and Forecasting:**

- A firm relies on converting inputs into outputs and generates revenue from them. A clear and accurate estimation of demand ensures a continuous efficiency of the firm. Several external factors like price, income, affect the demand that need to be analyzed.
- Upon analyzing these factors affecting the demand for a product, managers can decide on the production. After estimating the current demands, manager's move ahead to predict future demands for the product. This is referred to as demand forecasting.

2. **Cost and Production Analysis:**

- Cost Analysis is yet another function of Managerial economics.
- A company makes a profit in two ways: by **increasing the demand or by reducing the cost**.
- The determinants of assessing costs, the connection between cost and yield, the gauge of cost and benefit are indispensable to a firm.

3. **Pricing Decisions, Policies, and Practices:**

- Among the 4Ps of marketing, Price finds an important place. For any firm, Pricing is a very important aspect of Managerial Economics as a firm's revenue earnings largely depend on its pricing policy. However, it is a bit challenging as other players are competing in the same price segment.
- When pricing a product is done, the costs of production are also taken into account. Managerial Economics helps the management to go through all the analyses and then price a product. In an oligopoly market condition, the knowledge of pricing a product is essential.

4. **Capital Management:**

- Every asset a business owns is known as its capital. [Capital](#) management thus becomes an important practice.
- Planning and control of capital expenditures is a basic executive function. It involves the Equi-marginal principle.

- The prime objective is to ensure the sustainable use of capital. This means that funds should be kept at bay when the managerial returns are less than in other uses.

5. **Profit Management:**

- A business firm is an organization designed with an intention to make profits and profits reflect the success of a company. After all the analyses, it all rolls down to profits.
- To maximize profits a firm needs to manage certain things like pricing, cost aspects, resource allocation, and long-run decisions. This would mean that the firm should work from the very beginning, evaluate its investment decisions and frame the best capital budgeting policies. Profit management is considered as a difficult area of managerial economics.

OTHER:

- **Resource allocation:** Scarce resources have to be used with utmost efficiency to get optimal results. These include production programming, problem of transportation, etc.
- **Inventory and queuing problem:** Inventory problems involve decisions about holding of optimal levels of stocks of raw materials and finished goods over a period. These decisions are taken by considering demand and supply conditions. Queuing problems involve decisions about installation of additional machines or hiring of extra labour in order to balance the business lost by not undertaking these activities.
- **Pricing problems:** Fixing prices for the products of the firm is an important part of the decision making process. Pricing problems involve decisions regarding various methods of pricing to be adopted.
- **Investment problems:** Forward planning involves investment problems. These are problems of allocating scarce resources over time. For example, investing in new plants, how much to invest, sources of funds, etc.

Study of managerial economics essentially involves the analysis of certain major subjects like:

- Demand analysis and methods of forecasting
- Cost analysis
- Pricing theory and policies
- Profit analysis with special reference to break-even point

- Capital budgeting for investment decisions
- The business firm and objectives
- Competition.
- Inflation and economic conditions.

SIGNIFICANCE OF MANAGERIAL ECONOMICS:

- **Business Planning:** Managerial economics assists business organizations in formulating plans and better decision making. It helps in analyzing the demand and forecasting future business activities.
- **Cost Control:** Controlling the cost is another important role played by managerial economics. It properly analyses and decides production activities and the cost associated with them. Managerial economics ensure that all resources are efficiently utilized which reduces the overall cost.
- **Price Determination:** Setting the right price is one of the key decisions to be taken by every business organization. Managerial economics supplies all relevant data to managers for deciding the right prices for products.
- **Business Prediction:** Managerial economics through the application of various economic tools and theories helps managers in predicting various future uncertainties. Timely detection of uncertainties helps in taking all possible steps to avoid them.
- **Profit Planning and Control:** Managerial economics enables in planning and managing the profit of the business. It makes an accurate estimate of all cost and revenue which helps in earning the desired profit.
- **Inventory Management:** Proper management of inventory is a must for ensuring the continuity of business activities. It helps in analyzing the demand and accordingly, production activities are performed. Managers can arrange and ensure that the proper quantity of inventory is always available within the business organization.
- **Manages Capital:** Managerial economics helps in taking all decisions relating to the firm's capital. It properly analyses investment avenues before investing any amount into it to ensure the profitability of an investment.
- Assist in Decision Making
- Optimization of Resources

USES OF MANAGERIAL ECONOMICS:

1. **Production Decisions**: Production theory explains the principles in which the business has to take decisions on how much of each commodity it sells and how much it produces and also how much of raw material ie., fixed capital and labour it employs and how much it will use.
2. **Inventory Decisions**: Inventory decisions have a direct impact on production. For example, a decision to increase safety stock means that the production rate must increase until the desired level of safety stock is achieved
3. **Cost Decisions**: Its goal is to advise management on the most appropriate course of action based on the cost efficiency and capability. Cost accounting provides the detailed cost information that management needs to control current operations and plan for the future.
4. **Marketing Decisions**: Marketing decisions are those decisions that are taken by a marketing manager to achieve the ultimate marketing objectives of the organization.
5. **Investment Decisions**: Investment decision refers to selecting and acquiring the long-term and short-term assets in which funds will be invested by the business.
6. **Personnel Decisions**: Personnel decisions are decisions made in organizations that affect people's work lives, such as selection, placement, and discharge. • All business organizations must make personnel decisions about their employees. Some organizations use less formal and scientifically based methods than others.

MANAGERIAL ECONOMIST:

- A Managerial Economist is also termed as an economic advisor or business economist.
- He is responsible for analyzing various internal and external environmental forces that influence the functioning of business organizations.
- Managerial economist makes several successful business forecasts and updates the management team regarding the economic trends from time to time

ROLES AND RESPONSIBILITIES OF MANAGERIAL ECONOMIST:

A managerial economist helps the management by using his analytical skills and highly developed techniques in solving complex issues of successful decision-making and future advanced planning and assists the business planning process of a firm.

- **Studies Business Environment**

The managerial economist is responsible for analyzing the environment in which business operates. Proper study of all **external factors** that affect the functioning of organization is must for proper functioning. He studies various factors like **growth of national income,**

Competition level, price trends, phases of the business cycle and economy and updates the management regarding it from time to time.

● **Analyses Operations Of Business**

He analyses the internal operation of business and helps management in making better decisions in regard to internal workings. Managerial economist through his analytical and forecasting skills provides advice to managers for formulating policies regarding internal operations of the business.

● **Demand Forecasting and Estimation**

Proper estimation and forecasting of future trends helps the business in achieving desired profitability and growth. Managerial economist through proper study of all internal and external forces makes successful forecasting of future uncertainties or trends.

● **Production Planning**

Managerial economist is responsible for scheduling all production activities of business. He evaluates the capital budgets of organizations and accordingly helps in deciding timing and locating of various actions.

● **Economic Intelligence**

He provides economic intelligence services by communicating all economic information to management. Managerial economist keeps management always updated of all prevailing economic trends so that they can confidently talk in seminars and conferences.

● **Performing Investment Analysis**

A managerial economist analyzes various investment avenues and chooses the most appropriate one. He studies and discovers new possible fields of business for earning better returns.

● **Focuses On Earning Reasonable Profit**

He assists management in earning a reasonable rate of profit on capital employed in the business. Managerial economist monitors activities of organizations to check whether all operations are running efficiently as per the plans and policies.

● Maintaining Better Relations

A managerial economist maintains better relations with all internal and external individuals connected with the business. It is his duty to develop a peaceful and cooperative environment within the organization and aims to reduce any opposition taking place.

THEORY OF THE FIRM:

The theory of the firm is the microeconomic concept founded in neoclassical economics that states that a firm exists and makes decisions to maximize profits. The theory holds that the overall nature of companies is to maximize profits meaning to create as much of a gap between revenue and costs. The firm's goal is to determine pricing and demand within the market and allocate resources to maximize net profits.

Theory of the firm is related to comprehending how firms come into being, what are their objectives, how they behave and improve their performance and how they establish their credentials and standing in society or an economy and so on.

The theory of the firm aims at answering the following questions:

- Existence – why do firms emerge and exist, why are not all transactions in the economy mediated over the market?
- Which of their transactions are performed internally and which are negotiated in the market?
- Organization – why are firms structured in such a specific way? What is the interplay of formal and informal relationships?
- Heterogeneity of firm actions/performances – what drives different actions and performances of firms?

FIRM AND INDUSTRY

A **Firm** is a commercial enterprise, a company that buys and sells goods and services to consumers with the aim of making a profit. A business entity such as a corporation, limited liability company, public limited company, sole proprietorship, or partnership that has goods or services for sale is categorized as a firm.

An **Industry** is an economic activity concerned with the processing of raw materials and manufacture of goods. It can also be said as a sector that produces goods or related services within an economy.

OBJECTIVES OF THE FIRM

1. Profit maximization
2. Sales maximization
3. Utility maximization
4. Increase market share/market dominance
5. Social/environmental concerns
6. Profit satisficing
7. Co-operatives

1. **Profit maximization:** Considering any business that exists, they are usually concerned with maximizing profit. Higher profit means:

- Higher dividends for shareholders.
- More profit can be used to finance research and development. Higher profit makes the firm less vulnerable to takeover.
- Higher profit enables higher salaries for workers.

2. **Sales Maximization:** Firms often seek to increase their market share, even if it means less profit. This could occur for various reasons:

Increased market share increases monopoly power and may enable the firm to put up prices and make more profit in the long run.

Managers prefer to work for bigger companies as it leads to greater prestige and higher salaries.

Increasing market share may force rivals out of business. E.g. the growth of supermarkets has led to the demise of many local shops. Some firms may actually engage in predatory pricing which involves making a loss to force a rival out of business.

3. **Utility maximization:** It means “that managers get satisfaction from using some of the firm's potential profits for unnecessary spending on items from which they personally benefit.” To pursue his goal of utility maximization, the manager directs the firm's resources many ways.

4. **Increase Market Share/ Market Dominance:** This is similar to sales maximization and may involve mergers and takeovers. With this objective, the firm may be willing to make lower levels of profit in order to increase in size and gain more market share. More market share increases its monopoly power and ability to be a price setter.

5. **Social Concern:** A firm may incur extra expense to choose products which don't harm the environment or

products not tested on animals. Alternatively, firms may be concerned about local community / charitable concerns.

- Some firms may adopt social/environmental concerns as part of their branding. This can ultimately help profitability as the brand becomes more attractive to consumers.
 - Some firms may adopt social/environmental concerns on principal alone – even if it does little to improve sales/brand image.
6. **Profit Satisficing:** Profit satisficing is a situation where there is a separation of ownership and control. As a result, the owners are likely to have different objectives to the managers and workers.
7. **Co-operatives:** Co-operatives may have completely different objectives to a typical business. A co-operative is run to maximize the welfare of all stakeholders – especially workers. Any profit the co-operative makes will be shared amongst all members.

ALTERNATE OBJECTIVES OF THE FIRM:

- **Economic objectives:**
 - Maximize growth rate
 - Desire for liquidity
- **Non-economic objectives:**
 - Survival
 - Retention of Customers
 - Innovation
 - Recognition
 - Optimum Utilization of Resources
 - Supplying desired goods at reasonable prices
 - Social welfare
 - Building up public confidence for the public

MANAGERIAL THEORIES OF THE FIRM:

Managerial theories of the firm place emphasis on various incentive mechanisms in explaining the behavior of managers and the implications of this conduct for their companies and the wider economy.

According to traditional theories, the firm is controlled by its owners and thus wishes to maximize short run profits. The more contemporary managerial theories of the firm examine the possibility that the firm is controlled not by its owners, but by its managers, and therefore does not aim to maximize profits. Although

profit plays an important role in these theories as well, it is no longer seen as the sole or dominating goal of the firm. The other possible aims might be sale revenue maximization or growth.

MANAGERIAL THEORIES OF THE FIRM

- Baumol's Theory of Sales Revenue Maximization
- Marris Growth Maximization Model
- Williamson's Managerial Discretionary Theory

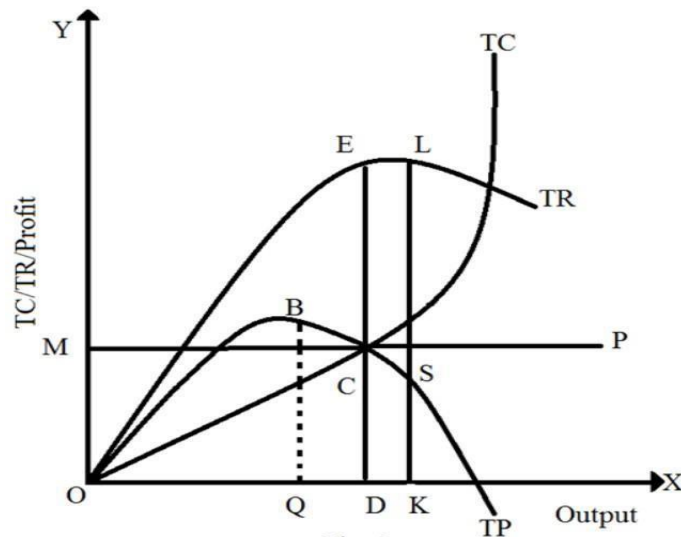
1. BAUMOL'S THEORY OF SALES REVENUE MAXIMISATION:

Baumol's Model: Baumol's theory of sales revenue maximization was created by American economist William Jack Baumol. It's based on the theory that, once a company has reached an acceptable level of profit for a good or service, the aim should shift away from increasing profit to focus on increasing revenue from sales.

W.J.Baumol suggested Sales Revenue maximization as an alternative goal to profit maximization. Managers only ensure acceptable level of profit, pursuing a goal which enhances their own utility.

Assumption of the Theory:

1. There is a single period time horizon of the firm.
2. The firm aims at maximizing its total sales revenue in the long run subject to a profit constraint.
3. The firm's minimum profit constraint is set competitively in terms of the current market value of its shares.
4. The firm is oligopolistic whose cost curves are U-shaped and the demand curve is downward sloping. Its total cost and revenue curves are also of the conventional type.



From the above graph considering Output in X axis and TR/TC/profit in Y axis, where TC is the total cost curve, TR the total revenue curve, TP the total profit curve and MP the minimum profit or profit constraint line. The firm maximizes its profits at OQ level of output corresponding to the highest point B on the TP curve. But the aim of the firm is to maximize its sales rather than profits. The sales maximization output is OK where the total revenue KL is the maximum at the sales maximization output OK is higher than the profit maximization output OQ. But sales maximization is subject to minimum profit constraint highest point of TR. This sales maximization output OK is higher than the profit maximization output OQ. But sales maximization is subject to minimum profit constraint. The output OK will not maximize sales as the minimum profits OM are not being covered by total profits KS hence total revenue gets decrease from L to E showing the quantity level showing the quantity of output 'OD'. By the above Baumol justified that sales revenue has to be optimally increased.

ARGUMENTS IN FAVOUR OF MAXIMISATION OF SALES GOAL:

Baumol's argument to justify sales revenue importance.

1. If the sales of a firm are declining then the banks, creditors and the capital market are not prepared to provide finance to the firm anymore.
2. Its own distributors and dealers might stop showing interest on the firm's product in future.
3. Consumers might not buy its product because of its unpopularity and there is a more chance of competitors acquiring the consumers.

4. Firm reduces its managerial and other staff with fall in sales.
5. But if firm's sales are large, there are economies of scale and the firm expands and earns large profits.
6. Salaries of workers and management also depend to a large extent on more sales and the firm gives them bonus and other facilities.

Conclusion: This theory states that the sales maximization is to increase the total revenue by money where, Sales can increase up to the point of profit maximization where the marginal cost equals marginal revenue. If sales are increased beyond this point money sales may increase at the expense of profits. If sales are increased beyond this point money sales may increase at the expense of profits.

2. MARRIS'S GROWTH MAXIMIZATION MODEL:

Robin Marris in his book The Economic Theory of 'Managerial' Capitalism (1964) has developed a dynamic balanced growth maximizing theory of the firm. He concentrates on the proposition that modern big firms are managed by managers and the shareholders are the owners who decide about the management of the firms.

The managers aim at the maximization of the growth rate of the firm and the shareholders aim at the maximization of their dividends and share prices. To establish a link between such a growth rate and the share prices of the firm, Marris develops a balanced growth model in which the manager chooses a constant growth rate at which the firm's sales, profits, assets, etc., grow.

"In Corporate firms, there is structural division of ownership and management which allows managers to set goals which do not necessarily conform to those of the owners.

The shareholders are the owners. Their utility function includes variables such as

- Profits,
- Size of output,
- Size of capital,
- Market share and
- Public image.

The Managers have other ideas. Their utility functions are:

- Salaries,
- Job security,

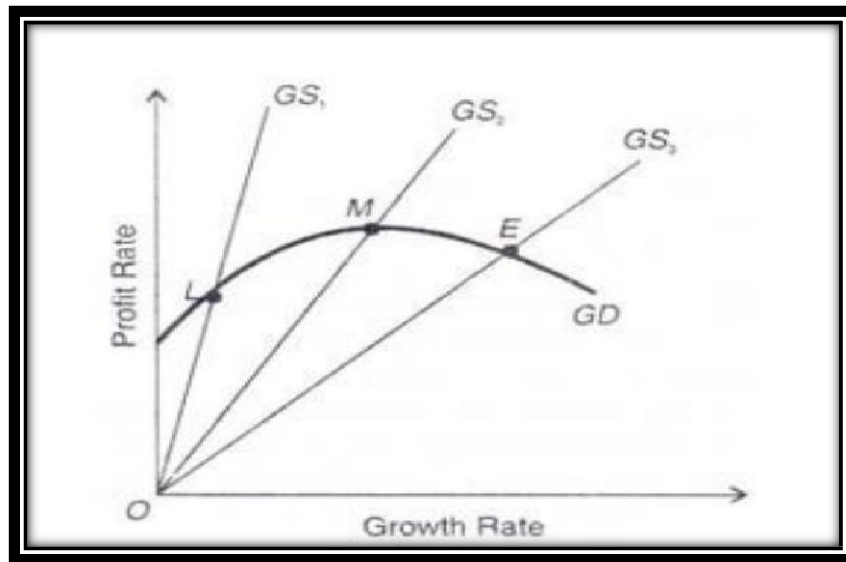
- Power and status

The Marris's model is based on the following assumptions:

- It assumes a given price structure.
- Production costs are given.
- There is no oligopolistic interdependence.
- Factor prices are constant.
- Firms are assumed to grow through diversification.
- All major variables such as profits, sales and costs are assumed to increase at the same rate.
- The objective of the firm is to maximize its balanced growth rate.

The Growth itself depends on two factors: First, the rate of growth of demand for the firm's product - GD; and second, the rate of growth of capital supply - GS.

All major variables such as profits, sales and costs are assumed to increase at the same rate.



- According to Marris, there are two different utility functions for the manager and the owner of the firm. The utility function of the manager consists of his emoluments, status, power, job security, etc. On the other hand, the utility function of the owner includes profits, capital, output, market share, etc.
- The firm may grow in size through the creation of new products which create new demands. Marris calls it differentiated diversification. The introduction of new products depends upon the rate of diversification, advertising expenses, R&D expenditures, etc.

- Marris establishes the relationship between growth and profits on the demand side through diversification into new products. The links between growth and profits are different at different levels of growth. In this growth-profits relationship, growth determines profits. When the rate of growth of the firm is low, the relationship is a positive one.
- As new products are introduced, the firm expands (grows) and profits increase. With the further increase in the growth rate due to greater diversification into new products, the growth-profits relationship becomes negative. This is because there is the managerial constraint which sets a limit on the rate of managerial growth that restricts the growth of the firm.

The firms' managerial ability to cope with a great number of changes at once is limited. It is not possible to develop a larger management team for the development and marketing of new products.

- The higher rate of diversification requires higher expenditures on advertising and R &D. As a result, beyond a certain growth rate, the higher growth rate leads to a lower rate of profit. This is illustrated in Figure 4 where the GD curve first rises, reaches the highest point M and then starts falling.
- The growth-supply curve will be very steep as shown by GS1 curve. The firm's equilibrium will be at point L where the GS1 curve intersects the GD curve. This is again not the optimal equilibrium point of the firm because here the growth rate is low and profits are below the maximum level.
- Larger retained profits are required by managers to invest larger funds for the growth of the firm. These raise the retention ratio which, in turn, leads to higher profits and higher growth rates until point M of maximum profits is reached.
- This is again not the optimum equilibrium point of the firm because the managers feel that this combination of higher growth rate and higher profits is approved by the shareholders and there is no threat to their job security. They will, therefore, be encouraged to raise the retention ratio further, invest more funds, expand and increase the growth rate of the firm.
- As a result, the growth-supply curve will become flatter and take the shape of GS3 curve as in the figure where it intersects the DS curve at point E. At this point, distributed profits to shareholders fall. But they are adequate to satisfy the shareholders so that there is no fear of fall in the prices of shares and of the threat of take-overs. There is also job security for managers.
- Thus point E is the optimal equilibrium point of the firm. If the managers adopt a higher retention ratio than this, the distributed profits will fall further and the shareholders will not be satisfied which will endanger the job security of managers. The existing shareholders may decide to replace the managers. If the distribution of low profits to shareholders brings a fall in the market prices of shares, it may lead to take-over of the firm.

- Criticism:
- Marris assumes a given price structure for the firms. He, therefore, does not explain how prices of products are determined in the market. This is a serious weakness of his model.
- . Another defect of this model is that it ignores the problem of oligopolistic interdependence of firms in non-collusive market.
- The assumption that all major variables such as profits, sales and costs increase at the same rate is highly unrealistic.
- It is also doubtful that a firm would continue to grow at a constant rate, as assumed by Marris. The firm might grow faster now and slowly later on.
- Marris lumps together advertising and R&D expenses in his model. This is a serious shortcoming of the model because the effectiveness of these two variables is not the same in any given period.

Conclusion:

- Thus the manager of a firm aims at maximizing his utility, and his utility depends upon the rate of growth of the firm. Though promoting the growth of the firm is the main aim of the manager, yet he is also motivated by his job security. The manager's job security depends upon the satisfaction of shareholders who are concerned to keep the firm's share prices and dividends as high as possible.
- Thus the manager aims at maximizing the rate of growth of the firm and the shareholders (owners) aim at maximizing their profits in the form of dividends and share prices. Marris analyses the means by which the firm tries to achieve its growth-maximization goal.

3. WILLIAMSON'S MANAGERIAL DISCRETIONARY THEORY:

- **Oliver E. Williamson found** (1964) that profit maximization would not be the objective of the managers of a company.
- This theory assumes that utility maximization is a manager's sole objective. However it is only in a corporate form of business organization that a self-interest seeking manager
Maximize his/her own utility, since there exists a separation of ownership and control.
- The managers can use their 'discretion' to frame and execute policies which would maximize their own utilities rather than maximizing the shareholders' utilities.
- This is essentially the principal-agent problem. This could however threaten their job security, if a

minimum level of profit is not attained by the firm to distribute among the shareholders.

- **Utility function** or "**expense preference**"^[8] of a manager can be given by:

$$U=U(S,M,Id)$$

U denotes the Utility function,

S denotes the "monetary expenditure on the staff" (**not only the manager's salary and other forms of monetary compensation received by him from the business firm**)

M stands for "Management Slack" (**non-essential management perquisites such as entertainment expenses, lavishly furnished offices, luxurious cars, large expense accounts, etc. which are above minimum to retain the managers in the firm**) and

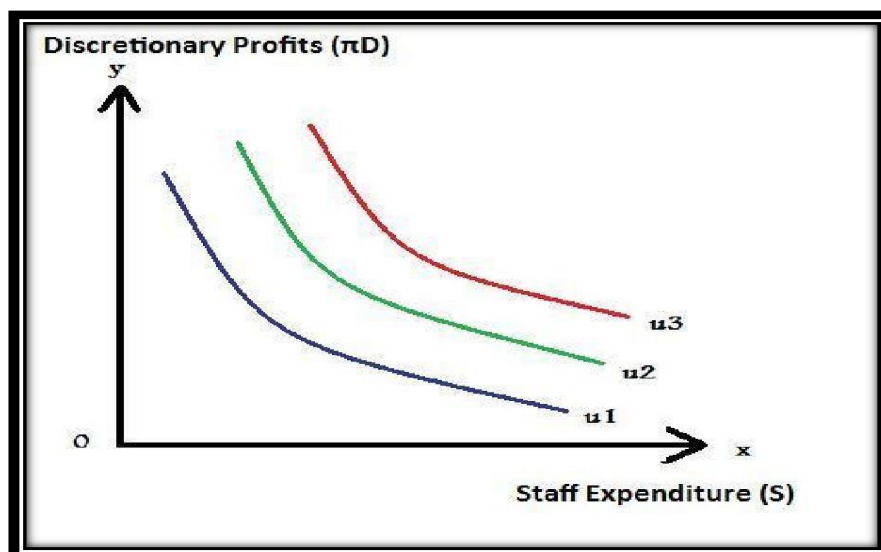
ID stands for amount of "Discretionary Investment". (The amount of resources left at a manager's disposal, to be able to spend at his own discretion. For example, spending on latest equipment, furniture, decoration material, etc.)

Managerial utility function:

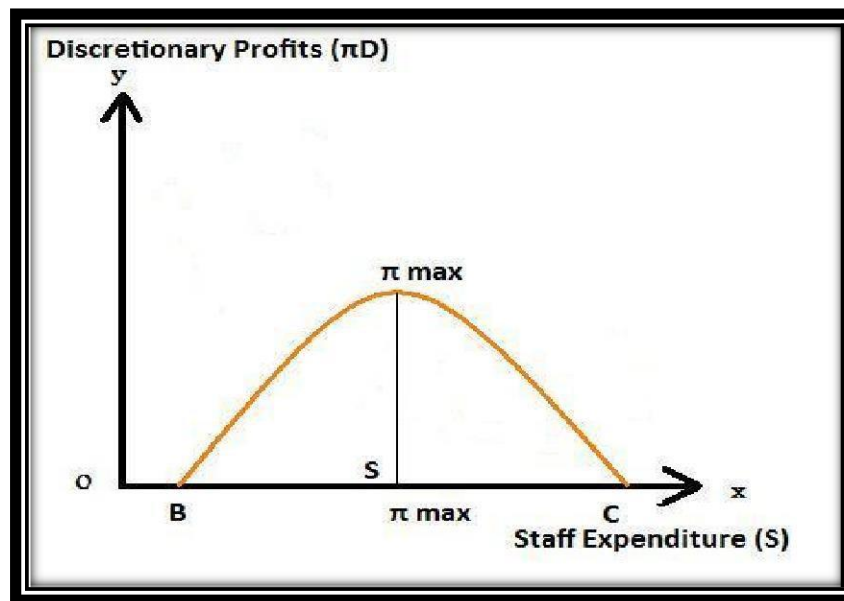
The managerial utility function includes variables such as salary, job security, power, status, dominance, prestige and professional excellence of managers.

The basic assumptions of the model are:

- Imperfect competition in the markets.
- Divorce of ownership and management.
- A minimum profit constraint exists for the firms to be able to pay dividends to their shareholders.



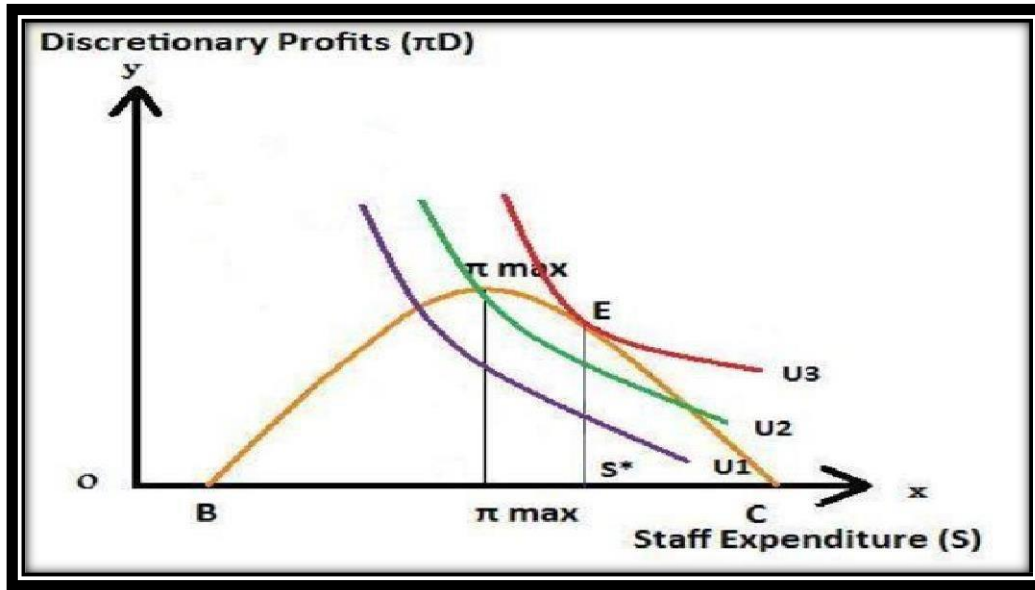
- Fig 1. Shows the various levels of utility (U_1 , U_2 , U_3) derived by the manager by combining different amounts of discretionary profits and staff expenditure. Higher the indifference curve, higher is the level of utility derived by the manager. Hence the manager would try to be on the highest level of indifference curve possible given the constraints. Staff expenditure is plotted on the x-axis and discretionary profits on the y-axis.
- The discretionary profit in this simplified model is equal to the discretionary investment. The indifference curves are downward sloping and convex to the origin. This shows diminishing marginal rate of substitution of staff expenditure for discretionary profits. The curves are [asymptotic](#) in nature which implies that at any point of time and under any given circumstance the manager will choose positive amounts of both discretionary profits and staff expenditure



- Assuming that the firm is producing an optimum level of output and the market environment is given, the discretionary profits curve is generated, shown in Fig 2. It gives the relationship between staff expenditure and discretionary profits.
- It can be seen from the figure that profit will be **positive** in the region between the points B and
- Initially with increase in profits, the staff expenditure the discretionary profits also increase, but this is only till the point $\Pi_{\max}$ that is, till S level of staff expenditure. Beyond this if staff expenditure is increased due

to increase in output, and then a fall in the discretionary profits is noticed.

- Staff expenditure of less than B and more than C is not feasible as it wouldn't satisfy the Minimum profit constraint and would in turn threaten the [job security](#) of managers.



- To find the equilibrium in the model, Fig 1. Is superimposed on Fig 2.
- The equilibrium point is the point where the discretionary profit curve is tangent to the highest possible indifference curve of the manager, which is point E in Fig 3. Staying at the highest profit point would require the manager to be at a lower indifference curve U2.
- In this case the highest attainable level of utility is U3. At equilibrium, the level of profits would be lower but staff expenditure S* is higher than the staff expenditure made at the maximum profit point.
- As indifference curve is downward sloping, the equilibrium point would always be on the right of the maximum profit point. **Thus the model shows the higher preference of managers for staff expenditure as compared to the discretionary investments**

Arguments:

- In the Williamson theory or model argues that managers have discretion in pursuing policies which maximize their own utility rather than attempting the maximization of profits which maximizes the utility of owner and shareholders.

- Profit acts as a constraint to this managerial behavior in that the financial market and the Shareholders require a minimum profit to be paid out in the form of dividends, otherwise the job Security of managers is endangered.
- In this theory Williamson considered the two important factors namely, staff expenditures on emoluments (slack payments), and funds available for discretionary investment give to managers a positive satisfaction (utility) because these expenditures are a source of security and reflect the power, status, prestige and professional achievement of managers.
- Being the head of a large staff is a symbol of power, status and prestige, as well as a measure of Professional success, because a progressive and increasing staff implies successful expansion of the particular activity for which a manager is responsible within a firm.

Criticism:

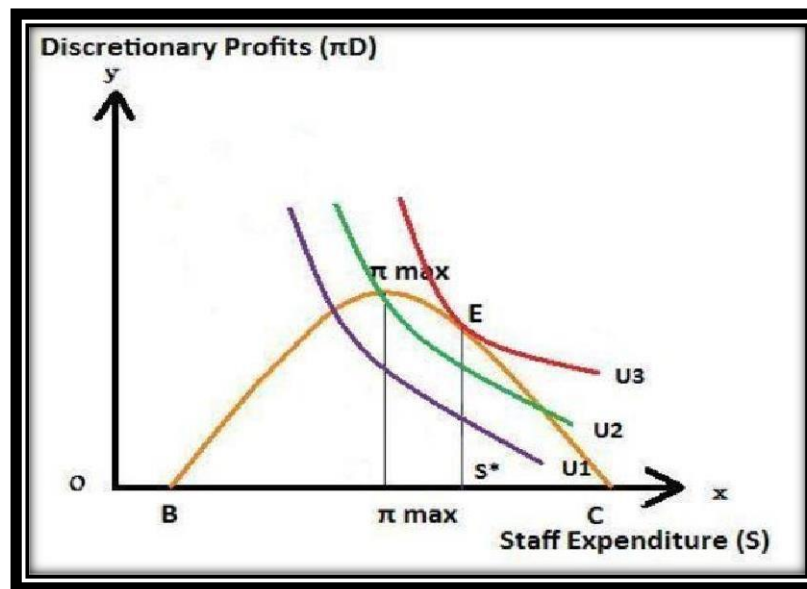
- This model does not clarify the basis of the derivation of his feasibility curve. In particular, he fails to indicate the constraint in the profit-staff relation, as shown by the shape of the feasibility curve.
- It lumps together staff and manager's emoluments in the utility curve. This mixing up of non-Pecuniary and pecuniary benefits of the manager make the utility function ambiguous.
- This model does not deal with oligopolistic interdependence and of oligopolistic rivalry.

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From the above graph, taking Discretion Profit in Y-axis and Staff Expenditure in X- axis, where U_1 , U_2 , U_3 shows the utility level of the managers with facilities provided by the firm. With the different combination of factors the level of utility changes. Hence the profit of the firm relies on the ideal combination of factors such as Discretion Profit and Staff Expenditure.

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QUESTION BANK:

1. Define managerial economics.3m
2. What is Managerial Revenue and Managerial Cost?3m
3. Explain the concepts of Micro and Macroeconomics.7m
4. Discuss the scope of managerial economics.7m/10m
5. Explain the significance and uses of managerial economics.10m
6. Who is Managerial Economist? Elaborate the roles and responsibilities of managerial economist.10m
7. Discuss the objectives of the firm.7m
8. Explain the Baumol's sales revenue maximization model, with a suitable graph.10m
9. Elaborate the Marris's model? Explain with graph 10m
10. Enumerate the Williamson's Discretionary model with a graph? 1

MODULE -2

DEMAND ANALYSIS AND DEMAND FORECASTING

Law of Demand, Exceptions to the Law of Demand, Elasticity of Demand – Classification of Price, Income & Cross elasticity, Advertising and promotional elasticity of demand. Uses of elasticity of demand for Managerial decision making, Measurement of elasticity of demand. Law of supply, Elasticity of supply, Demand forecasting: Meaning & Significance, Methods of demand forecasting.

INTRODUCTION:

The economy relies on the willingness of consumers to make purchases and the ability of companies to supply them. When consumers make more purchases, inflation and interest rates decrease. When consumers decrease their purchases or if producers are unable to supply, inflation and interest rates increase. Consumer demand drives production and supports a thriving economy. In this article, we provide the demand definition in economics, explore the different types of demand and explain the factors that influence it.

MEANING OF DEMAND:

Demand is an economic principle referring to a consumer's desire to purchase goods and services and willingness to pay a price for a specific good or service.

Demand refers to the willingness and ability of consumers to purchase a given quantity of a good or service at a given point in time or over a period in time. Demand is a consumer want or a need supported by an ability to pay.

DEFINITION OF DEMAND:

- Demand refers to consumers' desire to purchase goods and services at given prices.
- Demand can mean either market demand for a specific good or aggregate demand for the total of all goods in an economy.

TYPES OF DEMAND:

- Joint demand
- Composite demand
- Short-run and long-run demand
- Price demand

- Income demand
- Competitive demand
- Direct and derived demand

1. Joint demand

Joint demand is the demand for complementary products and services. These can be products that are accessories for others or that people commonly purchase together. Foreexample, cereal and milk or peanut butter and jelly. The two are linked but demand for one is not necessarily dependent on the demand for the other.

2. Composite demand

Composite demand happens when there are multiple uses for a single product. For example, corn can be used as animal feed, ethanol and food in its whole form. The rise in demand for any of these products leads to a shortage in supply for the others. This shortage can lead to a rise in price.

3. Short-run and long-run demand

Short-run demand refers to how people will immediately react to price changes while elements are fixed. For example, if the demand for a product drastically decreases and a manufacturer has high overhead costs, they have no choice but to absorb the profits lost. Over time, or in the long run, companies have a chance to adjust to the new situation by decreasing labor or increasing price and supplies.

4. Price demand

Price demand relates to the amount a consumer is willing to spend on a product at a given price. Businesses use this information to determine at what price point a new product should enter the market. Consumers will buy items based on their perception of that product's value. Price elasticity refers to how the demand will change with fluctuations in price.

5. Income demand

As consumers make more income, quantity demand increases. This means people will buy more overall when they earn more income. Tastes and expectations also change with an increase in income, reducing the size of one market and increasing the size of another. Consumers will often buy a product or service because it is what they can afford but may deem lower quality. The demand for those lower-quality products will decrease as income increases.

6. Competitive demand

Competitive demand occurs when there are alternative services or products a customer can choose from. From a business's perspective, they can use fluctuations in the price of their

competitors to determine how their own will sell. An example of this is between name-brand and store-brand medicine. If a consumer prefers a name brand but it is out of stock or the price increases significantly, the store brand will see a rise in sales.

7. Direct and derived demand

- ☐ Direct demand is the demand for a final good. Food, clothing and cell phones are an example of this. Also called autonomous demand, it's independent of the demand for other products.
- ☐ Derived demand is the demand for a product that comes from the usage of others. For example, the demand for pencils will result in the demand for wood, graphite, paint and eraser materials. In this example, the demand for wood is dependent on the demand for its uses.

FACTORS THAT INFLUENCE DEMAND:

Demand is influenced by the activities of consumers and businesses. Businesses attempt to drive demand through marketing efforts. Consumers drive demand through their tastes, income levels and resistance to price increases.

1. Price:

Price is the value of product or service expressed in monetary terms.

In other words, the price is the amount of money whose payment is expected or required for buying a product or accessing a service. A price of a product has a maximum effect on its demand.

If other factors remaining constant (do not change) then the relationship between price and demand is as follows:

If a product's price is low, then its demand is high.

If a product's price is high, then there is low demand for it.

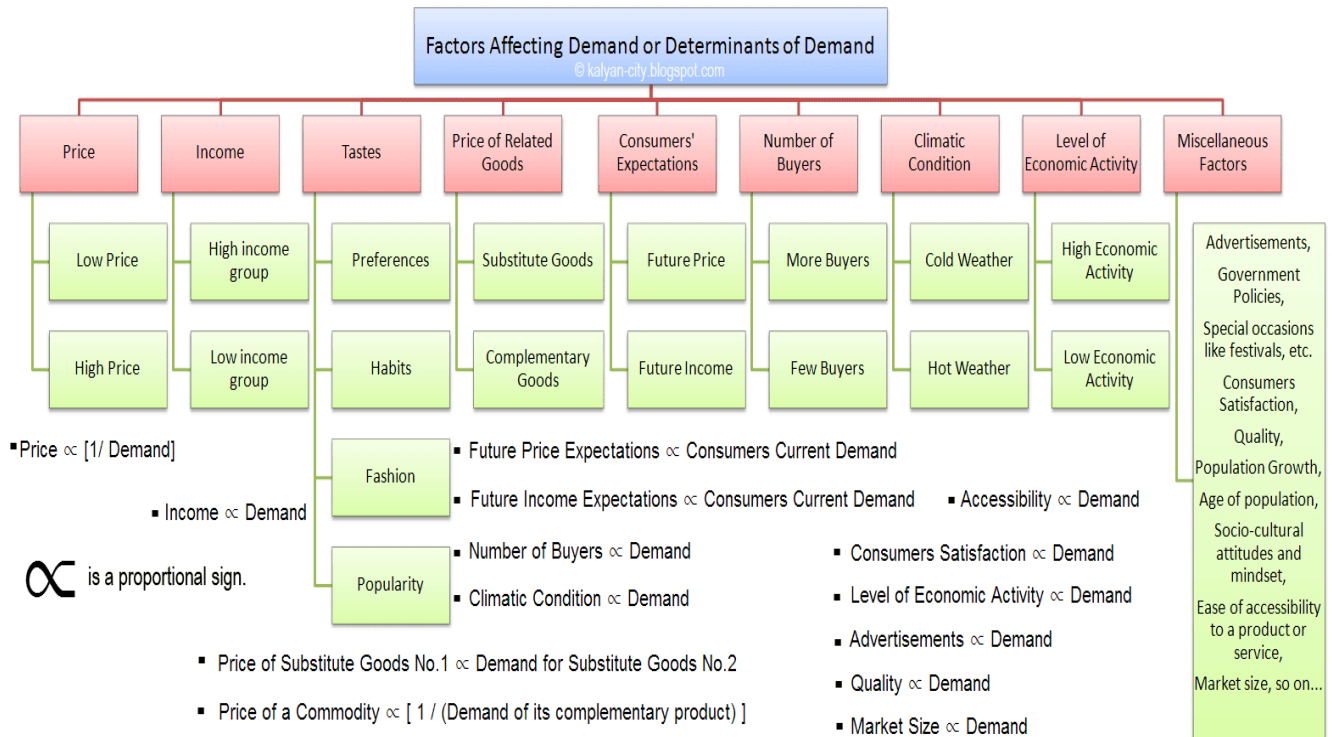
Hence, it can be said that price and demand are inversely proportional to each other, provided other determinants remain constant or do not change. In other words, if price increases then demand falls and vice-versa.

2. Income

Income of an individual is his capacity to earn money. A person with a higher income makes more money and vice-versa. After the determinant of price, income is the second most significant factor that affects demand. The relation between income and demand is as follows:

If income is high, then demand is also high.

If income is low, then demand is minimum. So, we can say, income is directly proportional to demand.



3. Tastes

Tastes, Preferences, Habits, Fashion, and Popularity, also has an impact on the demand. Tastes of people i.e. their likes and dislikes affect demand: If people like a product, then its demand is high even when charged at higher prices.

If people dislike a product, then its demand is low even when charged at lower prices.

Fashion or current trend in the **market** has an impact on the demand for a type of product. For example, now, young generation in India prefers to wear fashionable and comfortable western apparels over traditional and cheaper Indian clothes.

Popularity or fame also affects the demand for a product. For example, people prefer to buy regularly advertised and popular branded products over lesser known alternatives.

4. Price of Related Goods

Related Goods exhibit some relationship with each other.

These are of two types:

Substitute Goods, and

Complementary Goods.

Substitute Goods are products that can easily replace each other. That is, they can take each other's place. In other words, they act as an alternative to each other to satisfy a similar want or desire. They are even called Supplementary Goods because they readily supplement each other's role or function just

in case of unavailability, scarcity, higher market price, etc.

5. Consumers' Expectations

Consumers' expectations arise out of their predictions about:

Future Price, and

Future Income.

If there is an expectation of a rise in the future price of an essential commodity, then current demand for it increases. If consumers are expecting a fall in the future price of an essential commodity, then current demand for it will also fall.

If consumers are expecting a rise in their future income, then their current demand will increase. If consumers are expecting a fall in their future income, then their current demand will decrease.

6. Number of Buyers

The number of buyers or consumers' population is another determinant of demand:

If buyers are more, then their demand will also be more.

If buyers are few, then their demand will also be small.

So, the population of consumers is directly proportional to the demand they generate in the market.

7. Climatic Condition

The climatic condition or weather of an area is also a determinant of demand. For example:

In colder regions, there is a high demand for woolen clothes.

In hotter areas, there is more demand for cotton clothes.

8. Level of Economic Activity

The level of economic activity is a determinant of demand: A high level of economic activity usually comprises of huge investments in infrastructure development, high rate of employment, high consumption power, rising standard of living of people, etc. It increases demand. Contrarily, a low level of economic activity displays characteristics like lack of investments, increase in the unemployment rate, low purchasing power, plunging living standards of people, so on. It decreases demand.

If economic activity increases, demand also increases and vice-versa.

DEMAND ANALYSIS:

Demand analysis is the process of understanding the customer demand for a product or service in a target market. Companies use demand analysis techniques to determine if they can successfully enter a market and generate expected profits to expand their business operations. It also gives a better understanding of the high-demand markets for the company's offerings, using which

businesses can determine the viability of investing in each of these markets.

Steps in market demand analysis:

- ❖ Market identification
- ❖ Business cycle
- ❖ Product Niche
- ❖ Evaluate competition

LAW OF DEMAND:

The Law of demand explains the functional relationship between price of a commodity and the quantity demanded of the commodity. It is observed that the price and the demand are inversely related which means that the two move in the opposite direction.

An increase in the price leads to a fall in quantity demanded and vice versa. This relationship can be stated as “Other things being equal, the demand for a commodity varies inversely as the price”.

Chief Characteristics of the Law of Demand

The following are the chief characteristics of the Law of Demand.

- 1. Inverse Relationship.** The relationship between price and the demand of a particular commodity is inverse i.e., the demand of a commodity will fall with the increase in the price of the commodity or it will increase with the fall in the price.
- 2. Price an Independent Variable and Demand a Dependent Variable.** In the Law of Demand, price is regarded as an independent variable that affects the demand inversely. Thus, it is the effect of price on demand that is to be examined and not the effect of demand on price.
- 3. It is a Qualitative Statement.** The Law of Demand simply explains the direction of change in the demand with the increase or decrease in the price of a commodity. It does not explain the quantum of change. The law is thus, a qualitative statement and not a quantitative statement.
- 4. Other thing remains the same.** The Law of Demand applies only when other things remain the same. In other words, there should be no change in factors influencing demand except price.

Exceptions to the law of demand:

- 1. Giffen Goods:** Giffen goods are the inferior goods whose demand increases with the increase in its prices. There are several inferior commodities, much cheaper than the superior substitutes often consumed by the poor households as an essential commodity. Whenever the price of the Giffen goods increases its quantity demanded also increases because, with an increase in the price, and the income remaining the same, the poor people cut the consumption of superior substitute and buy more quantities of Giffen goods to meet their basic needs.

2. **Veblen Goods:** Another exception to the law of demand is given by the economist Thorstein Veblen, who proposed the concept of “**Conspicuous Consumption.**” According to Veblen, there are a certain group of people who measure the utility of the commodity purely by its price, which means, they think that higher priced goods and services derive more utility than the lesser priced commodities.
3. **Expectation of Price Change in Future:** When the consumer expects that the price of a commodity is likely to further increase in the future, then he will buy more of it despite its increased price in order to escape himself from the pinch of much higher price in the future.
On the other hand, if the consumer expects the price of the commodity to further fall in the future, then he will likely postpone his purchase despite less price of the commodity in order to avail the benefits of much lower prices in the future.
4. **Ignorance:** Often people are misconceived as high-priced commodities are better than the low-priced commodities and rest their purchase decision on such a notion. They buy those commodities whose price are relatively higher than the substitutes.
5. **Emergencies:** During emergencies such as war, natural calamity- flood, drought, earthquake, etc., the law of demand becomes ineffective. In such situations, people often fear the shortage of the essentials and hence demand more goods and services even at higher prices.
6. **Change in fashion and Tastes & Preferences:** The change in fashion trend and tastes and preferences of the consumers negates the effect of law of demand. The consumer tends to buy those commodities which are very much ‘in’ in the market even at higher prices.
7. **Conspicuous Necessities:** There are certain commodities which have become essentials of the modern life. These are the goods which consumer buys irrespective of an increase in the price. For example TV, refrigerator, automobiles, washing machines, air conditioners, etc.
8. **Bandwagon Effect:** This is the most common type of exception to the law of demand wherein the consumer tries to purchase those commodities which are bought by his friends, relatives or neighbors. Here, the person tries to emulate the buying behavior and patterns of the group to which he belongs irrespective of the price of the commodity.

ASSUMPTIONS OF THE LAW OF DEMAND:

The Law of Demand is based on the following assumptions :

- (1) No change in taste, habits, preferences : It is assumed that there is no change in the taste, habits, preferences of a rational consumer. Thus, consumers' choice of product must remain the same.
- (2) No change in the income level: If the consumer's income rises, he will demand more though the prices of commodities rise. In such a situation, the law will not hold good.
- (3) No change in population : The law is based on the assumption that there should be no change in population, size, sex ratio, age composition, etc.

- (4) No change in prices of related goods : The law assumes that the prices of close substitutes and the complementary products should remain constant.
- (5) No expectation of future change in the price: If the consumers expect high rise in the price in future, they demand more though current price is high. In such condition, the Law of Demand cannot be verified.
- (6) No change in taxation : It is assumed that the structure of direct and indirect taxes remain constant. Thus, the disposable income of a consumer should remain the same.
- (7) No introduction of new product: It is assumed that there is no introduction of a new product in the market. Thus, the consumer's taste, habits and preferences remain constant.
- (8) No change in technology : The law assumes that the present technology of production remains constant.

Elasticity of Demand:

- ▶ Elasticity is a measure of a variable's sensitivity to a change in other variables—[or a single variable](#). Most commonly this sensitivity is the change in quantity demanded relative to changes in other factors, such as price.
- ▶ Elasticity is defined as a ratio of the percentage change in the dependent variable to the percentage change in the independent variable

1. Price Elasticity of Demand:

Price Elasticity of Demand: The price elasticity of demand, commonly known as the elasticity of demand refers to the responsiveness and sensitiveness of demand for a product to the changes in its price.

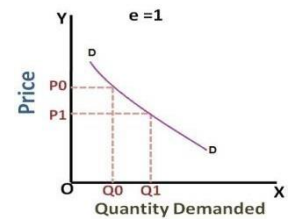
The response of the consumers to a change in the price of a commodity is measured by the price elasticity of the commodity demand.


$$\text{Elasticity of Demand Formula} = \frac{\text{Percentage Change in Demand}}{\text{Percentage Change in Price}}$$

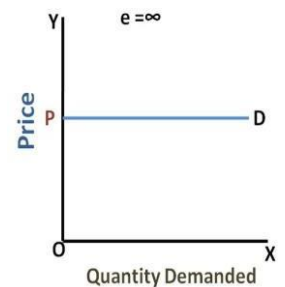

Types of price Elasticity of Demand

The following are the main types of price elasticity of demand:

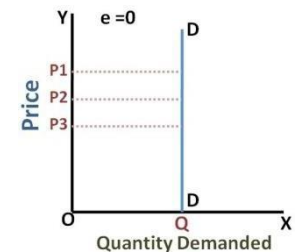
1. **Perfectly Elastic Demand ($E_p = \infty$):** The demand is said to be perfectly elastic when a slight change in the price of a commodity causes a major change in its quantity demanded. Such as, even a small rise in the price of a commodity can result into fall in demand even to zero. Whereas a little fall in the price can result in the increase in demand to infinity. In perfectly elastic demand the demand curve is a **straight horizontal line** which shows, the flatter the demand curve the higher is the elasticity of demand.



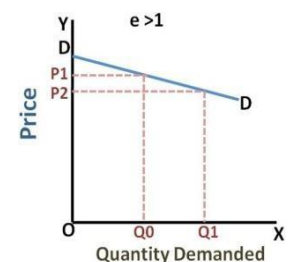
2. **Perfectly Inelastic Demand ($E_p = 0$):** When there is no change in the demand for a product due to the change in the price, then the demand is said to be perfectly inelastic. Here, the demand curve is a **straight vertical line** which shows that the demand remains unchanged irrespective of change in the price., i.e. quantity OQ remains unchanged at different prices, P1, P2, and P3.



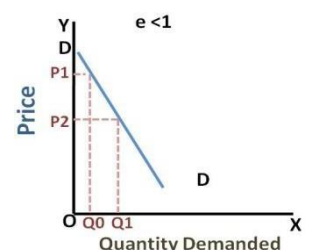
3. **Relatively Elastic Demand (1 to ∞):** The demand is relatively elastic when the proportionate change in the demand for a commodity is greater than the proportionate change in its price. Here, the demand curve is **gradually sloping** which shows that a proportionate change in quantity from OQ0 to OQ1 is greater than the proportionate change in the price from OP1 to OP2.



4. **Relatively Inelastic Demand (0-1):** When the proportionate change in the demand for a product is less than the proportionate change in the price, the demand is said to be relatively inelastic demand. It is also called as the elasticity less than unity, i.e. 1. Here the demand curve is **rapidly sloping**, which shows that the change in the quantity from OQ0 to OQ1 is relatively smaller than the change in the price from OP1 to OP2.



5. **Unitary Elastic Demand ($E_p = 1$):** The demand is unitary elastic when the proportionate change in the price of a product results in the same change in the quantity demanded. Here the shape of the demand curve is a **rectangular hyperbola**, which shows that area under the curve is equal to one.



FACTORS AFFECTING PRICE ELASTICITY OF DEMAND

The Factors affecting Price Elasticity of Demand are as follows:

1. Nature Of Commodity

Nature of the commodity is the most important factor that affects the price elasticity of demand. There are three types of commodities: Necessaries, Comforts and luxuries. Necessaries are those goods which are mandatory for survival of human being. Comforts are those which make our life smooth and happy living. Luxuries are the goods considered as status symbol. The price elasticity of demand of these three kinds of goods is:

NATURE OF GOODS	PRICE ELASTICITY OF DEMAND
Necessaries	Less elastic
Comforts	Unitary elastic
Luxuries	Highly elastic

2. Availability Of Substitutes

Substitutes are those which can be used in place of one another. Example: Tea and Coffee, Pepsi and Coke etc. The price elasticity in case of availability of substitutes is:

Easy Availability of Substitutes	Highly Elastic Demand
Less Substitutes Available	Less Elastic Demand

3. Alternative Or Multiple Uses

There are some commodities which can be used alternatively or these commodities have multiple uses like milk, coal etc. The demand for these goods rises as the price falls and vice versa. The price elasticity of demand in this case is as follows:

Goods having Multiple uses	Highly Elastic Demand
Goods having Single use	Less Elastic Demand

4. Time Period

In Economics, the time is divided into two horizons: Short Period and Long Period. The elasticity of demand is also get affected by the duration of time.

Long time period	Highly elastic demand
Short time period	Less elastic demand

5. Income Of Consumer

Elasticity of demand for a commodity also depends upon the income level of the consumers. If the buyers are high end consumers i.e. rich, they will not care for the price. Accordingly, elasticity of demand is expected to be low. Example: Demand for cars by multi-billionaires. On the other hand, if income level of the buyers is low, elasticity of demand is expected to be high. Example:

Demand for small cars by the middle class people in India.

6. Level Of Price

Elasticity of demand also depends upon the level of price of the commodity. The commodities having the highest or lowest price carry less elasticity of demand. On the contrary, the commodities with medium ranged price have highly elastic demand.

High price	Less elastic demand
Medium price	Highly elastic demand
Low price	Less elastic demand

7. Standard Of Living

The society where the standard of living of the people is high, the elasticity of demand is low and where the standard of living is low, the elasticity of demand is high.

8. Postponement Of Use Of Commodity

The commodities whose demand can be postponed in near future have more elastic demand. Example: Demand for luxury furniture or car etc. Whereas for those goods whose demand cannot be postponed have less elastic or inelastic demand. Example: Demand for food.

9. Habits Of Consumer

Commodities which have become habits of consumer have inelastic demand because demand does not get affected by change in price.

10. Proportion Of Income Spent On A Commodity

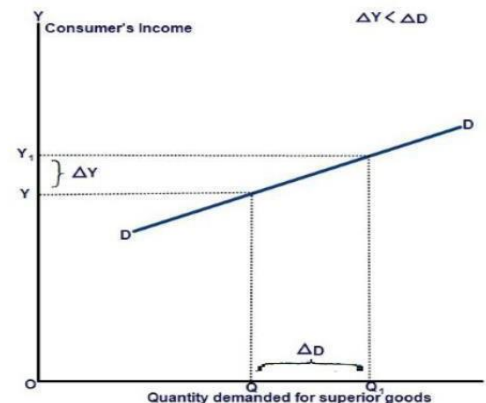
Goods on which consumer spend a small proportion of their income like toothpaste, newspaper etc. will have an inelastic demand. On the other hand, goods on which the consumers spend a large proportion of their income like clothes, scooter etc. tends to have elastic demand.

INCOME ELASTICITY OF DEMAND:

It measures the responsiveness of demand after a change in income level of individuals. Income elasticity of demand refers to the sensitivity of the quantity demanded for a certain good to a change in real income of consumers who buy this good, keeping all other things constant.

“It is the ratio of percentage change in quantity demanded over the percentage change in income level of individuals”.

$$\text{Income Elasticity of Demand} = \frac{\text{Percentage Change in Quantity Demand } (\Delta D/D)}{\text{Percentage Change in Income } (\Delta I/I)}$$



In the given figure, quantity demanded and consumer's income is measured along X-axis and Y-axis respectively. The small rise in income from OY to OY1 has caused greater rise in the quantity demanded from OQ to OQ1 and vice versa. Thus, the demand curve, DD shows income elasticity greater than unity.

CROSS ELASTICITY OF DEMAND:

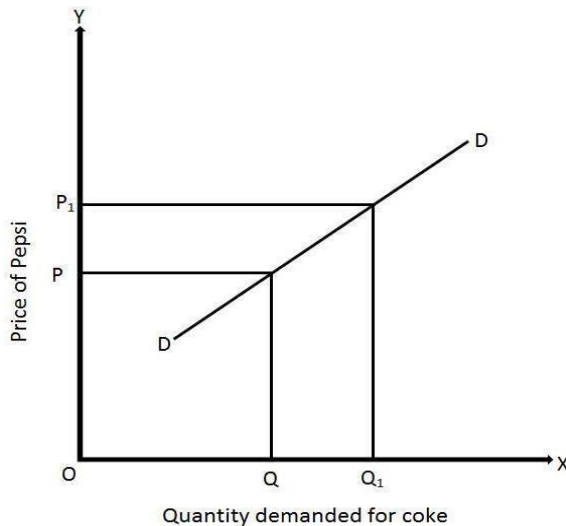
It measures the responsiveness of demand of one product, after a change in price level of other product. “It is the ratio of percentage change in quantity demanded of any one product (X) over the percentage change in price level of other product (Y)”.

The cross elasticity of demand for substitute goods and complimentary goods is always positive because the demand for one good increases when the price for the substitute goods and complimentary goods increases.

For example: if there is an increase in the price of tea by 10%. and the quantity demanded for coffee increases by 2%, then the cross elasticity of demand = $2/10 = +0.2$.

If two commodities are unrelated goods, the increase in the price of one good does not result in any change in the demand for the other goods.

$$\text{Cross Elasticity of Demand} = \frac{\% \text{ increase in quantity demanded of product A}}{\% \text{ increase in price of product B}}$$



In the above figure, quantity demanded for Coke and price of Pepsi are measured along X- axis and Y-axis respectively. When the price of Pepsi increases from OP to OP1, quantity demanded for coke rises from OQ to OQ1 and vice versa. Thus, the demand curve DD shows positive cross elasticity of demand.

ADVERTISING AND PROMOTIONAL ELASTICITY OF DEMAND:

In the modern competitive or partial competitive market economy, advertising has a great significance. Under advertising, various visible or verbal activities are done by the firm for the purpose of creating or increasing demand for its goods or services. Informative advertising is very helpful for the consumer in making rational purchase decisions.

“It is a ratio of percentage change in quantity demanded of any goods and services over the percentage change in expenses incurred for advertising and promotion”.

Following is the formula for advertising elasticity. $EA = \text{Percentage } \Delta Q / \text{Percentage } \Delta A$

$$\text{Cross Elasticity of Demand} = \frac{\% \text{ increase in quantity demanded}}{\% \text{ increase in advertisement expenses}}$$

The extension of demand through advertising can be measured by advertising or promotional elasticity of demand (EA) which measures the expected changes in demand as a result of change in other promotional expenses. The demand for some goods is affected more by advertising.

- Advertising elasticity of demand (AED) measures the impact advertising expenditure has in generating new sales for a company.
- Companies want a positive AED because this indicates their advertising efforts are resulting in an increased demand for their goods and services.
- AED may not be the most accurate predictor of advertising's impact on sales because it does not take into account other factors that affect demand, such as changes in consumer tastes and spending habits.

Consumer demand can also be impacted by the price of products and the availability of lower-priced substitutes.

USES OF ELASTICITY OF DEMAND FOR MANAGERIAL DECISION MAKING:

- **Determination of Price policy:** Determination of Prices means to determine the cost of goods sold and services rendered in the free market. A manufacturer has to consider the elasticity of demand for the product.
- **Price discrimination:** it is an act of selling same products at different prices to different section of customers or in different sub-markets. A monopolist adopts a price discrimination policy only when the elasticity of demand of different consumers or sub- markets is different. Consumers whose demand is inelastic can be charged a higher price than those with more elastic demand.
- **Public utility pricing:** In case of public utilities which are run as monopoly undertakings e.g. elasticity of water supply, railways, postal services, price discrimination is generally practiced, charging higher prices from consumers or users with inelastic demand and lower prices in case of elastic demand.
- **Shifting of tax burden:** It is possible for a business to shift a commodity tax in case of inelastic demand to his customers. But if the demand is elastic, he will have to bear the tax burden himself, otherwise demand for his goods will go down sharply.
- **Pricing of Joint supply products:** Certain goods, being products of the same process are jointly supplied, e.g. wool and mutton. Here if the demand for wool is inelastic compared to the demand for mutton, a higher price for wool can be charged with advantage.
- **Super Markets:** Super-markets are a combined set of shops run by a single organization selling a wide range of goods. They are supposed to sell commodities at lower prices than charged by shopkeepers in the bazaar. Hence, price policy adopted is to charge slightly lower price for goods with elastic demand.
- **Use of machines on employment:** Workers often oppose use of machines out of fear of unemployment. Machines need not always reduce demand for labor as this depends on price elasticity of demand for the commodity produced. When machines reduce costs and hence price of products, if the products demand is elastic, the demand will go up, production will have to be increased and more workers may be employed for the product

is inelastic, machines will lead to unemployment as lower prices will not increase the demand.
- **Factor pricing:** The factors having price inelastic demand can obtain a higher price than those with elastic demand. Workers producing products having inelastic demand can easily get their

wages raised.

- **Taxation policy:** Government can easily raise tax revenue by taxing commodities which are price inelastic.

DEMAND FORECASTING:

A forecast is a prediction or estimation of a future situation, under given conditions.

Demand forecasting is a technique that is used for the estimation of what can be the demand for the upcoming product or services in the future.

Demand forecasting is a field of predictive analytics which tries to understand and predict customer demand to optimize supply decisions by corporate supply chain and business management.

Demand estimation and forecasting means when, how, where, by whom and how much will be the demand for a product or service in near future. The process of demand estimation/forecasting can be broken into two parts i.e. analysis of the past conditions and analysis of current conditions with reference to a probable future trend. It helps in estimating the most likely demand of a good or service under given business conditions.

Significance of demand forecasting is shown in the following points

i. Fulfilling objectives:

- ▶ Implies that every business unit starts with certain pre-decided objectives. Demand forecasting helps in fulfilling these objectives. An organization estimates the current demand for its products and services in the market and move forward to achieve the set goals.
- ▶ For example, an organization has set a target of selling 50, 000 units of its products. In such a case, the organization would perform demand forecasting for its products. If the demand for the organization's products is low, the organization would take corrective actions, so that the set objective can be achieved.

ii. Preparing the budget:

Plays a crucial role in making budget by estimating costs and expected revenues. For instance, an organization has forecasted that the demand for its product, which is priced at Rs. 10, would be 10, 00, 00 units. In such a case, the total expected revenue would be $10 \times 100000 = \text{Rs. } 10, 00, 000$. In this way, demand forecasting enables organizations to prepare their budget.

iii. Stabilizing employment and production:

Helps an organization to control its production and recruitment activities. Producing according to the forecasted demand of products helps in avoiding the wastage of the resources of an organization. This further helps an organization to hire human resource according to requirement. For example, if an organization expects a rise in the demand for its products, it may opt for extra labor to fulfill the increased demand.

iv. Expanding organizations:

Implies that demand forecasting helps in deciding about the expansion of the business of the organization. If the expected demand for products is higher, then the organization may plan to expand further. On the other hand, if the demand for products is expected to fall, the organization may cut down the investment in the business.

v. Taking Management Decisions:

Helps in making critical decisions, such as deciding the plant capacity, determining the requirement

of raw material, and ensuring the availability of labor and capital.

vi. Evaluating Performance:

Helps in making corrections. For example, if the demand for an organization's products is less, it may take corrective actions and improve the level of demand by enhancing the quality of its products or spending more on advertisements.

vii. Helping Government:

Enables the government to coordinate import and export activities and plan international trade.

Methods of demand forecasting:

1. Opinion Polling Method/Qualitative methods

In this method, the opinion of the buyers, sales force and experts could be gathered to determine the emerging trend in the market.

The opinion polling methods of demand forecasting are of three kinds:

(a) Consumer's Survey Method or Survey of Buyer's Intentions:

In this method, the consumers are directly approached to disclose their future purchase plans. This is done by interviewing all consumers or a selected group of consumers out of the relevant population. This is the direct method of estimating demand in the short run.

(i) Complete Enumeration Survey:

Under the Complete Enumeration Survey, the firm has to go for a door to door survey for the forecast period by contacting all the households in the area. This method has an advantage of first hand, unbiased information, yet it has its share of disadvantages also. The major limitation of this method is that it requires lot of resources, manpower and time.

(ii) Sample Survey and Test Marketing:

Under this method some representative households are selected on random basis as samples and their opinion is taken as the generalised opinion. This method is based on the basic assumption that the sample truly represents the population. If the sample is the true representative, there is likely to be no significant difference in the results obtained by the survey. Apart from that, this method is less tedious and less costly.

A variant of sample survey technique is test marketing. Product testing essentially involves placing the product with a number of users for a set period. Their reactions to the product are noted after a period of time and an estimate of likely demand is made from the result.

(iii) End Use Method or Input-Output Method:

This method is quite useful for industries which are mainly producers goods. In this method, the sale of the product under consideration is projected as the basis of demand survey of the industries using this product as and intermediate product, that is, the demand for the final product is the end use demand of the intermediate product used in the production of this final product.

(b) Sales Force Opinion Method:

This is also known as collective opinion method. In this method, instead of consumers, the opinion of the salesmen is sought. Sales person in the company to make an individual forecast for his or her particular sales territory. These individual forecasts are discussed and agreed with the sales manager. The composite of all forecasts then constitutes the sales forecast for the organisation.

(c) Experts Opinion Method:

This method is also known as “Delphi Technique” of investigation. The Delphi method requires a panel of experts, who are interrogated through a sequence of questionnaires in which the responses to one questionnaire are used to produce the next questionnaire. Thus any information available to some experts and not to others is passed on, enabling all the experts to have access to all the information for forecasting.

2. Statistical Method/ Quantitative methods

Statistical methods have proved to be immensely useful in demand forecasting. In order to maintain objectivity, that is, by consideration of all implications and viewing the problem from an external point of view, the statistical methods are used.

(i) Trend Projection Method:

A firm existing for a long time will have its own data regarding sales for past years. Such data when arranged chronologically yield what is referred to as ‘time series’. Time series shows the past sales with effective demand for a particular product under normal conditions. Such data can be given in a tabular or graphic form for further analysis. This is the most popular method among business firms, partly because it is simple and inexpensive and partly because time series data often exhibit a persistent growth trend.

The trend can be estimated by using any one of the following methods:

a) Graphical Method:

This is the most simple technique to determine the trend. All values of output or sale for different years are plotted on a graph and a smooth free hand curve is drawn passing through as many points as possible. The direction of this free hand curve—upward or downward— shows the trend.

(b) Least Square Method:

Under the least square method, a trend line can be fitted to the time series data with the help of statistical techniques such as least square regression. When the trend in sales over time is given by straight line, the equation of this line is of the form: $y = a + bx$.

(ii) Barometric Technique:

This method is based on the notion that “the future can be predicted from certain happenings in the present.” In other words, barometric techniques are based on the idea that certain events of the present can be used to predict the directions of change in the future. This is accomplished by the use of economic and statistical indicators which serve as barometers of economic change.

(iii) Regression Analysis:

It attempts to assess the relationship between at least two variables (one or more independent and one dependent), the purpose being to predict the value of the dependent variable from the specific value of the independent variable.

The basis of this prediction generally is historical data. This method starts from the assumption that a basic relationship exists between two variables. An interactive statistical analysis computer package is used to formulate the mathematical relationship which exists.

(iv) Econometric Models:

Econometric models are an extension of the regression technique whereby a system of independent regression equation is solved. The requirements for satisfactory use of the econometric model in forecasting is under three heads: variables, equations and data.

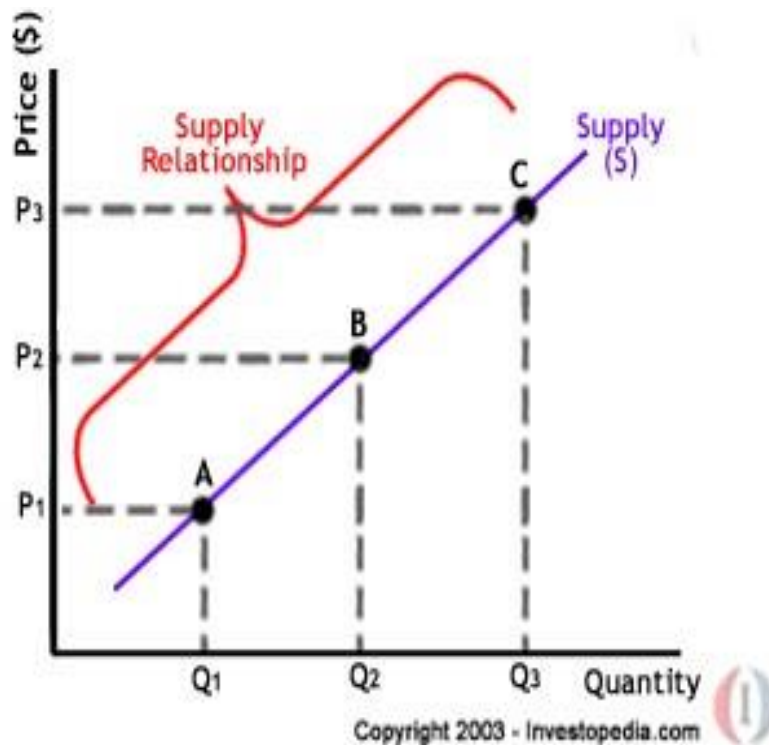
The appropriate procedure in forecasting by econometric methods is model building. Econometrics attempts to express economic theories in mathematical terms in such a way that they can be verified by statistical methods and to measure the impact of one economic variable upon another so as to be able to predict future events.

LAW OF SUPPLY

The law of supply is the microeconomic law that states that, all other factors being equal, as the price of a good or service increases, the quantity of goods or services that suppliers offer will increase, and vice versa.

The law of supply says that as the price of an item goes up, suppliers will attempt to maximize their profits by increasing the number of items for sale.





- An economic law stating that as the price of a good or service increases, the quantity supplied increases, and vice versa .
- The law of supply says that a higher price will induce producers to supply a higher quantity to the market.
- Because businesses seek to increase revenue, when they expect to receive a higher price for something, they will produce more of it.
- When college students learn that computer engineering jobs pay more than English professor jobs, the supply of students with majors in computer engineering will increase

FACTORS AFFECTING SUPPLY:

There are many factors affecting the supply of a commodity in the market including

1. Input costs,
2. Price of the commodity,
3. The state of technology at a given time, taxation,
4. Prices of other goods,
5. Objective of the seller,
6. Number of firms selling the same commodity among others.

EXCEPTIONS TO THE LAW OF SUPPLY:

- ☐ The law will not apply **if there are future expectations for further** change in prices.
- ☐ If **sellers expect further fall in prices** in future, they would be ready to sell more even at low prices.
- ☐ For **perishable goods like milk, vegetables, fish, eggs, etc. the supply is not affected by their prices**. Sellers cannot hold these goods for long.

Even for **agricultural goods, the supply depends more on natural factors** such as drought, floods, natural calamities etc. and less on their prices.

COST ANALYSIS & PRODUCTION ANALYSIS

Concepts of Production, production function with one variable input - Law of Variable Proportion, Laws of returns to scale, Indifference Curves, ISO-Quants & ISO-Cost line, Economies of scale, Diseconomies of scale. Types of cost, Cost curves, Cost – Output Relationship in the short run and in the long run, Long- Run Average Cost (LAC) curve

Break Even Analysis—Meaning, Assumptions, Determination of BEA, Limitations, Margin of safety, Uses of BEA In Managerial decisions (Theory and simple Problems).

INTRODUCTION OR PRODUCTION CONCEPTS:

Production can be defined as an organized activity of transforming physical inputs into output which will satisfy the products needs of the society.

or

Production is an act of creating value that satisfies the wants of the individuals.

Production refers to the transformation of inputs or resources into outputs or goods and services.

Production is a process in which economic resources or inputs (composed of natural resources like labour, land and capital equipment) are combined by entrepreneurs to create economic goods and services (outputs or products).

PRODUCTION ANALYSIS:

Production Analysis refers to analyzing the inputs or resources that are used to produce a firm's final product. It defines the relationships between the prices of the commodities and productive factors on one hand and the quantities of these commodities and productive factors that are produced on the other hand.

Production analysis basically is concerned with the analysis in which the resources such as land, labor, and capital are employed to produce a firm's final product. To produce these goods the basic inputs are classified into two divisions

Inputs : Fixed inputs and Variable inputs

The factors of production that carry out the production is called inputs.

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Fixed inputs	Variable inputs
☐ Remain the same in the short period . remain the same. Cost ☐ Examples:- Building, Land etc ☐ (In the long run fixed inputs are become varies)	☐ In the long run all factors of production are varies according to the volume of outputs. ☐ The cost of variable inputs is called Variable Cost ☐ Example:- Raw materials, labour, etc

What is Production Function?

Production Function signifies the technical relationship between the physical inputs and physical outputs of the firm for given production technology.

The basic relationship between the factors of production and the output is referred to as a Production Function.

The firm's production function for a particular good (q) shows the maximum amount of the good that can be produced using alternative combinations of capital (K) and labor (L)

The production function expresses a functional relationship between physical inputs and physical outputs of a firm at any particular time period. The output is thus a function of inputs. Mathematically production function can be written as

$$Q = f(L_1, L_2, C, O, T)$$

Importance:

1. When inputs are specified in physical units, production function helps to estimate the level of production.
2. It becomes is equates when different combinations of inputs yield the same level of output.
3. It indicates the manner in which the firm can substitute on input for another without altering the total output.
4. When price is taken into consideration, the production function helps to select the least combination of inputs for the desired output.
5. It considers two types" input-output relationships namely „law of variable proportions" and „law of returns to scale". Law of variable propositions explains the pattern of output in the short-run as the units of variable inputs are increased to increase the output. On the other hand law of returns to scale explains the pattern of output in the long run as all the units of inputs are increased.

6. The production function explains the maximum quantity of output, which can be produced, from any chosen quantities of various inputs or the minimum quantities of various inputs that are required to produce a given quantity of output.

Assumptions:

Production function has the following assumptions.

1. The production function is related to a particular period of time.
2. There is no change in technology.
3. The producer is using the best techniques available.
4. The factors of production are divisible.
5. Production function can be fitted to a short run or to long run.

Types of production function:-

These two types of relationships have been explained in the form of laws.

- i) Law of variable proportions (short run production function)
- ii) Law of returns to scale (long run production function)

I. Law of variable proportions:

It is also called law of diminishing returns or law of returns to variable factor.

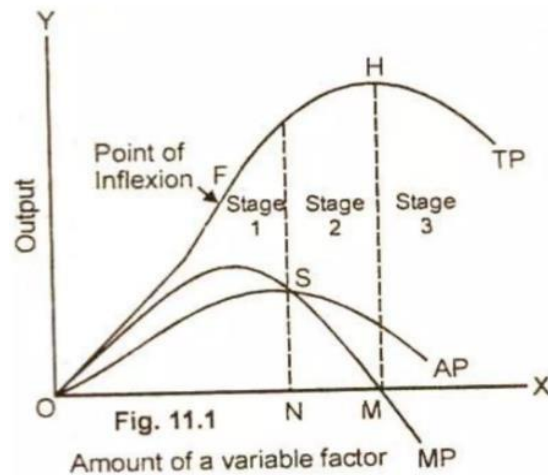
The law of variable proportions which is a new name given to old classical concept of “Law of diminishing returns has played a vital role in the modern economics theory. Assume that a firms production function consists of fixed quantities of all inputs (land, equipment, etc.) except labour which is a variable input when the firm expands output by employing more and more labour it alters the proportion between fixed and the variable inputs. The law can be stated as follows:

“When total output or production of a commodity is increased by adding units of a variable input while the quantities of other inputs are held constant, the increase in total production becomes after some point, smaller and smaller”

“If equal increments of one input are added, the inputs of other production services being held constant, beyond a certain point the resulting increments of product will decrease i.e. the marginal product will diminish”. (G. Stigler)

“As the proportion of one factor in a combination of factors is increased, after a point, first the marginal and then the average product of that factor will diminish”. (F. Benham)

The law of variable proportions refers to the behaviour of output as the quantity of one Factor is increased Keeping the quantity of other factors fixed and further it states that the marginal product and average product will eventually do cline. This law states three types of productivity an input factor – Total, average and marginal physical productivity.



Assumptions of the Law: The law is based upon the following assumptions:

- i) The state of technology remains constant. If there is any improvement in technology, the average and marginal output will not decrease but increase.
- ii) Only one factor of input is made variable and other factors are kept constant. This law does not apply to those cases where the factors must be used in rigidly fixed proportions.
- iii) All units of the variable factors are homogenous.

Three stages of law:

The behaviors of the Output when the varying quantity of one factor is combines with a fixed quantity of the other can be divided in to three district stages. The three stages can be better understood by following the table.

Table 11.1: Production Schedule

Units of Variable Input (Labour) (n)	Total Product (TP)	Average Product (AP) (TP_n/n)	Marginal Product (MP) ($TP_n - TP_{n-1}$)	
1	20	20	20	Stage I
2	50	25	30	
3	90	30	40	
4	120	30	30	
5	135	27	15	Stage II
6	144	24	9	
7	147	21	3	
8	148	18.5	1	
9	148	16.4	0	Stage III
10	145	14.5	- 3	

In the first stage, total product increases at an increasing rate. The marginal product in this stage increases at an increasing rate resulting in a greater increase in total product. The average product also increases. This stage continues up to the point where average product is

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equal to marginal product. The law of increasing returns is in operation at this stage. The law of diminishing returns starts operating from the second stage onwards. At the second stage total product increases only at a diminishing rate. The average product also declines.

The second stage comes to an end where total product becomes maximum and marginal product becomes zero. The marginal product becomes negative

In the third stage. So the total product also declines. The average product continues to decline.

We can sum up the above relationship thus when „A.P.“ is rising, „M. P.“ rises more than „A. P.“; When „A. P.“ is maximum and constant, „M. P.“ becomes equal to „A. P.“ when „A. P.“ starts falling, „M. P.“ falls faster than „A. P.“ Thus, the total product, marginal product and average product pass through three phases, viz., increasing diminishing and negative returns stage. The law of variable proportion is nothing but the combination of the law of increasing and diminishing returns.

Total Product	Marginal Product	Average Product
Stage I First increases at increasing rate then at diminishing rate.	Increases in the beginning then reaches a maximum and begins to decrease.	First increases, continues to increase and becomes maximum.
Stage II Continues to increase at diminishing rate and becomes maximum.	Continues to diminish and becomes equal to zero.	Becomes equal to MP and then begins to diminish.
Stage III Diminishes	Becomes negative.	Continues to diminish but will always be greater than zero.

PRODUCTION FUNCTION WITH TWO VARIABLE INPUTS:

Refers to the relationship between the output resulting in production by varying the two or more inputs. There may be various technical possibilities of producing a given output by using different factor combinations. Which particular factor combination will be actually selected by the firm depends both on the technical possibilities of factor substitution as well as on the prices of the factors of production.

Production Function with two Variable Inputs explains the production behavior of the firm with all variable factors. We are restricting our analysis to two variable inputs because it simply allows us the scope for graphical analysis.\

II. Law of Returns of Scale:

The law of returns to scale explains the behavior of the total output in response to change in the scale of the firm, i.e., in response to a simultaneous to changes in the scale of the firm, i.e., in response to a simultaneous and proportional increase in all the inputs. More precisely, the Law of returns to scale explains how a simultaneous and proportionate increase in all the inputs affects the total output at its various levels.

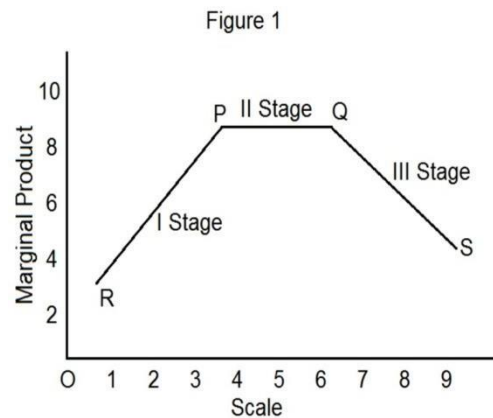
When a firm expands, its scale increases all its inputs proportionally, then technically there are three possibilities. (i) The total output may increase proportionately (ii) The total output may increase more than proportionately and (iii) The total output may increase less than proportionately.

Assumptions of Law of returns to scale:

- All inputs or factors of production are variable
- Production technology remains constant
- Product produce is measure in physical units

Unit	Scale of Production	Total Returns	Marginal Returns
1	1 labor + 2 Acres of Land	4	4 (Stage I - Increasing Returns)
2	2 Labor + 4 Acres of Land	10	6
3	3 Labor + 6 Acres of Land	18	8
4	4 Labor + 8 Acres of Land	28	10 (Stage II - Constant Returns)
5	5 Labor + 10 Acres of Land	38	10
6	6 Labor + 12 Acres of Land	48	10
7	7 Labor + 14 Acres of Land	56	8 (Stage III - Decreasing Returns)
8	8 Labor + 16 Acres of Land	62	6

Stage 1: Increasing Returns to Scale In figure, stage I represents increasing returns to scale. During this stage, the firm enjoys various internal and external economies such as technical economies, managerial economies and marketing economies. Economies simply mean advantages for the firm. Due to these economies, the firm realizes increasing returns to scale. This stage is also explained saying increasing returns in terms of



“increased efficiency” of labor and capital in the improved organization with the expanding scale of output and employment factor unit. It is referred to as the economy of organization in the earlier stages of production.

Increasing Return to Scale: If increase in the output is greater than the proportional increase in the inputs, it means increasing return to scale

Stage 2: Constant Returns to Scale

In figure, the stage II represents constant returns to scale. During this stage, the economies accrued during the first stage start vanishing and diseconomies arise. Diseconomies refers to the limiting factors for the firm’s expansion. Emergence of diseconomies is a natural process when a firm expands beyond certain stage. In the stage II, the economies and diseconomies of scale are exactly in balance over a particular range of output. When a firm is at constant returns to scale, an increase in all inputs leads to a proportionate increase in output but to an extent.

Constant returns to scale: If increase in the total output is proportional to the increase in input, it means constant returns to scale.

Stage 3: Diminishing Returns to Scale

In figure, the stage III represents diminishing returns or decreasing returns. This situation arises when a firm expands its operation even after the point of constant returns. Decreasing returns mean that increase in the total output is not proportionate according to the increase in the input. Because of this, the marginal output starts decreasing. Important factors that determine diminishing returns are managerial inefficiency and technical constraints.

Diminishing Returns to Scale: If increase in the output is less than proportional increase in the inputs, it means diminishing returns to scale.

ISOQUANTS:

The term Isoquants is derived from the words „iso“ and „quant“ – „Iso“ means equal and „quant“ implies quantity. Isoquant therefore, means equal quantity. A family of iso-product curves or isoquants or production difference curves can represent a production function with two variable inputs, which are substitutable for one another within limits.

Isoquants are the curves, which represent the different combinations of inputs producing a particular quantity of output. Any combination on the Isoquant represents the same level of output.

For a given output level firm's production becomes,

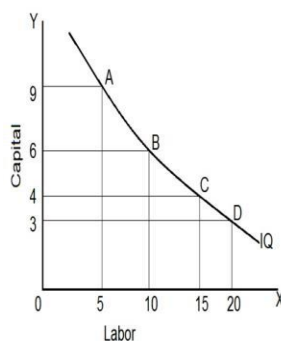
$$Q = f(L, K)$$

Where „Q“, is the units of output is a function of the quantity of two inputs „L“ and „K“.

Thus an Isoquant shows all possible combinations of two inputs, which are capable of producing equal or a given level of output. Since each combination yields same output, the producer becomes indifferent towards these combinations.

Assumptions:

1. There are only two factors of production, viz. labour and capital.
2. The two factors can substitute each other up to a certain limit
3. The shape of the Isoquant depends upon the extent of substitutability of the two inputs.
4. The technology is given over a period.



The above table is based on the assumption that only two factors of production, namely, Labor and Capital are used for producing 100 meters of cloth.

Combination A = $5L + 9K = 100$ meters of cloth

Combination B = $10L + 6K = 100$ meters of cloth

Combination C = $15L + 4K = 100$ meters of cloth

Combination D = $20L + 3K = 100$ meters of cloth

Marginal Rate of Technical Substitution:

Marginal rate of technical substitution in the theory of production is similar to the concept of marginal rate of substitution in the indifference curve analysis of consumer's demand. Marginal rate of technical substitution indicates the rate at which factors can be substituted at the margin without altering the level of output.

PROPERTIES OF ISOQUANTS

Property 1: An isoquant curve slopes downward, or is negatively sloped. This means that the same level of production only occurs when increasing units of input are offset with lesser

units of another input factor. This property falls in line with the principle of the Marginal Rate of Technical Substitution (MRTS). As an example, the same level of output could be achieved by a company when capital inputs increase, but labor inputs decrease.

Property 2: An isoquant curve, because of the MRTS effect, **is convex to its origin**. This indicates that factors of production may be substituted with one another. The increase in one factor, however, must still be used in conjunction with the decrease of another input factor.

Property 3: Isoquant curves cannot be tangent or intersect one another. Curves that intersect are incorrect and produce results that are invalid, as a common factor combination on each of the curves will reveal the same level of output, which is not possible.

Property 4: Isoquant curves in the **upper portions of the chart yield higher outputs**. This is because, at a higher curve, factors of production are more heavily employed. Either more capital or more labor input factors result in a greater level of production.

Property 5: An isoquant curve **should not touch the X or Y axis on the graph**. If it does, the rate of technical substitution is void, as it will indicate that one factor is responsible for producing the given level of output without the involvement of any other input factors.

Property 6: Isoquant curves **do not have to be parallel to one another**; the rate of technical substitution between factors may have variations.

Property 7: Isoquant curves are oval-shaped, allowing firms to determine the most efficient factors of production.

TYPES OF ISOQUANT:

Isoquants are differentiated on the basis of substitutability of the factors of production and they are as below

1. Linear Isoquant:

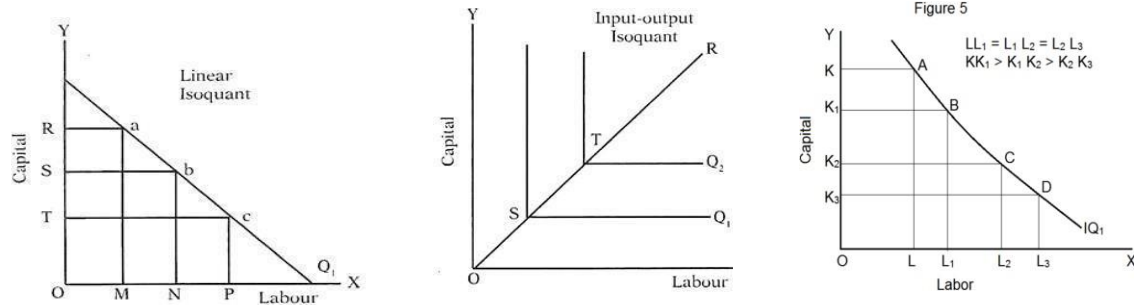
This type of isoquant are depicted by a straight line sloping downward from left to right, as shown in Figure-(a). It indicated a perfect and unlimited substitutability between two factors implying that the product may be produced even by using only capital or labour or by infinite combinations of the two factors.

2. Right angled Isoquant:

are L-shaped curve Figure-(b) and also known as Leontief isoquants. They assume a perfect complementary nature between factors implying zero substitutability. Factors are jointly used in a fixed proportion. It means that there is only one method of production to produce a commodity. Hence, to increase output, both factors are to be increased holding the proportion constant.

3. Convex Isoquant:

In this the inputs can be substituted but not perfectly. The curve of this type is convex to origin as we always consider for the isoquant curve.



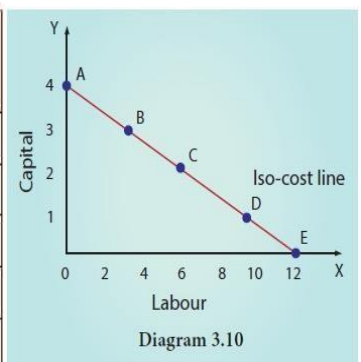
ISO-COST CURVE:

Isocost line shows all combinations of inputs which cost the same total amount. The use of the Isocost line pertains to cost-minimization in production, as opposed to utility-maximization. An Isocost line shows the maximum amount which a firm is willing to expend on production.

Slope of iso cost line: With the change in the factor prices the slope of iso cost line will change. If the price of labour falls the firm could buy more of labour and the line will shift away from the origin. The slope depends on the prices of factors of production and the amount of money which the firm spends on the factors. When the amount of money spent by the firm changes, the isocost line may shift but its slope remains the same. A change in factor price makes changes in the slope of isocost lines as shown in the figure.

Table 3.3 The Iso-cost

Combinations	Units of Capital Price = ₹30	Units of Labour Price = ₹10	Total Expenditure (in Rupees)
A	4	0	120
B	3	3	120
C	2	6	120
D	1	9	120
E	0	12	120

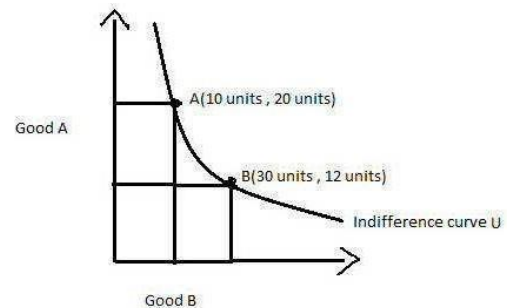


INDIFFERENCE CURVE: An indifference curve connects points on a graph representing different quantities of two goods, points between which a consumer is indifferent. An indifference curve is a graph showing combination of two goods that give the consumer equal

satisfaction and utility. Each point on an indifference curve indicates that a consumer is indifferent between the two and all points give him the same utility.

Graphically, the indifference curve is drawn as a downward sloping convex to the origin. The graph shows a combination of two goods that the consumer consumes. The above diagram shows the U indifference curve showing

bundles of goods A and B. To the consumer, bundle A and B are the same as both of them give him the equal satisfaction. In other words, point A gives as much utility as point B to the individual. The consumer will be satisfied at any point along the curve assuming that other things are constant.



LEAST COST COMBINATION FACTOR:

Producer's equilibrium or optimization occurs when he earns maximum profit with optimal combination of factors. A profit maximization firm faces two choices of optimal combination of factors

1. To Maximize its output for a given cost

2. To minimize its cost for given output

Thus the least cost combination of factors refers to a firm producing the largest volume of output from a given cost and producing a given level of output with the minimum cost when the factors are combined in an optimum manner.

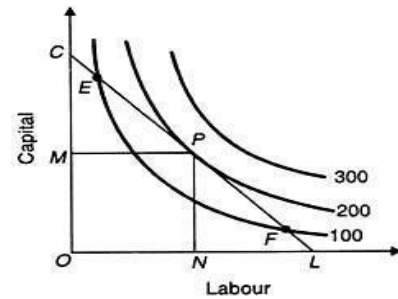
Assumptions made for to understand this concept are:

1. There are two factors, labour and capital.
2. All units of labour and capital are homogeneous.
3. The prices of units of labour (w) and that of capital (r) are given and constant.
4. The cost outlay is given.
5. The firm produces a single product.
6. The price of the product is given and constant.
7. The firm aims at profit maximisation.
8. There is perfect competition in the factor market.

1. To maximize output for given cost:

The firm maximizes its profits by maximizing its output, given its cost outlays, and the prices of the two factors. This analysis is based on the same assumptions, as given above From the

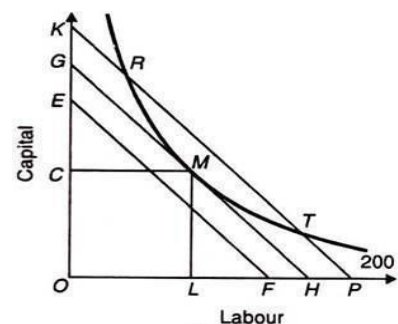
fig, the firm can reach the optimal factor combination level of maximum output by moving along the iso-cost line CL from either point E or F to point P. This movement involves no extra cost because the firm remains on the same iso-cost line the firm is maximising its output level of 200 units by employing the optimal combination of OM of capital and ON of labour, given its cost outlay CL.



But it cannot be at points E or F on the iso-cost line CL, since both points give a smaller quantity of output, being on the isoquant 100, than on the isoquant 200. The firm cannot attain a higher level of output such as isoquant 300 because of the cost constraint. The firm is in equilibrium at point P where the isoquant curve 200 is tangent to the iso-cost line CL. Thus the equilibrium point has to be P with optimal factor combination OM + ON.

2. To minimize cost for given output:

Given these assumptions, the point of least-cost combination of factors for a given level of output is where the isoquant curve is tangent to an iso-cost line. The iso-cost line GH is tangent to the isoquant 200 at point M. The firm employs the combination of OC of capital and OL of labour to produce 200 units of output at point M with the given cost-outlay GH.



At this point, the firm is minimising its cost for producing 200 units.

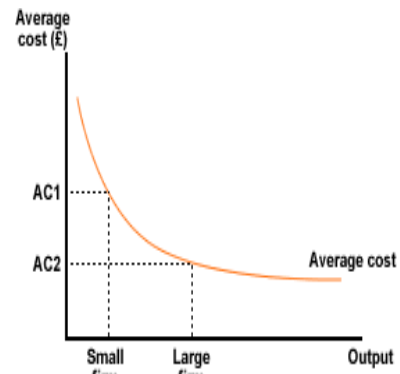
Any other combination on the isoquant 200, such as R or T, is on the higher iso-cost line KP which shows higher cost of production. The iso-cost line EF shows lower cost but output 200 cannot be attained with it. Therefore, the firm will choose the minimum cost point M which is the least-cost factor combination for producing 200 units of output.

M is thus the optimal combination for the firm. The point of tangency between the iso-cost line and the isoquant is an important first order condition but not a necessary condition for the producer's equilibrium.

ECONOMIES & DISECONOMIES OF SCALE: Economies of scale are defined as the cost advantages that an organization can achieve by expanding its production in the long run. In other words, these are the advantages of large scale production of the organization. The cost advantages are achieved in the form of lower average costs per unit.

It is a long term concept. Economies of scale are achieved when there is an increase in the sales of an organization. Economies of scale are the cost advantages that enterprises obtain due to their scale of operation, with cost per unit of output decreasing which causes scale increasing.

Figure illustrates that average cost falls as output increases, with the result that large firms may enjoy lower costs than smaller competitors. This competitive cost advantage allows large firms to have larger profit margins and have more options in pricing policy.



Reason for Economies of Scale:

1. Managerial - managers are on a fixed salary
2. Marketing - advertising, endorsements promotional events do not directly depend on quantity produced
3. Technical - machinery, buildings etc are paid for as a fixed amount
4. Bulk buying - remember it is the cost per unit of buying in bulk not the total cost (Great example is supermarkets and local shop)
5. Financial - similar in principle to buying in bulk but this time interest rates a more favorable.

The economies of scale are divided in to internal economies and external economies:

INTERNAL ECONOMIES:

As a firm increases its scale of production, the firm enjoys several economies named as internal economies. Basically, internal economies are those which are special to each firm. For example, one firm will enjoy the advantage of good management; the other may have the advantage of specialisation in the techniques of production and so on.

“Internal economies are those which are open to a single factory, or a single firm independently of the action of other firms. These result from an increase in the scale of output of a firm and cannot be achieved unless output increases.” Cairncross

Refer to real economies which arise from the expansion of the plant size of the organization. These economies arise from the growth of the organization itself.

A. Technical Economies of Scale: Technical economies have their influence on the size of the firm. Generally, these economies accrue to large firms which enjoy higher efficiency from capital goods or machinery. Bigger firms having more resources at their disposal are

able to install the most suitable machinery. Therefore, a firm producing on large scale can enjoy economies by the use of superior techniques.

Technical economies are of three kinds:

(i) Economies of Dimension:

A firm by increasing the scale of production can enjoy the technical economies. When a firm increases its scale of production, average cost of production falls but its average return will be more.

(ii) Economies of Linked Process:

A big firm can also enjoy the economies of linked process. A big firm carries all productive activities. These activities get economies. These linked activities save time and transport costs to the firm.

(iii) Economies of the Use of By-Products:

All the large sized firms are in a position to use its by-products and waste-material to produce another material and thus, supplement to their income. For instance, sugar industries make power, alcohol out of the molasses.

B. Marketing Economies of Scale:

When the scale of production of a firm is increased, it enjoys numerous selling or marketing economies. In the marketing economies, we include advertisement economies, opening up of show rooms, appointment of sole distributors etc. Moreover, a large firm can conduct its own research to effect improvement in the quality of the product and to reduce the cost of production. The other economies of scale are advertising economies, economies from special arrangements with exclusive dealers. In this way, all these acts lead to economies of large scale production.

C. Labor Economies of Scale:

As the scale of production is expanded they accrue many labour economies, like new inventions, specialization, time saving production etc. A large firm employs large number of workers. Each worker is given the kind of job he is fit for. The personnel officer evaluates the working efficiency of the labour if possible. Workers are skilled in their operations which save production, time and simultaneously encourage new ideas.

D. Managerial Economies of Scale:

Managerial economies refer to production in managerial costs and proper management of large scale firm. Under this, work is divided and subdivided into different departments. Each department is headed by an expert who keeps a vigil on the minute details of his department.

A small firm cannot afford this specialisation. Experts are able to reduce the costs of production under their supervision. These also arise due to specialization of management and mechanisation of managerial functions.

E. Economies of Transport and Storage:

A firm producing on large scale enjoys the economies of transport and storage. A big firm can have its own means of transportation to carry finished as well as raw material from one place to another. Moreover, big firms also enjoy the economies of storage facilities. The big firm also has its own storage and go down facilities. Therefore, these firms can store their products when prices are unfavorable in the market.

EXTERNAL ECONOMIES:

□ External economies refer to all those benefits which accrue to all the firms operating in a given industry. Generally, these economies accrue due to the expansion of industry and other facilities expanded by the Government.

□ According to Cairncross, “External economies are those benefits which are shared in by a number of firms or industries when the scale of production in any industry increases.” Moreover, the simplest case of an external economy arises when the scale of production function of a firm contains as an implicit variable the output of the industry. A good example is that of coal mines in a locality.

□ Occur outside the organization. These economies occur within the industries which benefit organizations. When an industry expands, organizations may benefit from better transportation network, infrastructure, and other facilities. This helps in decreasing the cost of an organization.

A. Economies of Concentration:

As the number of firms in an area increases each firm enjoys some benefits like, transport and communication, availability of raw materials, research and invention etc. Further, financial assistance from banks and non-bank institutions easily accrue to firm. We can, therefore, conclude that concentration of industries lead to economies of concentration.

B. Economies of Information:

When the number of firms in an industry expands they become mutually dependent on each other. In other words, they do not feel the need of independent research on individual basis. Many scientific and trade journals are published. These journals provide information to all the firms which relates to new markets, sources of raw materials, latest techniques of production etc.

C. Economies of Disintegration

As an industry develops, all the firms engaged in it decide to divide and sub-divide the process of production among themselves. Each firm specializes in its own process. For instance, in case of moped industry, some firms specialize in rims, hubs and still others in chains, pedals, tires etc. It is of two types-horizontal disintegration and vertical disintegration. In case of horizontal disintegration each firm in the industry tries to specialize in one particular item whereas, under vertical disintegration every firm endeavors to specialize in different types of items. Material of one firm may be available and useable as raw materials in the other firms. Thus, wastes are converted into by-products.

The selling firms reduce their costs of production by realizing something for their wastes. The buying firms gain by getting other firms' wastes as raw materials at cheaper rates. As a result of this, the average cost of production declines.

SIGNIFICANCE OF ECONOMIES OF SCALE:

The significance of economies of scale is discussed as under:

(a). Nature of the Industry:

The foremost significance of economies of scale is that it plays an important role in determining the nature of the industry i.e. increasing cost industry, constant cost industry or decreasing cost industry.

(b). Analysis of Cost of Production:

When an industry expands in response to an increase in demand for its products, it experiences some external economies as well as some external diseconomies. The external economies tend to reduce the costs of production and thereby causing an upward shift in the long period average cost curve, whereas the external diseconomies tend to raise the costs and thereby causing an upward shift in the long period average cost curve. If external diseconomies outweigh the external economies, that is, when there are net external diseconomies, the industry would be an Increasing cost industry.

Diseconomies of scale are when the cost per unit of production (Average cost) increases because the output (sales) increases. **Reasons for diseconomies of scale**

1. Communication - becomes more complex
2. Coordination - between departments
3. X- Inefficiency - management costs increase (non-productive costs)
4. Principle agent problem - delegating to employees who are not as committed as the owner

Cost analysis:

Cost Analysis refers to the measure of the cost – output relationship, i.e. the economists are concerned with determining the cost incurred in hiring the inputs and how well these can be re- arranged to increase the productivity (output) of the firm.

Cost refers to the money value that is incurred in acquiring the resources and producing the product.

Determinants of Cost:

Level of output: The cost of production varies according to the quantum of output. If the size of production is large then the cost of production will also be more.

Price of input factors: A rise in the cost of input factors will increase the total cost of production.

Productivities of factors of production: When the productivity of the input factors is high then the cost of production will fall.

Size of plant: The cost of production will be low in large plants due to mass production with mechanization.

Time period: In the long run cost of production will be stable.

Technology: When the organization follows advanced technology in their process then the cost of production will be low.

TYPES OF COST

Fixed Costs (FC): The costs which don't vary with changing output. Fixed costs might include the cost of building a factory, insurance and legal bills. Even if your output changes or you don't produce anything, your fixed costs stay the same.

Costs (VC): Costs which depend on the output produced. For example, if you produce more cars, you have to use more raw materials such as metal. This is a variable cost.

Marginal Costs (MC): is the cost that incurred due to the production of an extra or additional unit.

Opportunity Cost: Opportunity cost is the next best alternative foregone. "It is cost incurred due to one's own choice", i.e the opportunity an individual loose by choosing other.

Incremental cost: Incremental cost is the total **cost** incurred due to an additional unit of product being produced.

Explicit costs: these are costs that a firm directly pays for and can be seen on the accounting sheet. Explicit costs can be variable or fixed and it is a clear amount.

Implicit costs: these are opportunity costs, which do not necessarily appear on its balance sheet but affect the firm. For example the assets of the owner used for the purpose of business.

Sunk Cost: these are costs that have been incurred and cannot be recouped. Depreciation is considered as one of the best example for sunk cost. Example if you spend money on advertising to enter an industry, you can never claim these costs back.

Future Cost: It is a type of cost, where the cost incurred to estimate the future data or forecasting of the organization. It is a type of cost which cannot be exceptional in nature.

Social Cost: This is the total cost to society. It will include the private costs plus also the external cost (cost incurred by a third party). May also be referred to as „True costs“ where the company spends on betterment of the society.

COST CURVES:

Total fixed costs: Given that *total fixed costs* (TFC) are constant as output increases, the curve is a horizontal line on the cost graph.

In economic analysis, the following types of costs are considered in studying cost data of a firm:

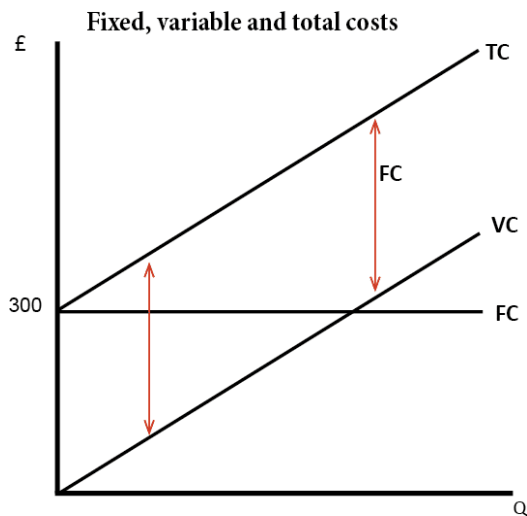
- ☐ Total Cost (TC),
- ☐ Total Fixed Cost (TFC),
- ☐ Total Variable Cost (TVC),
- ☐ Average Fixed Cost (AFC),
- ☐ Average Variable Cost (AVC),
- ☐ Average Total Cost (ATC), and
- ☐ Marginal Cost (MC).

Total Fixed Cost (TFC). It is the cost pertaining to all fixed inputs like machinery, etc., at any given level of output.

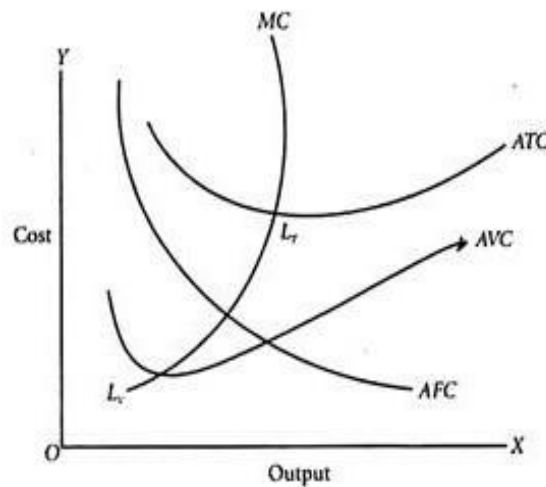
☐ **Total Variable Cost (TVC).** It is the cost pertaining to all variable inputs like raw materials, etc., at any given level of output.

Total Cost (TC). It is the cost pertaining to the entire factor inputs at any given level of output. It is the total cost of production derived by aggregating total fixed and variable costs together.

Thus, $TC = TFC + TVC$.



Short-run Output Cost Curves



Units of Output Q	Total fixed cost TFC	Total variable cost TVC	Total cost (TFC + TVC) TC	Average variable cost (TVC / Q) AVC	Average fixed cost (TFC / Q) AFC	Average cost (TC/Q) AC	Marginal cost MC
0	60	—	60	—	—	—	—
1	60	20	80	20	60	80	20
2	60	36	96	18	30	48	16
3	60	48	108	16	20	36	12
4	60	64	124	16	15	31	16
5	60	90	150	18	12	30	26
6	60	132	192	22	10	32	42

Behaviour of Total Costs

Examining cost schedules in Table 9.1, we may observe the following interesting points about the behaviour of various total costs:

- **TFC remains constant at all levels of output.** It is the same even when the output is nil. Fixed costs are thus independent of output.
- **TVC varies with the output.** It is nil when there is no output. Variable costs are, thus, direct costs of the output.

TVC does not change in the same proportion. Initially, it is increasing at a decreasing rate, but after a point, it increases at an increasing rate. This is due to the operation of the law of variable proportions or non-proportional output, which suggests that initially to obtain a

given amount of output relatively, variations in factors are needed in less proportion, but after a point when the diminishing phase operates, variable factors, are to be employed in a greater proportion to increase the same „level“ of output.

□ **TC varies in the same proportion as the TVC.** Thus, in the short period, the changes in total cost are entirely due to changes in the total

THE BEHAVIOUR OF SHORT-RUN AVERAGE COST CURVES

The behaviour patterns and relations of short-run unit costs become more explicit when we plot the cost data on a graph and draw the respective cost curves.

Average Fixed Cost in Short Run: The average fixed cost is the total fixed cost divided by the volume of output. There is an inverse relation between output and average fixed cost. With the increase in output average fixed cost decreases and with the decrease in output the average fixed cost will increase.

It is calculated from the following formula:

$$AFC = TFC/Q$$

- Q is volume of output
- AFC and TFC are average fixed cost and total fixed cost.

2. Average Variable Cost (AVC): The average variable cost is total variable cost divided by the volume of output. Average variable cost falls with the increase in output, reaches at its minimum and then starts rising. By the operation of law of increasing returns the AVC decreases, and by the operation of constant returns leads to constancy in AVC and the law of diminishing returns leads to increase in AVC. The shape of AVC is U-shaped because of the operation of the laws of returns during short period.

The AVC is calculated by the formula given below:

$$AVC = TVC/Q$$

AVC and TVC are average variable cost and total variable cost while Q is the volume of output.

3. Average Total Cost (ATC): Average Total Cost (ATC) is the aggregate of AFC and AVC. ATC can be calculated from total cost (TC) divided by the volume of output or by aggregating AVC and AFC.

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The ATC curve decreases with the increase in output and remains constant up to a point and thereafter it increases with the increase in output. Its shape is U-shaped because of the operation of the laws of return during short period.

The following is the formula of calculating AC:

$$ATC = TC/Q \text{ or } ATC = AFC + AVC$$

ATC and TC are average total cost and total cost while Q is the volume of output.

4. Marginal Cost (MC): It is an addition to total cost by producing an additional unit of output. It can be calculated as the change in total cost divided by an additional unit change in the output.

Marginal Cost changes with the change in average variable cost (AVC) and it is independent of average fixed cost (AFC). Initially, MC Falls, reaches the minimum and thereafter continuously increases. MC is also U-shaped curve. MC curve cuts the ATC and AVC curves at their minimum points.

$$MC = \text{Change in ATC} / \text{Change in Q}$$

****Note:**

- (1) AFC declines continuously, approaching both axes asymptotically (as shown by the decreasing distance between ATC and AVC) and is a rectangular hyperbola.
- (2) AVC first declines, reaches a minimum at Q₂ and rises thereafter. When AVC is at its minimum, MC equals AVC.
- (3) ATC first declines, reaches a minimum at Q₃, and rises thereafter. When ATC is at its minimum, MC equals ATC.
- (4) MC first declines, reaches a minimum at Q₁, and rises thereafter. MC equals both AVC and ATC when these curves are at their minimum values.

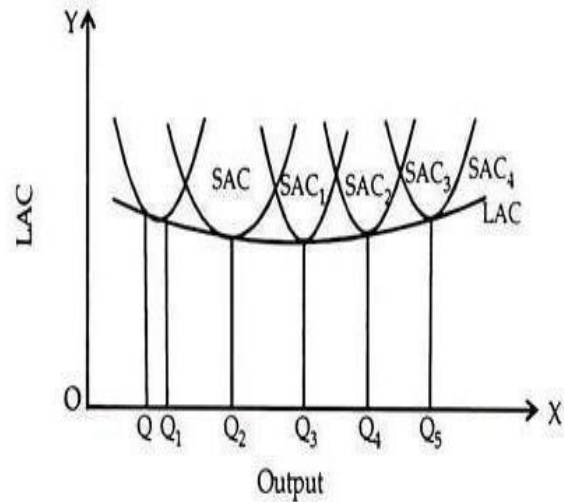
Cost – Output Relationship in Long Run:

Long period gives sufficient time to business managers to change even the scale of production. All the factors of production are variable. All the costs are variable costs and there is no fixed cost. The supply of goods can be adjusted to their demands because scale of production and factors of production can be changed.

Log Run Average Cost (LAC): In the long run, all the factors of production are variable and the firm has a variety of choices to select the size of the plants and the factors of production to be employed. Various short run average cost curves represent the various sizes of the plants available to a firm. We can get the long run average cost curve with the help of all the short run

average cost curves. The long run average cost curve envelopes all the short run average cost curves in it. It is also called an 'Envelope Curve' or 'Planning Curve'.

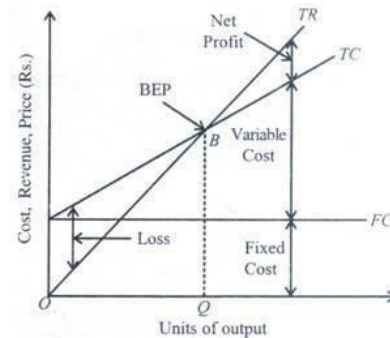
From the above figure long run cost represented on OY-axis and output on OX-axis. SAC, SAC₁, SAC₂, SAC₃ and SAC₄ are short run average cost curves which represent the different size of plants. LAC has been drawn by combining all those points of least cost of producing the corresponding output. The least per unit cost of production is OQ, OQ₁, OQ₂, OQ₃, OQ₄, and OQ₅ respectively.



BREAK-EVEN ANALYSIS

The *BEA* is an important technique to trace the relationship between costs, revenue and profits at the varying levels of output or sales

In *BEA*, the break-even point is located at that level of output or sales at which the net income or profit is zero. At this point, total cost is equal to total revenue. Hence, the breakeven point is the no-profit-no-loss zone.



However, the object of the *BEA* is not just to determine the break-even point (BEP), but to understand the functional relationship among cost, revenue and the rate of output.

ASSUMPTIONS OF BREAK-EVEN ANALYSIS

The validity of the *BEA* is conditioned by a number of assumptions as follows:

1. The cost function and the revenue function are linear.
2. The total cost is divided into fixed and variable costs.
3. The selling price is constant.
4. The volume of sales and the volume of production are identical.
5. Average and marginal productivity of factors are constant.
6. The product-mix is stable in the case of a multi-product firm.
7. Factor price is constant.

LIMITATIONS OF BEA

The break-even analysis has certain major limitations as follows:

- **It is static.** In the *BEA*, everything is assumed to be constant. This implies a static condition. It is not suited to a dynamic situation.
- **It is unrealistic.** It is based on many assumptions which do not hold good in practice. Linearity of cost and revenue function are true only for a limited range of output.
- **It has many shortcomings.** The *BEA* regards profit as a function of output only. It fails to consider the impact of technological change, better management, division of labour, improved productivity and such other factors influencing profits.
- **Its scope is limited to the short-run only.** The *SEA* is not an effective tool for a long-run analysis.
- **It assumes horizontal demand curve with the given price of the product.** But this is not so in the case of a monopoly firm.
- **It is difficult to handle selling costs in the BEA.** Selling costs do not vary with output. They manipulate sales and affect the volume of output.
- **The traditional BEA is very simple.** It makes no provision for corporate income-tax, etc.

Managerial Use of Break Even Analysis:

(i) **Safety Margin:** The break-even chart helps the management to know at a glance the profits generated at the various levels of sales. The safety margin refers to the extent to which the firm can afford a decline before it starts incurring losses.

The formula to determine the sales safety margin is:

$$\text{Safety Margin} = (\text{Sales} - \text{BEP}) / \text{Sales} \times 100$$

(ii) **Target Profit:** The break-even analysis can be utilised for the purpose of calculating the volume of sales necessary to achieve a target profit.

(iii) **Change in Price:** The management is often faced with a problem of whether to reduce prices or not. Before taking a decision on this question, the management will have to consider a profit. A reduction in price leads to a reduction in the contribution margin.

This means that the volume of sales will have to be increased even to maintain the previous level of profit. The higher the reduction in the contribution margin, the higher is the increase in sales needed to ensure the previous profit.

(iv) Change in Costs:

When costs undergo change, the selling price and the quantity produced and sold also undergo changes.

Changes in cost can be in two ways:

(i) Change in variable cost, and

(ii) Change in fixed cost.

(i) Variable Cost Change: An increase in variable costs leads to a reduction in the contribution margin. This reduction in the contribution margin will shift the break-even point downward. Conversely, with the fall in the proportion of variable costs, contribution margins increase and break-even point moves upwards.

(ii) Fixed Cost Change: An increase in fixed cost of a firm may be caused either due to a tax on assets or due to an increase in remuneration of management, etc. It will increase the contribution margin and thus push the break-even point upwards. Again to maintain the earlier level of profits, a new level of sales volume or new price has to be found out.

(v) Decision on Choice of Technique of Production: The breakeven analysis is the most simple and helpful in the case of decision on a choice of technique of production. For low levels of output, some conventional methods may be most probable as they require minimum fixed cost. For high levels of output, only automatic machines may be most profitable. By showing the cost of different alternative techniques at different levels of output, the break-even analysis helps the decision of the choice among these techniques.

(vi) Make or Buy Decision: Firms often have the option of making certain components or for purchasing them from outside the concern. Break-even analysis can enable the firm to decide whether to make or buy.

(vii) Plant Expansion Decisions: The break-even analysis may be adopted to reveal the effect of an actual or proposed change in operation condition. This may be illustrated by showing the impact of a proposed plant on expansion on costs, volume and profits. Through the break-even analysis, it would be possible to examine the various implications of this proposal.

(viii) Plant Shut Down Decisions: In the shut down decisions, a distinction should be made between out of pocket and sunk costs. Out of pocket costs include all the variable costs plus the fixed cost which do not vary with output. Sunk fixed costs are the expenditures previously made but from which benefits still remain to be obtained e.g. depreciation.

(ix) Advertising and Promotion Mix Decisions: The break-even point concept helps the management to know about the circumstances. It enables him not only to take appropriate decision but by showing how these additional fixed cost would influence BEPs. The advertisement pushes up the total cost curve by the amount of advertisement expenditure.

(x) Decision Regarding Addition or Deletion of Product Line: If a product has outlive utility in the market immediately, the production must be abandoned by the management and examined what would be its consequent effect on revenue and cost. Alternatively, the management may like to add a product to its existing product line because it expects the product as a potential profit spinner. The break-even analysis helps in such a decision.

Practical problems on break even analysis

MODULE 4

MARKET STRUCTURE & PRICING PRACTICES

Market is a place where the various sellers and buyers sell and buy the products respectively. In this module we will be discussing about various market structure.

Perfect Competition: Perfect market is a market condition with large buyers and sellers dealing in homogenous commodity or identical product and in this market condition, all buyers and sellers are aware of the price of the product that indicates the price of the product throughout the market.

Perfect competition describes a market structure where competition is at its greatest possible level. To make it more clear, a market which exhibits the following characteristics in its structure is said to show perfect competition:

1. Large number of buyers and sellers
2. Homogenous product is produced by every firm
3. Free entry and exit of firms

Features of Perfect Competition:

1. Large number of sellers and buyers: In a perfectly competitive market, there are large numbers of buyers and seller each demanding a small part of the total market demand and supply of the product respectively. As a result, no single seller or buyer is in a position to influence the market price determined by the forces of market demand and supply.

2. Homogeneous Product: In a perfectly competitive market, all the firms produce and supply the identical products. It means that the products of all the firms are perfect substitutes of each other. As a result of this, the price elasticity of demand for a firm's product is infinite.

3. Free entry and exit: In a perfectly competitive market, there are no restrictions on the entry of new firms into market or on the exit of existing firms from the market.

4. Perfect knowledge: In a perfectly competitive market, the firms and the buyers possess perfect information about the market. It implies that no buyer or firm is ignorant about the price prevailing in the market.

5. Perfect mobility of factors of production: In a perfectly competitive market, the factors of production are completely mobile leading to factor-price equalization throughout the market

6. Free and Perfect Competition among buyers and sellers: In a perfect market, there are no checks either on the buyers or sellers. They are free to buy or to sell to any person. It means there are no monopolies.

7. Absence of Price control: In a perfectly competitive market, each buyer is aware of product and its price in the market likewise the seller is also aware of the pricing of the product in the market. Thus no individual in the market can influence the market.

8. Absence of transport cost: In a perfectly competitive market, it is assumed that there are no transport costs. If transport costs are incurred, prices should be different in different sectors of the market.

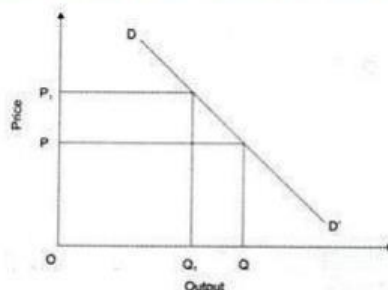
9. One price of Commodity: As there are large sellers in the market, all the sellers sell the product at same price. Every firm is a price taker. It takes the price as decided by the forces of demand and supply. No firm can influence the price of the product.

10. Independent Relationship among buyers and sellers: In a perfectly competitive market, it is considered to be no seller is related to buyer. Thus there is no chance for buyer to ask the product for lesser price and buyer will not sell the products at lesser price.

Determination of Price in Perfect Competition:

In perfect competition, the price of a product is determined at a point at which the demand and supply curve intersect each other. This point is known as **equilibrium point** as well as the price is known as **equilibrium price**. In addition, at this point, the quantity demanded and supplied is called **equilibrium quantity**.

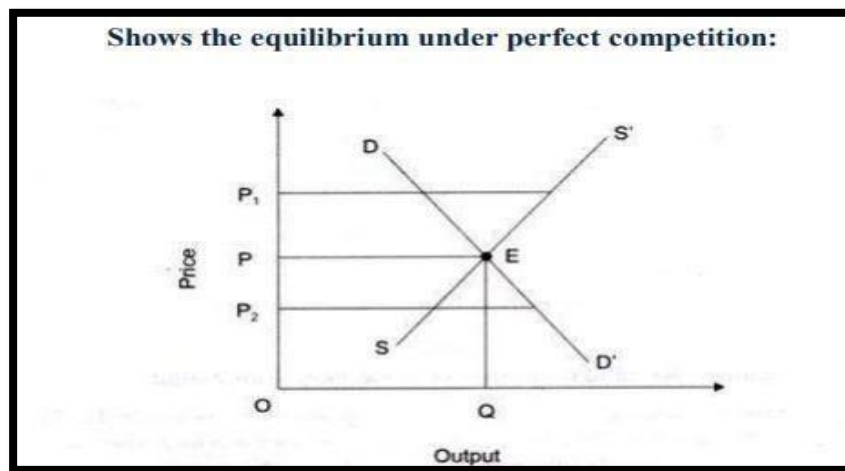
Represents the demand curve under perfect competition:



From the above figure when price is OP , the quantity demanded is OQ . On the other hand, when price increases to OP_1 , the quantity demanded reduces to OQ_1 . Therefore, under perfect competition, the demands curve (DD') slopes downward

From the above figure the quantity supplied is OQ at price OP . When price increases to OP_1 , the quantity supplied increases to OQ_1 . This is because the producers are able to earn large profits by supplying products at higher price. Therefore, under perfect competition, the supply curves (SS') slopes upward.

From the Below figure, it can be seen that at price OP_1 , supply is more than the demand.



Therefore, prices will fall down to OP . Similarly, at price OP_2 , demand is more than the supply. Similarly, in such a case, the prices will rise to OP . Thus, E is the equilibrium at which equilibrium price is OP and equilibrium quantity is OQ .

The price at which demand and supply are equal is known as equilibrium price and the quantity bought and sold at the equilibrium price is known as equilibrium output.

In the figure, equilibrium price is determined at the point E where both demand and supply are equal. The upper limit of the price of a product is determined by the demand. The lower limit of the price is determined by the production cost. The point E can be regarded as the position of stable equilibrium.

1. Very Short Period
2. Short Period
3. Long Period

1. Very Short Period:

Refers to a time period in which quantity supplied of a product cannot be increased with increase in its demand.

In simple terms, in very short period of time, the supply of a product is fixed. For example, a confectioner has 20 pastries at a particular time.

After an hour, a customer requires 40 pieces of pastries. In such a case, the confectioner cannot prepare 20 more pastries in an hour and can only supply 20 pastries. Therefore, the supply is fixed, which is 20 in the given example. The price determined in very short period is known as market price.

2. Short Period:

Refers to a time period in which the level of supply of a particular product can be increased, but only as per the production capacity of an organization. For example, an organization can produce 50 mobile phones in a day. This is the maximum production capacity of the organization.

Suppose the demand of mobile phones increases to 150 per day for three days. In such a scenario, the organization cannot install new machines or hire more labor in three days to meet the additional demand. Thus, the supply is fixed even in short period and the price determined in this period is known as sub-normal price.

3. Long Period:

Refers to a period in which the supply of a product can be increased or decreased with the changing level of demand. In this period, organizations can install new machines or hire more labor to meet the supply requirements.

Generally, in the long period, demand is determined by change in the size of population and change in customer's tastes and preferences. On the other hand, organizations can reduce production level if there is decrease in demand. Therefore, it can be said that in the long period, the price of a product is influenced by supply. The price in the long period is called normal price.

MONOPOLY:

A market structure characterized by a single seller, selling a unique product in the market. In a monopoly market, the seller faces no competition, as he is the sole seller of goods with no close substitute.

"Monopoly is a market situation in which there is a single seller. There are no close substitutes of the commodity it produces, there are barriers to entry". -Koutsoyiannis

Features:

- 1. One Seller and Large Number of Buyers:** The monopolist's firm is the only firm; it is an industry. But the number of buyers is assumed to be large.
- 2. No Close Substitutes:** There shall not be any close substitutes for the product sold by the monopolist. The cross elasticity of demand between the product of the monopolist and others must be negligible or zero.
- 3. Difficulty of Entry of New Firms:** There are either natural or artificial restrictions on the entry of firms into the industry, even when the firm is making abnormal profits.
- 4. Monopoly is also an Industry:** Under monopoly there is only one firm which constitutes the industry. Difference between firm and industry comes to an end.
- 5. Price Maker:** Under monopoly, monopolist has full control over the supply of the commodity. But due to large number of buyers, demand of any one buyer constitutes an infinitely small part of the total demand. Therefore, buyers have to pay the price fixed by the monopolist.
- 6. Price discrimination:** A company that is operating in a monopolistic market can change the price and quantity of the product or service. Price discrimination occurs when the company sells the same product to different buyers at different prices.

Considering that the market is elastic, the company will sell a higher quantity of the product if the price is low and will sell a lesser quantity if the price is high.

7. No Competition: As there is only one firm in monopoly there are no competition in this type of markets. Thus, this market is competition free.

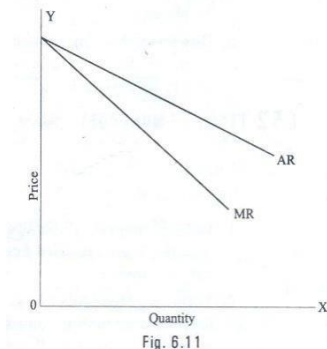
8. Full Control over Price: As a single firm itself acts as the industry, the firm operating in this market will always have an hedge of pricing the products at its will. Thus it controls the price completely.

9. Full Control over Supply: In this market the products or services do not have close substitutes and thus the producers have the control over the supply of their products.

Pricing or Price-Output Determination Under Monopoly:

Pricing under Monopoly

Monopoly refers to a market situation where there is only one seller. He has complete control over the supply of a commodity. He is therefore in a position to fix any price. Under monopoly there is no distinction between a firm and an industry. This is because the entire industry consists of a single firm.



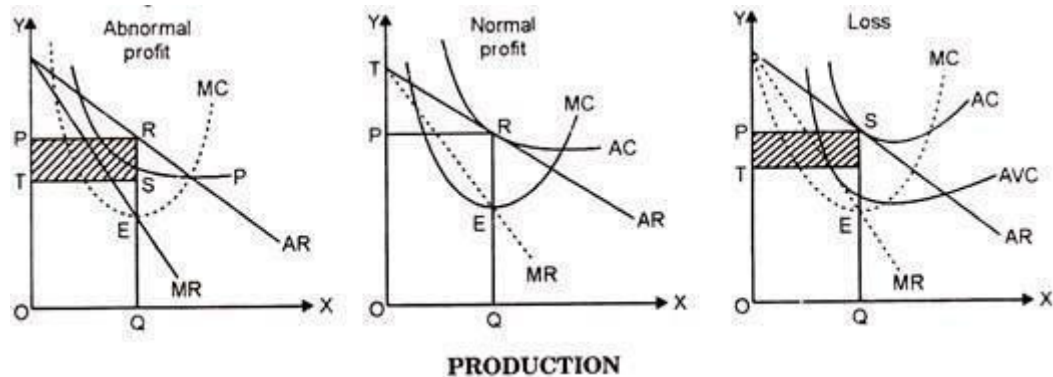
Determination of Price in the SHORT Period:

If, in a short period, the cost of production of a commodity is zero, he will go on producing it to the extent or so long the marginal revenue from the sale of that commodity does not fall to zero. As soon as the marginal revenue is zero he will not increase its supply.

Some economists think that, in a short period, three different situations may arise before the monopolist:

- (i) When the monopolist earns abnormal profits,
- (ii) When he gets only normal profits, and
- (iii) When he suffers losses.

The explanation and diagrams of these situations are given below:



On the point E the firm is in equilibrium when $MC = MR$. Thereafter MC curve starts to rise. Under the condition, OP is the price and OQ is the 'total production' of the commodity so determined. In order to calculate profits or losses, we will have to measure the difference between AR and AC. If $AR > AC$, the difference between the two is profit per unit and by multiply it with total number of units produced we can get total profit.

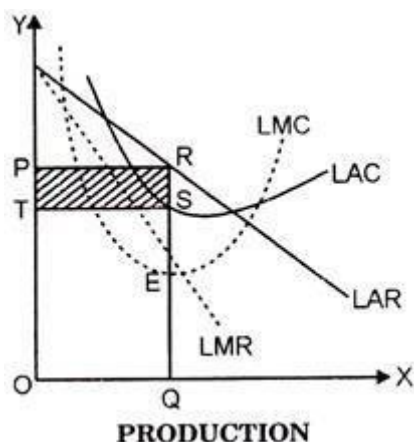
In the first figure $RQ = OP$ is the price, TO is the cost of production per unit. Thus, $RS = PT$ is unit for profit. On the OQ quantity of production, total profit is PTSR shaded area which is abnormal profit. In the second figure $RQ = OP$ is the determined price and RQ is the average cost. Under this condition, there will be only normal profit.

In the figure three also price per unit is $RQ = OP$ but cost per unit is SQ. Thus, SR (TP) is loss per unit. As a result TPRS shaded area will be the total loss. But this loss is only short period phenomenon. In the long period, this loss will disappear, under that condition and situation, only profit will be earned.

Determination of Price in the Long Period:

In the long period the monopolist introduces changes in his equipment's and techniques of production. During this period in order to gain excess profit, he will change efficiency and capacity of his resources according to his need. But the determination of the quantity of production follows, the same line as under short period.

This is clear from the following figure:



In this figure LMC and LMR intersect each other at the point E and after that LMC goes on

rising. Thus OQ production is determined and OP is the price. But average cost is SQ. So profit per unit is RS and at OQ output the total profit is PTRS.

Under Price Competition $AR = MR$, whereas under Monopoly $MR < AR$.

Under perfect competition price is determined by the interaction of total demand and supply. This price is acceptable to all the firms in the industry. No firm can change this price. So, average revenue and marginal revenue, at every level of production, will be constant and equal. Their curves are parallel to X-axis.

Under Monopoly, to sell every additional unit of the commodity price will have to be lower. In this way, with the sale of every additional unit, average and marginal income goes on falling. But the decrease in average revenue is relatively less sharp than the decrease in marginal revenue. It is because marginal revenue is limited to one unit, whereas in case of average revenue, the decrease price is divided by the number of units. Therefore, the fall in average revenue has relatively less slope. That is the reason why marginal revenue is less than average revenue.

PRICE DISCRIMINATION:

In monopoly, there is a single seller of a product called monopolist. The monopolist has control over pricing, demand, and supply decisions, thus, sets prices in a way, so that maximum profit can be earned. The monopolist often charges different prices from different consumers for the same product. This practice of charging different prices for identical product is called price discrimination.

According to Robinson, "Price discrimination is charging different prices for the same product or same price for the differentiated product."

Different price discrimination are as follows:

First Degree Price Discrimination
Second Degree Price Discrimination
Third Degree Price Discrimination

- i First-degree Price Discrimination:** Refers to a price discrimination in which a monopolist charges the maximum price that each buyer is willing to pay. This is also known as perfect price
- ii discrimination** as it involves maximum exploitation of consumers. In this, consumers fail to enjoy any consumer surplus. First degree is practiced by lawyers and doctors.

iii. Second-degree Price Discrimination: Refers to a price discrimination in which buyers are divided into different groups and different prices are charged from these groups depending upon what they are willing to pay. Railways and airlines practice this type of price discrimination.

iv. Third-degree Price Discrimination: Refers to a price discrimination in which the monopolist divides the entire market into submarkets and different prices are charged in each submarket. Therefore, third-degree price discrimination is also termed as market segmentation.

In this type of price discrimination, the monopolist is required to segment market in a manner, so that products sold in one market cannot be resold in another market. Moreover, he/she should identify the price elasticity of demand of different submarkets. The groups are divided according to age, sex, and location. For instance, railways charge lower fares from senior citizens. Students get discount in cinemas, museums, and historical monuments.

MONOPOLISTIC COMPETITION

Monopolistic competition is a form of market structure having large number of sellers and buyers dealing in similar products or close substitute products where there are no entry or exit barriers.

Features of Monopolistic Competition:

1. Large Number of Sellers:

There are large numbers of firms selling closely related, but not homogeneous products. Each firm acts independently and has a limited share of the market. So, an individual firm has limited control over the market price. Large number of firms leads to competition in the market.

2. Product Differentiation:

Each firm is in a position to exercise some degree of monopoly (in spite of large number of sellers) through product differentiation. Product differentiation refers to differentiating the products on the basis of brand, size, colour, shape, etc. The product of a firm is close, but not perfect substitute of other firm.

Implication of 'Product differentiation' is that buyers of a product differentiate between the same products produced by different firms. Therefore, they are also willing to pay different prices for the same product produced by different firms. This gives some monopoly power to an individual firm to influence market price of its product.

3. Selling costs on Advertisements:

Under monopolistic competition, products are differentiated and these differences are made known to the buyers through selling costs. Selling costs refer to the expenses incurred on marketing, sales promotion and advertisement of the product. Such costs are incurred to persuade the buyers to buy a particular brand of the product in preference to competitor's brand. Due to this reason, selling costs constitute a substantial part of the total cost under monopolistic competition.

4. Freedom of Entry and Exit:

Under monopolistic competition, firms are free to enter into or exit from the industry at any time they wish. It ensures that there are neither abnormal profits nor any abnormal losses to a firm in the long run. However, it must be noted that entry under monopolistic competition is not as easy and free as under perfect competition.

5. Lack of Perfect Knowledge:

Buyers and sellers do not have perfect knowledge about the market conditions. Selling costs create artificial superiority in the minds of the consumers and it becomes very difficult for a consumer to evaluate different products available in the market. As a result, a particular product (although highly priced) is preferred by the consumers even if other less priced products are of same quality.

6. Pricing Maker:

A firm under monopolistic competition is neither a price-taker nor a price-maker. However, by producing a unique product or establishing a particular reputation, each firm has partial control over the price. The extent of power to control price depends upon how strongly the buyers are attached to his brand.

7. Non-Price Competition:

In addition to price competition, non-price competition also exists under monopolistic competition. Non-Price Competition refers to competing with other firms by offering free gifts, making favourable credit terms, etc., without changing prices of their own products.

Firms under monopolistic competition compete in a number of ways to attract customers. They use both Price Competition (competing with other firms by reducing price of the product) and Non-Price Competition to promote their sales.

8. Demand curve is downward sloping:

Under monopolistic competition, large number of firms selling closely related but differentiated products makes the demand curve downward sloping. It implies that a firm can sell more output only by reducing the price of its product.

9. No Close Substitute products:

Products in this market do not have close substitutes to the competitors' products.

Price-Determination under Monopolistic Market in Short Run:

During short period there may be three situations of the firms under monopolistic competition as given below:

1. Profit Making Situation: $AR > AC$

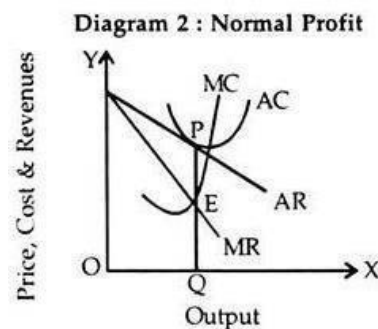
Profit making situation will be when individual firm's revenue is greater than its cost ($AR > AC$).

Profit making situation is also called abnormal or super profit situation as given in Diagram 1.

Price, costs and revenues are shown on OY-axis while output on OX-axis. The point of equilibrium is E where marginal cost is equal to marginal revenue ($MC=MR$) of the firm. The price is OP, output is OQ. The average profit ($AR-AC$) is TS and total profit is PLST.

2. Normal Profit: $AR=AC$

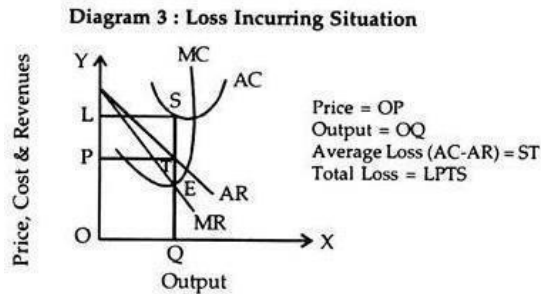
When the firm's average revenue is equal to its average cost the situation is called normal profit.



In the diagram output is shown on OX-axis, price, costs and revenue are shown on OY-axis. The point of equilibrium is E where firm's marginal cost is equal to its marginal revenue ($MC=MR$). Price is PQ and output is OQ. At P the average revenue is equal to average cost hence the firm is earning normal profit only.

3. Loss Making Situation: $AR < AC$

Under monopolistic situation during short period a firm will earn loss when its cost is greater than its revenue ($AC > AR$). It can be explained with the help of the Diagram 3.



Price, cost and revenue are shown on OY-axis while output is shown on OX-axis respectively. The point of equilibrium ($MC=MR$) is E. Price is OP and output is OQ. Average loss ($AC-AR$) is ST and total loss is LPTS.

Price-Determination under Monopolistic Market in long Run:

In monopolistic competition, since the product is differentiated between firms, each firm does not have a perfectly elastic demand for its products. In such a market, all firms determine the price of their own products. Therefore, it faces a downward sloping demand curve.

Conditions for the Equilibrium of an individual firm

The conditions for price-output determination and equilibrium of an individual firm are as follows:

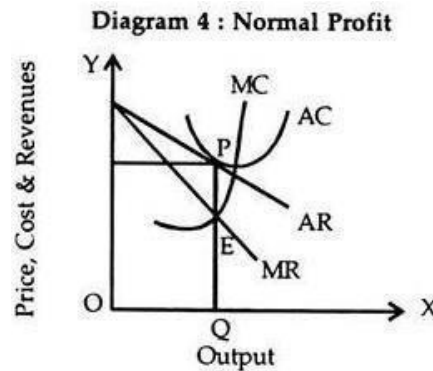
1. $MC = MR$
2. The MC curve cuts the MR curve from below.

Price and Output Determination during Long Period:

Under monopolistic competition, firms have freedom to enter and exit the industry. In the long run if firms are earning profit new firms are attracted and it will increase the output and consequently prices will fall leading to conversion of profit making situation into normal profit situation.

Contrary to it when firms are incurring losses during long period they will leave the industry. It will reduce the volume of output, prices will increase and the loss making situation will be converted into normal profit. Thus, the firms will earn normal profit only during long period. It can be seen from Diagram 4.

Price, costs and revenue are shown on OY-axis and output on OX-axis. Point of equilibrium ($MC=MR$) is E. Price is PQ and output is OQ. At P point average cost is equal to average revenue ($AC=AR$). Hence, the firm is earning normal profit only during long period.



PRODUCT DIFFERENTIATION:

A firm may have convinced consumers that its product is significantly better than the product of new entrants. The new firm may be forced to sell at lower price and reduce profit though the existing product may not essentially be superior.

The real differentiation refers to the technical features like the product's technical life and performance, durability, cost of operation and maintenance, etc.

On the other hand, the non-technical differentiation may take the form of brand names, trademark, packing, shape, size, etc. The non-technical differentiation adds a subjective appeal to a product inducing buyers to increase its demand or pay more for it.

Price: Price can be used to differentiate a product in two ways. Companies can charge the lowest price compared to competitors to attract cost-conscious buyers—the retailer Costco is an example. However, companies can also charge high prices to imply quality and that a product is a luxury or high-end item, such as a Bugatti sports car.

Performance and Reliability: Products can be differentiated based on their reliability and durability. Some batteries, for example, are reputed to have a longer life than other batteries, and consumers will buy them based on this factor.

Location and Service: Local businesses can differentiate themselves from their larger national competitors by emphasizing that they support the local community. A local restaurant, for example, will hire locally and may source its food and ingredients from local farmers and purveyors.

Quality: How does the quality, reliability, and ruggedness of your product compare to others on the market?

Design: Have you done something different with your design? Is it minimalistic and sleek? Easy-to-navigate?

Customization: Can you customize parts of the product that competitors cannot?

Similarly we can differentiate the products on basis of color, brand, size etc.

OLIGOPOLY:

Oligopoly market is a situation where in there are few number of sellers and large buyers dealing in homogenous product or differentiated products and there are entry-exit barriers for new firm.

Features of Oligopoly:

1 Few firms: Under oligopoly, there are few large firms. The exact number of firms is not defined. Each firm produces a significant portion of the total output. There exists severe competition among different firms and each firm try to manipulate both prices and volume of production to outsmart each other. For example, the market for automobiles in India is an oligopolist structure as there are only few producers of automobiles.

2 Interdependence: Firms under oligopoly are interdependent. Interdependence means that actions of one firm affect the actions of other firms. A firm considers the action and reaction of the rival firms while determining its price and output levels. A change in output or price by one firm evokes reaction from other firms operating in the market.

3 Nature of the Product: The firms under oligopoly may produce homogeneous or differentiated product.

If the firms produce a homogeneous product, like cement or steel, the industry is called a pure or perfect oligopoly.

If the firms produce a differentiated product, like automobiles, the industry is called differentiated or imperfect oligopoly.

4 Barriers to Entry of Firms: The main reason for few firms under oligopoly is the barriers, which prevent entry of new firms into the industry. Patents, requirement of large capital, control over crucial raw materials, etc, are some of the reasons, which prevent new firms from entering into industry. Only those firms enter into the industry, which is able to cross these barriers. As a result, firms can earn abnormal profits in the long run.

5 Non-Price Competition: Under oligopoly, firms are in a position to influence the prices. However, they try to avoid price competition for the fear of price war. They follow the policy of price rigidity. Price rigidity refers to a situation in which price tends to stay fixed irrespective of changes in demand and supply conditions. Firms use other methods like advertising, better services to customers, etc. to compete with each other.

If a firm tries to reduce the price, the rivals will also react by reducing their prices. However, if it tries to raise the price, other firms might not do so. It will lead to loss of customers for the firm, which intended to raise the price. So, firms prefer non- price competition instead of price competition.

6 Lack of Uniformity: Another feature of oligopoly market is the lack of uniformity in the size of firms. Firms differ considerably in size. Some may be small, others very large. Such a situation is asymmetrical. This is very common in the American economy. A symmetrical situation with firms of a uniform size is rare.

7. Existence of Price Rigidity: In oligopoly situation, each firm has to stick to its price. If any firm tries to reduce its price, the rival firms will retaliate by a higher reduction in their prices. This will lead to a situation of price war which benefits none. On the other hand, if any firm increases its price with a view to increase its profits; the other rival firms will not follow the same. Hence, no firm would like to reduce the price or to increase the price. The price rigidity will take place.

8 Advertising: As the firms follow non-price competition by keeping the prices close to rival firms; to increase the sales and profit of the firm they have to involve themselves in marketing and promotions of the product.

9 Selling Costs: Since firms try to avoid price competition and there is a huge interdependence among firms, selling costs are highly important for competing against rival firms for a larger market share.

Price Determination under Oligopoly:

Oligopoly can be either collusive or non-collusive. Collusive oligopoly is a market situation wherein the firms cooperate with each other in determining price or output or both. A non-collusive oligopoly refers to a market situation where the firms compete with each other rather than cooperating.

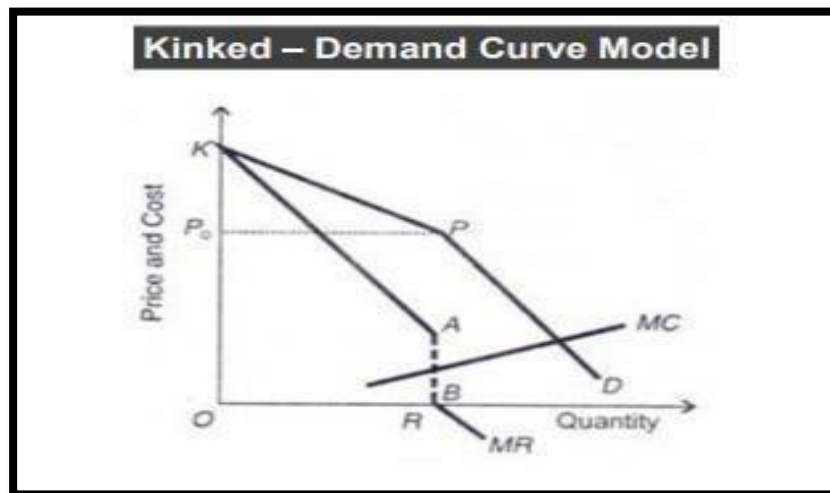
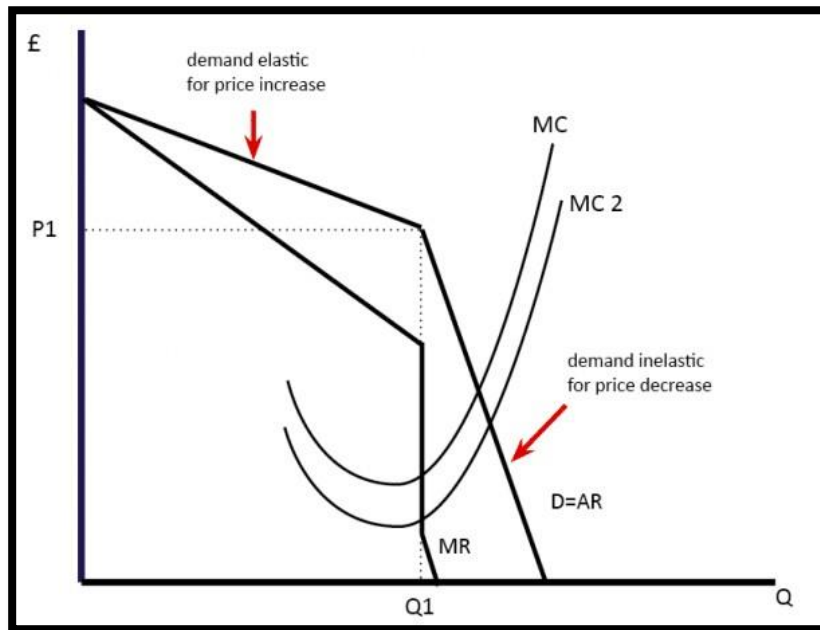
1. Price Determination in Non-Collusive Oligopoly:

In this case, each firm follows an independent price and output policy on the basis of its judgment about the reactions of his rivals. If the firms are producing homogeneous products, price war may occur. Each firm has to fix the price at the competitive level. On the contrary, in case of differentiated oligopoly, due to product differentiation, each firm has some monopoly control over the market and therefore charge near monopoly price.

Usually, in Oligopolistic markets, there are many price rigidities. In 1939, Paul Sweezy used an unconventional demand curve – the kinked demand curve to explain these rigidities.

Kinked-Demand Curve:

Like traditional demand curves, kinked demand curves are downward sloping. As the name suggests, the kinked demand curves have a 'kink'. This kink is nothing but a discontinuity at a concave bend and this kink is what sets it apart from the traditional demand curves. Now let's find out why these kinks exist in the first place.



MR= Marginal Revenue, MC= Marginal Cost

KPD= Deman Curve or Average Revenue, OR is output at price OP_0

Assumptions:

1. Firm selling their products at satisfactory price.
2. Rivals will tend not to raise prices in response to the firm's higher prices.
3. Rivals will not follow an attempted increase in price by one of the firms but will follow a decrease.

In the figure above, KPD is the is the kinked-demand curve and OP_0 is the prevailing price in the oligopoly market for the OR product of one seller. Starting from point P, corresponding to the point OP_1 , any increase in price above it will considerably reduce his sales as his rivals will not follow his price increase.

This is because the KP portion of the curve is elastic and the corresponding portion of the MR curve (KA) is positive. Therefore, any price increase will not just reduce the total sales but also his total revenue and profit. On the other hand, if the seller reduces the price of the product below OPQ (or P), his rivals will also reduce their prices.

However, even if his sales increase, his profits would be less than before. This is because the PD portion of the curve below P is less elastic and the corresponding part of the marginal revenue curve below R is negative. Therefore, in both price-raising and price-reducing situations, the seller is the loser. He will stick to the prevailing market price OP_0 which remains rigid.

This asymmetrical behavioral pattern results in a kink in the demand curve and hence there is price rigidity in oligopoly markets. The prices remain rigid at the kink (point P). In other words, the price will remain sticky at OP_0 and the output = OR at this price.

Due to the difference in the elasticities, the MR curve becomes discontinuous corresponding to the point of change in elasticity of the demand curve. The kink represents this. At the output $< OR$, the demand curve is KP and the corresponding MR curve is KA and $MR > MC$. For output $> OR$, the demand curve is PD and the corresponding MR curve is BMR and $MR < MC$.

2. Price Determination in Collusive Oligopoly:

Sometimes, firms may try to remove uncertainty related to acting independently and enter into price agreements with each other. This is collusion. Collusion is either formal or informal. It can take the form of cartel or price leadership.

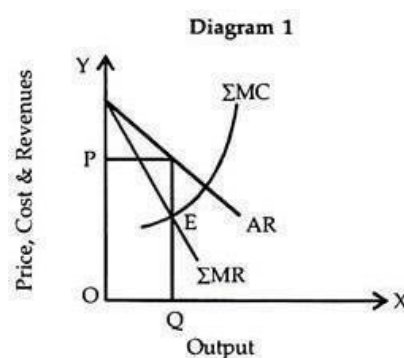
Price and Output under Perfect Collusion:

Under oligopoly, perfect collusion may be formed among different producers, sellers in two ways, namely, centralised cartel and market sharing cartel, and price and output are determined accordingly.

A cartel is an association of independent firms within the same industry which follow the common policies relating to price, output, sale, profit maximization, and the distribution of products.

(a) Centralised Cartel:

Under this type of collusion different firms of oligopoly market structure set up a centralised cartel. These firms transfer their function relating to managerial decisions and other activities to the centralised cartel to improve the volume of profit. Centralised cartel aims at maximisation of profit of all the firms. The cartel fixes price of the product, volume of output and production quota of individual firms.



In the diagram AR is the demand curve of industry and MR is the marginal revenue curve which has been drawn on the basis of AR or demand curve. ΣMC is the total marginal cost curve of the

industry which is the aggregate marginal cost of two firms engaged in production in that industry. Marginal revenue curve of industry is cut by the marginal cost curve of the industry at point E where $\sum MR$ is equal to $\sum MC$ ($\sum MR = \sum MC$). It is the equilibrium of the industry. Price is OP and output is OQ in the industry. Number of products to be produced by a firm is fixed on the basis of MR, that is when MR of firm is equal to combined MR of market.

(b) Market Sharing Cartel:

Another form of perfect collusion is the market sharing by the firms. This type of collusion can be effective and successful when all the firms are producing homogeneous product and production costs are similar. In order to explain this type of collusion we assume that there are two firms producing on the uniform cost of production and are ready to share market on 50:50 basis.

There are two main methods of market-sharing:

(a) Non-price competition; and (b) Quota system.

(a) Non-Price Competition Cartel:

The non-price competition agreement among oligopolistic firms is a loose form of cartel. Under this type of cartel, the low-cost firms press for a low price and the high-cost firms for a high price. But ultimately, they agree upon a common price below which they will not sell. Such a price must allow them some profits. The firms can compete with one another on a non-price basis by varying the colour, design, shape packing etc. of their product and having their own different advertising and other selling activities. Thus each firm shares the market on a non-prices basis while selling the product at the agreed common price.

(b) Market Sharing by Quota Agreement:

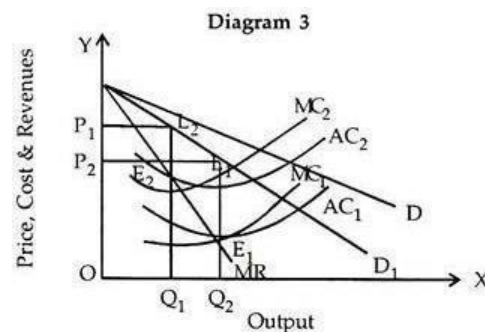
The second method of market sharing is the quota agreement among firms. (All firms in an oligopolistic industry enter into collusion for charging an agreed uniform price. But the main agreement relates to the sharing of the market equally among member firms so that each firm gets profits on its sales.

Price and Output under Imperfect Collusion: Price Leadership

Under oligopoly price and output can also be determined without any collusion among the firms. The firms may decide to follow a firm in price and output determination in the long run. Such sort of policy is called price leadership under oligopoly.

Price leadership is based on informed collusion. Under price leadership, one firm is a large or dominant firm and acts as the price leader who fixes the price for the products while the other firms allow it.

In the diagram output is shown on OX-axis while price, cost, and revenue are shown on OY-axis. DD is the market demand curve while DD₁ is the demand curve of the firm. AC₂ and MC₂ are average cost curve and marginal cost curve of the firm having high cost of production. Its point of equilibrium is E₂ where the price is OP₂ and the output is OQ₂.



The average cost curve (AC_1) and marginal cost curve (MC_1) are of a firm having least cost of production. Its point of equilibrium is E_1 where the price of the firm is OP_1 and output is OQ_1 . The marginal revenue curve (MR) of both the firms is equal to their MC_1 and MC_2 . The firm producing OQ_1 output with OP_1 price is the low cost producing firm and it will be the price leader in the market and the same price policy OP_1 will be following by other firm.

There are several types of price leadership. The following are the principal types:

- (a) **Price leadership of a dominant firm**, i.e., the firm which produces the bulk of the product of the industry. It sets the price and rest of the firms simply accepts this price.
- (b) **Barometric price leadership**, i.e., the price leadership of an old, experienced and the largest firm assumes the role of a leader, but undertakes also to protect the interest of all firms instead of promoting its own interests as in the case of price leadership of a dominant firm.

(c) **Exploitative or Aggressive price leadership**, i.e., one big firm built its supremacy in the market by following aggressive price leadership. It compels other firms to follow it and accept the price fixed by it. In case the other firms show any independence, this firm threatens them and coerces them to follow its leadership.

Descriptive Pricing Approaches:

Pricing is the art of converting the value of products or services into quantitative terms to customers at a point of time.

Factors determining Pricing:

(i) **Cost of Production:** Cost of production is the main component of price. No company can sell its product or services at less than the cost of production. Thus, before price fixation, it is necessary to compile data relating to the cost of production and keep that in mind.

(ii) **Demand for Product:** Intensive study of demand for product and services in the market be undertaken before price fixation. If demand is relatively more than supply, higher price can be fixed.

(iii) **Price of Competing Firms:** It is necessary to take into consideration prices of the products of the competing firms prior to fixing the price. In case of cut-throat competition it is desirable to keep prices low.

(iv) **Purchasing Power of Customers:** What are the purchasing power of the customers and at what price and how much they can purchase? It should also be taken into consideration.

(v) **Government Regulation:** If the price of the commodity and services is to be fixed as per the regulation of the government, it should also be borne in mind.

(vi) **Objective:** Usually, at the time of price fixation a certain amount of profit is added to the cost of production. If company's objective is to earn higher profit it may add higher amount of it.

(vii) **Marketing Method Used:** Price is also influenced by the marketing method used by the company, e.g., commission which is to be paid to the middlemen for sale of the goods is also added to the price. Similarly, if the customers are to be provided "after sale service" facility, then those expenses are also added to the price.

PRICING STRATEGIES ARE:

- Full cost pricing
- Product line pricing
- Price Skimming
- Penetration Pricing
- Loss leader pricing
- Peak Load pricing

1. Full cost pricing: Full cost pricing is a practice where the price of a product is calculated by a firm based on its direct costs per unit of output plus a markup to cover overhead costs and profits.

It is a price-setting method under which you add together the direct material cost, direct labor cost, selling and administrative costs, and overhead costs for a product, and add to it a markup percentage (to create a profit margin) in order to derive the price of the product. The pricing formula is:

$$\frac{(\text{Total production costs} + \text{Selling and administration costs} + \text{Markup})}{\text{Number of units expected to sell}}$$

Number of units expected to sell

2. Product line pricing: Product line pricing involves the separation of goods and services into cost categories in order to create various perceived quality levels in the minds of consumers. You might also hear product line pricing referred to as price lining, but they refer to the same practice.

3. Price Skimming: Price skimming is a product pricing strategy by which a firm charges the highest initial price that customers will pay and then lowers it over time. As the demand of the first customers is satisfied and competition enters the market, the firm lowers the price to attract another, more price sensitive segment of the population. The skimming strategy gets its name from "skimming" successive layers of cream, or customer segments, as prices are lowered over time.

4. Penetration Pricing: Penetration pricing is a marketing strategy used by businesses to attract customers to a new product or service by offering a lower price during its initial offering. The

lower price helps a new product or service penetrate the market and attract customers away from competitors. Market penetration pricing relies on the strategy of using low prices initially to make a wide number of customers aware of a new product.

The goal of a price penetration strategy is to entice customers to try a new product and build market share with the hope of keeping the new customers once prices rise back to normal levels.

5. Loss leader pricing: Loss leader pricing is an aggressive pricing strategy in which a store sells selected goods below cost in order to attract customers who will, according to the loss leader philosophy, make up for the losses on highlighted products with additional purchases of profitable goods. Loss leader pricing is employed by retail businesses; a somewhat similar strategy sometimes employed by manufacturers is known as penetration pricing.

6. Peak Load pricing: The Peak Load Pricing is the pricing strategy wherein the high price is charged for the goods and services during times when their demand is at peak. In other words, the high price charged during the high demand period is called as the peak load pricing.

This type of price discrimination is based on the efficiency, i.e. a firm discriminates based on high usage, high-traffic, high demand times and low demand times. The consumer who purchases the commodity during the high demand period has to pay more as compared to the one who buys during low demand periods.

Question Bank:

1. Define perfect competition.
2. Define cartels.
3. What is monopoly? Discuss its characteristics.
4. What is price discrimination? Mention the degrees of price discrimination.
5. Mention the characteristics of perfect competition.
6. Explain the concept of monopolistic competition. How prices are determined.
7. Elaborate the characteristics of oligopoly.
8. Define market structure. What are the different types of market structure?
9. Explain kinked demand curve along with graph and its assumptions.
10. How are prices determined under perfect competition? Explain with the help of suitable graphs.
11. Discuss the various descriptive pricing strategies.
12. Enumerate the factors governing pricing decisions.

MODULE 5

INDIAN BUSINESS ENVIRONMENT

Nature, Scope, Structure of Indian Business Environment, Internal and External Environment. Political and Legal Environment, Economic Environment, Socio– Cultural Environment, Global Environment. Private Sector, Growth, Problems and Prospects, SMEs, Significance in Indian economy, challenges and prospects. Fiscal policy and Monetary Policy: Meaning of Fiscal policy, three main types of fiscal policy – neutral policy, expansionary, and contractionary. Monetary policy: Meaning, Objectives of monetary policies: Controlling inflation, Managing employment levels, and Maintaining long-term interest rates. (Theory only)

Meaning of Business Environment :–

Business environment refers to those aspects of the surroundings of business enterprise which have influence on the functioning of business. An organization can survive and grow only when it continuously and quickly adopts to changing environment.

Acc. to Wheeler, Business environment refers to those aspects of the surroundings of business enterprise which have influence on the functioning of business. An organization can survive and grow only when it continuously and quickly adopts to changing environment.

Acc. to Keith Davis, “Business Environment is the aggregate of all conditions, events and influence that surround and affect the business.”

NATURE AND SCOPE OF BUSINESS ENVIRONMENT:

1. Complex

- Business environment is complex in nature. It consists of several factors, conditions and events which directly influence the functioning of business. These factors influence the productivity and profitability of business. It is quite difficult to understand how a particular factor affects the operations of business. Different factors affect differently the performance of business.

2. Dynamic

- Business environment is a dynamic concept and keeps on changing continuously. Various factors which constitute business environment are always changing from time to time.
- These changes are in terms of :
 - changes in customer preferences, technological improvements, new competitor's entry etc.
- These ever changing factors bring changes in character and shape of business environment

making it dynamic.

3. Interdependence

Factors of business environment are dependent on each other. Business environment include economic, social, legal, technological and political factors. Changes in any one of the factors will bring changes in several other factors. Like countries having better economy are able to enhance their technology by incurring sufficient expenditure on research and development.

4. Uncertain

Factors which constitute business environment are uncertain. It is quite difficult to predict several future conditions because no one knows what is going to happen in nature. These factors changes too frequently like changes in fashion, technology, demand and economic conditions.

5. Relativity

Business environment is relative in nature which means that it changes from one place to another and from one country to another. Economic conditions In Canada are different from those in India. Similarly, demand for women's shalwar kameez is high in India than in U.S.A.

6. System Approach

Business environment is a systematic approach which facilitates business in its functioning. Business is a system which manufactures the products and services. It satisfies the wants of customers by taking several inputs like raw materials, capital, labour etc. from environment and delivering the required goods and services. In original, business is a system by which it produces goods and services for the satisfaction of wants, by using several inputs, such as, raw material, capital, labor etc. from the environment.

7. Social Responsibility Approach

Social responsibility is another important characteristic. According to this, business exists for serving various members of society. These include shareholders, employees, customers, government etc In this approach business should fulfill its responsibility towards several categories of the society such as consumers, stockholders, employees, government etc.

8. Creative Approach:

As per this approach, business gives shape to the environment by facing the challenges and availing the opportunities in time. The business brings about changes in the society by giving attention to the needs of the people.

SCOPE OF BUSINESS ENVIRONMENT

- **Identifies Business Opportunities And Threats**

Business environment helps business in identification of various opportunities and threats. When business is able to detect market opportunities timely, they can easily take advantages of such opportunity at earliest. They can earn maximum returns by availing such opportunity before the competitors. By proper interaction between business and its environment all threats can be easily detected. It will enable business in taking corrective measures timely.

- **Helps In Planning And Policy Formulation**

Proper understanding of business environment helps in formulating better policies and strategies. It conveys all current information regarding market conditions to business. All opportunities and threats are scanned through the study of the business environment. Businessmen are properly aware of environment and thereby take all decisions according to it. Their entire plan can be changed effectively and efficiently through environmental awareness.

- **Provides Useful Resources**

Business depends on the environment in which they operate for several resources. Business environment supplies several inputs like raw materials, capital and labour which are used by the business for its operations. These inputs are converted into goods and services for satisfying the needs of the market. Without proper supply of inputs, business cannot continue its operations. It is fully dependent upon environment for taking inputs and delivering the required goods or services.

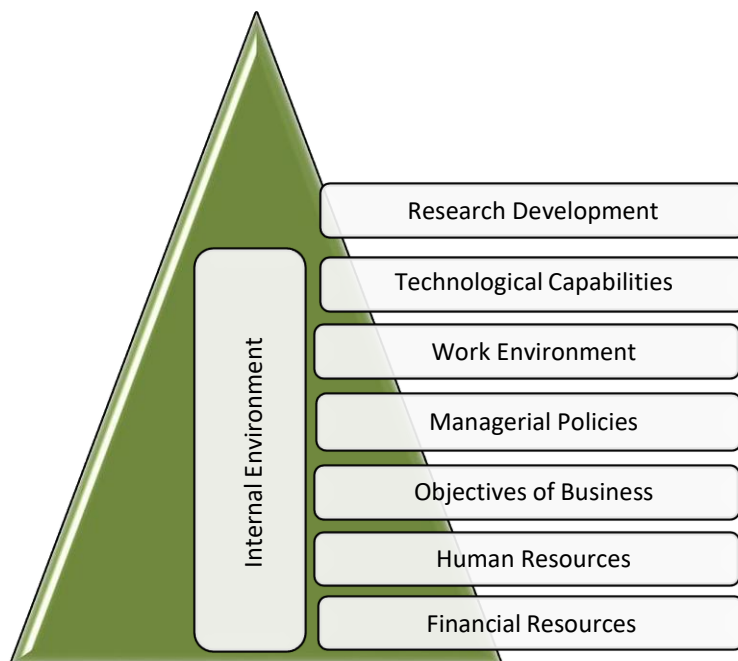
- **Improves Performance**

Business environment has an effective role in accelerating the overall performance of business organisations. Through continuous environmental awareness, managers update their knowledge and skills. Environmental study serves as the medium of educating management.

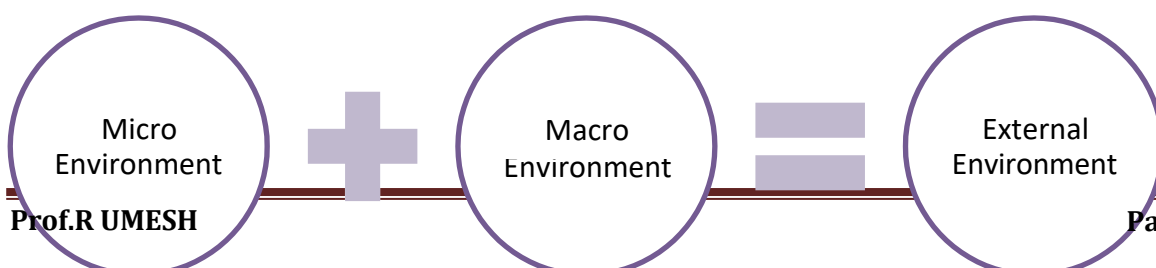
STRUCTURE /COMPONENTS OF BUSINESS ENVIRONMENT :-

Business Environment has two components :-

- a) **Internal Environment** :- Internal Environment refers to environment within the organization. It includes internal factors of the business which can be controlled by business. It includes objectives of business, managerial policies, management and employees of the organization, labor management relationship, brand image and corporate image, working conditions in the organization, technological and research and development capabilities etc. Internal environment includes 5 M's i.e. men, material, money, machinery, management available with business. These components are usually within the control of business.



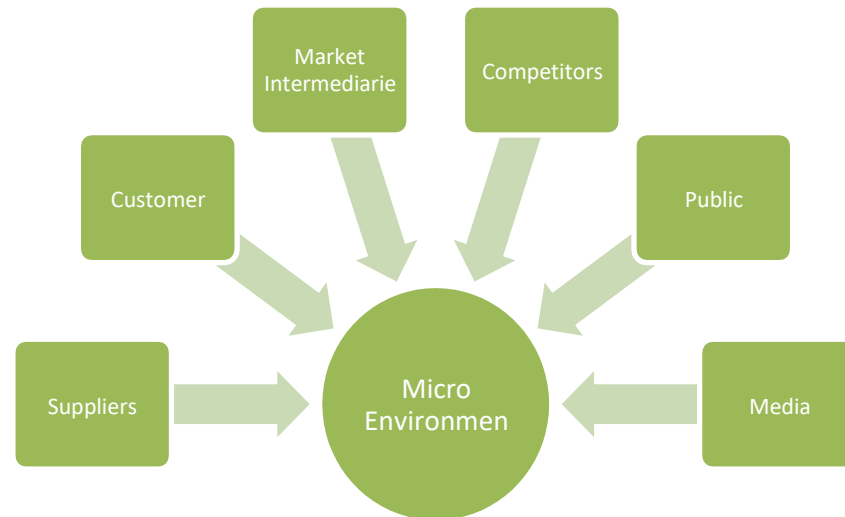
- b) **External Environment** :- External Environment refers to external aspects of the surroundings of business enterprise which have influence on the functioning of business. These factors beyond the control of business. External environment includes factors outside the firm which can provide opportunities or pose threats to the firm. External environment has two types :- i) Micro Environment ii) Macro Environment



Micro Environment

:- Macro

Environment :- The micro environment of a company consists of elements that directly affect the company. It includes suppliers, customers, market intermediaries,



competitors and public etc., which is explained as below :-

- i) **Suppliers** :Suppliers are those who supply raw materials and components of the company. Every business requires the suppliers. If our supplier is reliable, our business will run smoothly. If our supplier is not reliable, we have to maintain high inventories.
- ii) **Customer** :-Customer is the central point of the business. The success of a business organization depends upon the customers, their needs, tastes etc. Now a days the competition is growing so it is very essential to satisfy the customer. For attracting new customers companies conduct consumer research, provide after saleservices etc.
- iii) **Market Intermediaries** :-Market Intermediaries which include agents and brokers who help the company to find customers. It is a link between company and consumer. Market intermediaries help the company to promote sell and distribute its goods to final buyers. Market intermediaries include middlemen, marketing agencies, financial intermediaries, physical intermediaries etc.
- iv) **Competitors** :- Competitors means other business units which are producing similar products or a very close substitute of our product. Now a days competitionhas increased. No business units enjoys monopoly in the market. So the business has to satisfy the customer for the success in the market.
- v) **Public** :-Public is group that has actual or potential interest in the business. So public also affect the business. Media also affect the business.
- vi) **Media** :- It includes al newspaper, magazines, journals etc. Media also affects thereputation of the company.

Macro Environment :—Macro Environment means general environment of business. These factors are uncontrollable. These factors create opportunities and pose threats to the business. It includes economic, demographic, natural, technological, political and cultural environments.

i) **Economic Environment** :— Economic Environment refers to those economic factors which have impact on the working of business. Economic environment is very complex in nature. It is very dynamic. It has three elements :-

- a) **Economic Conditions** :— Economic conditions of the economy affect the business. Economic conditions include income level, distribution of income, demand and supply trends etc. If the economy is in boom conditions, it positively affects demand and market share. On the other hand if the economy is in depression, it will have a negative effect on the business.
- b) **Economic Policies** :— Economic policies are framed by government. These policies establish a relationship between business and government. The effect of these policies may be favorable or unfavorable.
- c) **Economic System** :— Different economic systems prevail in different countries. These systems affect the business. The economic system includes capitalism, socialism and mixed economy

ii)

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Political & Legal Environment :— Political Environment affects the different business units. A stable and dynamic political environment is very necessary for business growth. Political environment includes political stability in the country, relation of the government with other countries, welfare activities of government, centre state relationship, thinking of opposition parties towards business. If the political system is stable and efficient then the business grows. In the lack of political stability long term plans cannot be formulated. Thus business is adversely affected if the government is not stable. Similarly relations of government with other countries also affect business. If a country enjoys friendly relations with other nations, then it has a favorable effect on foreign trade.

Socio Cultural Environment :— Socio- Cultural Environment refers to social and cultural factors which are beyond the control of business unit. Such factors include attitude of people at work, family system, caste system, education, habits, language, religion etc. Socio-cultural environment is one of the important non-economic external components of business environment. Religion has considerable effect on business. Some religions restrict their

followers from doing a particular type of business, e.g. Jain religion does not allow its followers to engage in leather industry, wine making etc. Similarly difference in language is another problem area in national level and international level business. The businessman must be familiar with the local language of the place where business is to be operated.

iii) **Technological Environment** :– Technological environment is most important factor which affect the business enterprise. The faster changes in technology create problems for business enterprise. Product have shorter life span than the past because of rapid technological development. Technology provides a various advantage. Success in many industries depends on a innovation and research. For promote innovation and research some companies establish research and development department in their enterprise. For example Japanese industries have achieved a great success because of innovation and rapid technological up gradation

iv) **Natural Environment** :– Natural Environment refers to geographical and ecological factors which are beyond the control of the enterprise. It includes natural resources, weather and climatic conditions, landforms rainfall, environmental pollution etc. Climate and weather conditions affect the location certain industries like textile industry. Similarly environment pollution in the form of air pollution, water pollution and noise pollution have caused disturbances in ecological balance. Government has framed various Acts for the control of environmental pollution. The business enterprise must keep in mind these factors.

v) **Demographical Environment** :– Demographical environment affect the business externally. Demographic environment differs from country to country and from place to place within the same country. Demographic factors includes size of population and population growth, family size, age composition, sex composition, urban-rural population education level etc. Huge population size indicate cheap labor and more demand in the economy. If population size is large then there will be more demand for goods and services. It will have favorable effect on business. Similarly, Education level is also important demographic factor affecting business. If public is highly educated, supply of unskilled labor will decrease. On the other hand if education level is low then supply of unskilled labor will increase.

vi) **International Environment** :– International Environment is the important component of the business environment. International environment affect the business differently. International environment is very important for certain types of business. It is

particularly important for industries directly depending on imports or exports. Recession in foreign market may create difficulties for industries depending on exports. Liberalization of imports may help some industries but may adversely affect other industries. For eg. the entry of multinationals such as LG, Samsung in electronics industry has adversely affected the marketshare of domestic business firms.

DIFFERENCE BETWEEN MICRO AND MACRO ECONOMICS

Parameter	Micro Economics	Macro Economics
Economic Unit	It is the study of individual economic units of an economy	It is the study of economy as a whole and its aggregates .
Scope	It deals with individual income, individual prices and individual output, etc.	It deals with aggregates like national income, general price level and national output, etc.
Central Problem	Its Central problem is price determination and allocation of resources	Its central problem is determination of level of income and employment.
Main Tools	Its main tools are demand and supply of a particular commodity/factor.	Its main tools are aggregate demand and aggregate supply of economy as a whole.
Use	It helps to solve the central problem of what, how and for whom to produce in the economy	It helps to solve the central problem of full employment of resources in the economy.
Equilibrium Analysis	It discusses how equilibrium of a consumer, a producer or an industry is attained.	It is concerned with the determination of equilibrium level of income and employment of the economy
Determinants	Price is the main determinant of microeconomic problems	Price is the main determinant of microeconomic problems
Limitation	It is based on unrealistic assumptions, i.e. In microeconomics it is assumed that there is a full employment in the society which is not at all possible.	It has been analyzed that 'Fallacy of Composition' involves, which sometimes doesn't prove true because it is possible that what is true for aggregate may not be true for individuals too.
Approach	While analyzing any economy, microeconomics takes a bottom-up approach.	The macroeconomics takes a top-down approach into the consideration.
Examples	Examples are: individual income, individual savings, price determination of a commodity, individual firm's output, consumer's equilibrium	Examples are: National income, national savings, general price level, aggregate demand, aggregate supply, poverty, unemployment etc.

MEANING OF PRIVATE SECTOR ENTERPRISES

These undertakings are owned, controlled and financed by private businessmen. There is no government participation in them. The main motive of private sector undertakings is to earn profits.

CHARACTERISTICS OF PRIVATE ENTERPRISES

(a) Private Ownership and Control:

A private sector undertaking is fully owned and controlled by the private entrepreneurs. It may be owned by one individual or by a group of individuals jointly. When owned by one person, it is called Sole Proprietorship. A group of persons may jointly own the firm in the form of joint Hindu family business, partnership, Joint Stock Company or cooperative society.

(b) Profit Motive:

The main objective of private sector undertakings is earning profits. Profits provide the reward for the risk assumed and the required return on capital.

(c) No State Participation:

There is no participation by the Central or State Governments in the ownership and control of a private sector undertaking.

(d) Private Finance:

The capital of a private sector undertaking is arranged by its owners. The sole trader contributes the capital of a sole proprietorship. In case of partnership, capital is invested by the partners. A joint stock company raises capital by the issue of shares and debentures. A private sector undertaking can also raise loans to meet its long-term and short-term needs for funds.

(e) Independent Management:

A private sector undertaking is managed by its owners. In case of sole proprietorship and partnership, the owners directly manage the firm. The management of a joint stock company lies in the hands of directors who are the elected representatives of the shareholders.

The six major problems faced by private sector in India.

1. Regulatory Procedure and Related Delays:

Too many regulatory measures imposed by the Government on the private sector has resulted in lengthy procedure and delays in getting final clearance of a new industrial project. On the

Government level, decision making system is so poor that it normally takes 7 to 8 years for large investment project to complete its gestation period.

Delegation of decision making in the Government bureaucracy is so poor that even the simple decisions are rolled back to the top level leading avoidable procedural delays, huge cost escalation, increasing interest burden and higher burden on consumers.

2. Unnecessary Control:

From the beginning, the private sector of the country is subjected to unnecessary Government control. Price controls imposed by the Government on certain goods has resulted in disincentive to increase production. Rather competition among the rival producers can enlarge the production base and thereby can reduce the prices automatically.

But in India, under the conditions of shortage, price controls, dual pricing etc. has resulted in black marketing and hoarding of such commodities. Moreover, the system of licensing of capacity as a capacity restraint has also resulted in undesirable effects on the investors instead of preventing monopolistic tendencies. It is only since 1980, unnecessary controls on the utilization of excess capacity and on the creation of new capacities have been either abolished or liberalized.

3. Inadequate Diversification:

The private sector has been suffering from inadequate diversification as the Government did not allow them to participate in those basic, heavy and infrastructural sectors which were earlier reserved for the public sector. It is only in post-1991 period, some of these areas are now opened for the private sector participation.

4. Reservation for the Small Sector:

From the initial stage of development, the Government is providing necessary support to the small industrial sector in the form of reservation of certain products exclusively for the small sector so as to save it from unfair competition of large units and also by providing excise exemption or lower excise duties on the goods produced by the small sector. But for the proper development of the small sector, modernization of their production techniques, proper product-mix, updating of designs must be given adequate priority.

5. Lack of Finance and Credit:

Although the large scale industrial corporate units of the private sector are mobilizing their fund from banks, development financial institutions and from the market through sale of their equities or debentures but the small scale units are facing acute problem in raising fund for their expansion.

6. Low Ratio of Profit:

Another important problem of the private sector enterprises is the declining trend in its net profit ratio. Accordingly, the net profit to turnover ratio of these total Indian private sector enterprises has been declining from 6.1 per cent in 1994-95 to 3.2 per cent in 1996-97 and then to 2.3 per cent in 1997-98.

Moreover, the net profit to net worth (NP/NW) reflecting on return on investment, of the total private sector enterprises also declined considerably from 15.2 per cent in 1994-95 to 6.5 per cent in 1996-97 and then to 4.7 per cent in 1997-98 as compared to that of 5.4 per cent of the Central Public Sector Enterprises (CPSEs).

Prospects of Private Sector:

Since independence, the private sector was assigned with a secondary role to participate in the industrial activity of the country by the Government. It was only since the Sixth Plan, the Government started to assign much responsibility to the private sector by allocating about 47 per cent of the total planned investment in the private sector.

Side by side, the private sector has also improved its condition, showing sufficient buoyancy and registering higher rate of growth by raising higher volume of funds from the capital market and also by setting up many joint ventures in some other countries. In spite of that, the private sector of India was about 20 years behind the schedule as compared to that of other developing countries.

In order to overcome such gap, a series of measures has been taken by the Government so as to improve the conditions of the private sector. These measures include permitting automatic expansion of capacity to a good number of industries, providing special incentive to export oriented units, exemption from MRTP restrictions on industrial units producing export goods, granting licenses easily to industrial units located in “zero industry” districts, fastening the licensing procedures, liberalisation of import and pricing policies etc. All these measures which were introduced during the Sixth Plan were further liberalised and strengthened during the Seventh Plan.

Moreover, the Government has announced the New Industrial Policy, 1991, which has liberalised the private sector considerably. The main changes brought by this new policy include abolishing the system of industrial licensing for all industrial undertaking except for a short list of 18 industries, reducing the list of industries under public sector to 8 as against the 17 industries reserved earlier, inviting private sector participation in PSUs earning higher profit, abolishing the system of pre-entry scrutiny of investment decisions of the MRTP

companies, removing the asset limit of the MRTP companies, providing automatic permission for foreign technology agreement, removal of mandatory convertibility clause etc. Thus, the new policy has assigned a greater role on the private sector by removing restrictions and controls and introduced a liberalised regime for the private sector of the country.

Thus, the prospects of the private sector in the future development programme of the country are quite bright under the new liberalised regime. Private sector has now been allowed to participate in those basic, difficult and strategic areas to compete with the public sector of the country.

Thus, this is quite a challenging situation before the private sector of the country. It is expected that the private sector will be able to accept this challenge and prove its worth within the definite time frame.

Small Scale Industries (SSIs)

The definition of SSIs is based on the criterion of value of plant and machinery which has been revised over the years. At present the small industrial unit has been classified as follows:

- i) Small Scale Industrial Units (SSI) units having investment in plant and machinery upto Rs.1 crore.
- ii) Ancillary Industrial Units having investment in plant and machinery upto Rs.1 crore. Such an undertaking must sell not less than 50 percent of its output to other industrial undertakings.
- iii) Export-oriented Units having investment in plant and machinery upto 1 crore. The unit must export at least 30 percent of its output by the end of three years from the date of commencement of production.
- iv) Tiny Units having investment in plant and machinery upto Rs.25 lakhs irrespective of location.

These are the industrial undertakings having fixed investment in plant and machinery, whether held on ownership basis or lease basis or hire purchase basis not exceeding Rs. 1 crore. SSI is defined as an unit engaged in manufacturing, serving, repairing, processing and preservation of goods and having investment in plant and machinery at an original cost of not exceeding Rs. 1 crore.

Importance of SME:

1. Employment generation
2. Optimisation Of Capital
3. Balanced Regional Development
4. Mobilisation Of Local Resources
5. Export Promotion
6. Feeder To Large Scale Industries
7. Development Of Entrepreneurship

Problems faced by SME in India

1. Lack of finance
2. Lack of availability of raw materials
3. Problem of marketing
4. Low level of technology
5. Competition from large scale industries
6. Lack of managerial skills

Meaning of Fiscal policy

The fiscal policy is concerned with the raising of government revenue and incurring of government expenditure.

To generate revenue and to incur expenditure, the government frames a policy called **budgetary policy or fiscal policy**. So, the fiscal policy is concerned with government expenditure and government revenue.

Fiscal policy has to decide on the size and pattern of flow of expenditure from the government to the economy and from the economy back to the government.

In broad term fiscal policy refers to "**that segment of national economic policy which is primarily concerned with the receipts and expenditure of central government.**"

Fiscal Policy is a policy under which the government uses its revenue and expenditure programmes to produce desirable effects and avoid undesirable effects on national income, production and employment.

Government policy of changing its taxation and public expenditure programs intended to achieve its objective"

Objectives of fiscal policy

The fiscal policy is designed to achieve certain objectives as follows :-

1. Development by effective Mobilisation of Resources

The principal objective of fiscal policy is to ensure rapid economic growth and development. This objective of economic growth and development can be achieved by Mobilisation of Financial Resources.

The central and the state governments in India have used fiscal policy to mobilise resources.

The financial resources can be mobilised by :-

Taxation : Through effective fiscal policies, the government aims to mobilise resources by way of direct taxes as well as indirect taxes because most important source of resource mobilisation in India is taxation.

Public Savings : The resources can be mobilised through public savings by reducing government expenditure and increasing surpluses of public sector enterprises.

Private Savings : Through effective fiscal measures such as tax benefits, the government can raise resources from private sector and households. Resources can be mobilised through government borrowings by ways of treasury bills, issue of government bonds, etc., loans from domestic and foreign parties and by deficit financing.

2. Efficient allocation of Financial Resources

The central and state governments have tried to make efficient allocation of financial resources. These resources are allocated for Development Activities which includes expenditure on railways, infrastructure, etc. While Non-development Activities includes expenditure on defence, interest payments, subsidies, etc.

But generally the fiscal policy should ensure that the resources are allocated for generation of goods and services which are socially desirable. Therefore, India's fiscal policy is designed in such a manner so as to encourage production of desirable goods and discourage those goods which are socially undesirable.

3. Reduction in inequalities of Income and Wealth

Fiscal policy aims at achieving equity or social justice by reducing income inequalities among different sections of the society. The direct taxes such as income tax are charged more on the rich people as compared to lower income groups. Indirect taxes are also more in the case of semi-luxury and luxury items, which are mostly consumed by the upper middle class and the upper class. The government invests a significant proportion of its tax revenue in the implementation of Poverty Alleviation Programmes to improve the conditions of poor people in society.

4. Price Stability and Control of Inflation

One of the main objective of fiscal policy is to control inflation and stabilize price. Therefore, the government always aims to control the inflation by Reducing fiscal deficits, introducing tax savings schemes, Productive use of financial resources, etc.

5. Employment Generation

The government is making every possible effort to increase employment in the country through effective fiscal measure. Investment in infrastructure has resulted in direct and indirect employment. Lower taxes and duties on small-scale industrial (SSI) units encourage more investment and consequently generates more employment. Various rural employment programmes have been undertaken by the Government of India to solve problems in rural areas. Similarly, self employment scheme is taken to provide employment to technically qualified persons in the urban areas.

6. Balanced Regional Development

Another main objective of the fiscal policy is to bring about a balanced regional development. There are various incentives from the government for setting up projects in backward areas such as Cash subsidy, Concession in taxes and duties in the form of tax holidays, Finance at concessional interest rates, etc.

7. Reducing the Deficit in the Balance of Payment

Fiscal policy attempts to encourage more exports by way of fiscal measures like Exemption of income tax on export earnings, Exemption of central excise duties and customs, Exemption of sales tax and octroi, etc.

The foreign exchange is also conserved by Providing fiscal benefits to import substitute industries, Imposing customs duties on imports, etc.

The foreign exchange earned by way of exports and saved by way of import substitutes helps to solve balance of payments problem. In this way adverse balance of payment can be corrected either by imposing duties on imports or by giving subsidies to export.

8. Capital Formation

The objective of fiscal policy in India is also to increase the rate of capital formation so as to accelerate the rate of economic growth. An underdeveloped country is trapped in vicious (danger) circle of poverty mainly on account of capital deficiency. In order to increase the rate of capital formation, the fiscal policy must be efficiently designed to encourage savings and discourage and reduce spending.

9. Increasing National Income

The fiscal policy aims to increase the national income of a country. This is because fiscal policy facilitates the capital formation. This results in economic growth, which in turn increases the GDP, per capita income and national income of the country.

10. Development of Infrastructure

Government has placed emphasis on the infrastructure development for the purpose of achieving economic growth. The fiscal policy measure such as taxation generates revenue to the government. A part of the government's revenue is invested in the infrastructure development. Due to this, all sectors of the economy get a boost.

11. Foreign Exchange Earnings

Fiscal policy attempts to encourage more exports by way of Fiscal Measures like, exemption of income tax on export earnings, exemption of sales tax and octroi, etc. Foreign exchange provides fiscal benefits to import substitute industries. The foreign exchange earned by way of exports and saved by way of import substitutes helps to solve balance of payments problem.

5 Major Instruments of Fiscal Policy

Some of the major instruments of fiscal policy are as follows:

- A. Budget : The budget of a nation is a useful instrument to assess the fluctuations in an economy.
- B. Taxation : Taxation is a powerful instrument of fiscal policy in the hands of public authorities which greatly effect the changes in disposable income, consumption and investment.
- C. Public Expenditure : The increased public spending will have a multiple effect upon income, output and employment exactly in the same way as increased investment has its effect on them.
- D. Public Works : Keynes General Theory highlighted public works programme as the most significant anti-depression device.
- E. Public Debt.: Public debt is a sound fiscal weapon to fight against inflation and deflation. It brings about economic stability and full employment in an economy.

Role of Fiscal Policy in Economic Development

1. To Mobilize Resources:

The foremost aim of fiscal policy in underdeveloped countries is to mobilize resources in the private and public sectors. Generally, the national income and per capita income is very low due to low rate of savings. Therefore, the governments of such countries through forced savings pushes the rate of investment and capital formation which in turn accelerates the rate of economic development.

It also undertakes the policy of planned investment in the public sector. Private investments have the favourable effect of increasing investment, the curtailment of conspicuous consumption and investment in unproductive channels can help to check the inflationary trend in the economy. Moreover, these countries face the problem of foreign capital. Thus the remedy lies in increasing the incremental saving ratio, the marginal propensity to save through public finance, taxation and forced loans.

To some extent, progressive taxation, heavy duty on luxury imports, ban on the manufacture of luxury and semi-luxury goods are other measures which help to mobilize the resources, Therefore, progressive taxation on windfall gains, on unearned incomes on capital gains, on expenditure and real estates etc. can go a long way in equitable distribution of wealth.

2. To Accelerate the Rate of Growth:

Fiscal policy helps to accelerate the rate of economic growth by raising the rate of investment in public as well as private sectors. Therefore, various tools of fiscal policy as taxation, public borrowing, deficit financing and surpluses of public enterprises should be used in a combined manner so that they may not adversely affect the consumption, production and distribution of wealth.

3. To Encourage Socially Optimal Investment:

In underdeveloped countries, fiscal policy encourages the investment into those productive channels which are considered socially and economically desirable. This means optimal investment which promotes economic development and avoids wasteful and unproductive investment.

In short, aim of the fiscal policy should be to make investment on social and economic overheads such as transportation, communication, technical training, education, health and soil conservation. They tend to raise productivity and widen the market to enjoy external economies. At the same time, unproductive investment is checked and diverted towards productive and socially desirable channels.

4. Inducement to Investment and Capital Formation:

Fiscal policy plays crucial role in underdeveloped countries by making investment in strategic industries and services of public utility on one side and induces investment in private sector by giving assistance to new industries and introduces modern techniques of production. Thus, investment on social and economic overheads are helpful in increasing the social marginal productivity and thereby raising the marginal productivity of private investment and capital formation. Here, optimum pattern of investment can also go a long way to yield fruitful results of economic development.

Economic development is a most dynamic process which involves changes in the size and quality of population, tastes, knowledge and social institutions. Keeping all factors in mind, if social marginal productivity in socially desirable projects is low, fiscal policy should be framed to raise social marginal productivity and to divert resources to that productive channels where the social marginal productivity is the highest.

5. To Provide more Employment Opportunities: Since in less developed countries, population grows at a very fast rate, the aim of fiscal policy in such countries is to make high doses of expenditures which are helpful to raise employment opportunities.

6. Promotion of Economic Stability:

□ Still another role played by the fiscal policy in developing countries is of maintaining reasonable internal and external economic stability. Generally, a developing country is prone to the effects of international cyclical fluctuations. Such countries mainly export primary products and import manufactured and capital goods. However, in order to minimize the effects of international cyclical fluctuations, fiscal policy should be viewed from a longer perspective.

□ It must aim at the diversification of all sectors of the economy. For bringing balanced growth and reducing the effects of cyclical fluctuations, a contra-cyclical fiscal policy of deficit budgeting in depression and surplus budgeting in inflation are most suitable measures.

□ In a recession, public works programme through deficit financing brings fruitful results. No doubt, injection of additional purchasing power would tend to inflationary pressures which can be controlled with preventive measures. On the contrary, such a policy should be supplemented by appropriate monetary measures.

(5) To Check Inflationary Tendencies:

- Inflationary tendencies is one of the main problems of developing countries as these countries make heavy doses of investment for their development activities. Thus, there is always an imbalance between the demand for and the supply of real resources.
- With additional injection of purchasing power, the demand rises and supply remains inelastic on account of its structural rigidities, market imperfections and other bottlenecks which in turn lead to inflationary pressures on the economy. Aggregate demand as a result of rise in the income of the people exceeds the aggregate supply. Capital goods and consumption goods fail to keep pace with the rising income.

(6) **Reallocation of Resources:** Allocation of resources are not proper in the underdeveloped countries.

(7) . **Balanced Growth:** Private sector in these countries normally concentrates its production on those luxury goods which are consumed mostly by richer sections who live in the urban areas. Hence, backward areas will not be developed unless government interferes into the decision making relating to industrial location

(8) **Reduction of Inequality:** the fiscal measures directed to the reduction of the gap between poverty and prosperity

Limitations in U.D.C:

1. **Existence of Barter Economy:** In UDC's, there exists non-monetized sector i.e. barter system prevails in the economy. This sector remains unaffected by the fiscal policy.
2. **Lack of Elasticity:** Taxation system in underdeveloped countries is not modern, rational and elastic. Tax evasion leads to the generation of black market. It becomes difficult to earn sufficient revenue by the way of taxes which hinders the development activities.
3. **Inadequate Data:** Generally, in less developed countries, there is inadequate statistical data. In the absence of accurate data, the scope of fiscal policy is minimized.
4. **Illiteracy:** Lack of knowledge and proper understanding on account of illiteracy, the scope of fiscal policy becomes limited. The common people are unable to recognize the significance of fiscal policy.
5. **Lack of co-operation:** In U.D.Cs lack of confidence and non-cooperative attitude among people hinders the significance of fiscal policy.

MONETARY POLICY

- Monetary Policy is the process by which monetary authority of a country, generally a central bank controls the supply of money in the economy by exercising its control over interest rates in order to maintain price stability and achieve high economic growth. Government and central bank regulates quantity, use and cost of money to achieve pre-determined objectives.
- As Per Prof. Kent Monetary Policy means the system of expanding and contracting the quantity of money in circulation and also the credit to fulfill some specific objective.
- MONEY: Anything of value that serves as a (1) generally accepted medium of financial exchange, (2) legal tender for repayment of debt, (3) standard of value, (4) unit of accounting measure, and (5) means to save or store purchasing power. See also cash.

Objectives of Monetary Policy

Monetary Policy always has some pre-determined objective as per economic requirement of each country. These objectives may vary as per different nations like economic stability or high growth etc. In context of our country monetary policy has following objectives –

1. Price Stability
2. Growth
3. Social Justice
4. Controlled Expansion of Bank Credit
5. Promotion of Fixed Investment
6. Export Promotion
7. Desired Distribution of Credit

TYPES OF MONETARY POLICY

Monetary policy design changes as per the goals set for the monetary policy and the emerging economic scenario.

1. Expansionary Monetary Policy

Expansionary or easy monetary policy aims at encouraging spending on goods and services by expanding the supply of credit and money by lowering the policy rates (bank rate or repo rate), lowering the reserve requirements and purchasing the government securities from the market.

Key Actions implements:

- Decreasing the discount rate

- ▶ Purchasing government securities
- ▶ Reducing the reserve ratio

2. Contractionary Monetary Policy

Contractionary or tight monetary policy aims at preventing inflation by contracting the money supply. Contraction in money supply is achieved by increasing the policy rates, increasing the reserve requirements and purchasing the government securities from the market.

- The contractionary policy is utilized when the government wants to control inflation levels.
- Increase taxes.
- Reduce government spending
- Decrease gross domestic product (GDP)

3. Countercyclical Monetary Policy

Countercyclical policy aims at moderating the cyclical fluctuations in the economy and stabilizing the economy around its trend path by following countercyclical measures.

4. Rule Based Monetary Policy

- Under rule based policy money supply and related variables are controlled by predetermined rules, norms and standards.
- The central bank authorities cannot use their discretion to change the values of these variables.

5. Discretionary Monetary Policy

Discretionary Monetary Policy allows the central bank greater autonomy in the conduct of monetary policy. Under such a policy rather than getting constrained by the pre-set rule, the central banks, after assessing the emerging economic scenario and using its own judgment, can change the values of money supply and the related variables.

Indian Industrial Policy : New industrial policy 1991, **Production Linked Incentive** (PLI) scheme for Promoting manufacturing of Telecom & Networking Products in India, New economic initiatives proposed by Indian government for economic growth Private Sector-Growth- like Atma Nirbhar Bharath Abhiyan.

Industrial Policy

Industrial policy is a document that sets the tone in implementing, promoting the regulatory roles of the government. It was an effort to expand the industrialization and uplift the economy to its deserved heights. It signified the involvement of Indian government in the development of industrial sector.

With the introduction of new economic policies, the main aim of the government was to free the Indian industry from the chains of licensing. The regulatory roles of the Indian government refer to the policies towards industries, their establishments, their functioning, their expansion, their growth as well as their management.

Objectives of Industrial Policy

The main objectives of the Industrial Policy of Government in India are:

- The industrial policy helps keep the industry on track with growth, leading to improved productivity.
- Better industrial growth leads to more job opportunities and higher economic growth.
- It makes the industries more regulated & productive and improves human resource use.
- It will help accelerate the economic growth of the country.
- Better functioning of the industries will lead to better international standards, enabling the domestic market to compete with the global market.

Problems in industrial development during the plan period.

The following are some of the major problems and obstacles that are being faced in the process of industrialization of the country:

1. Poor Capital Formation:

Poor rate of capital formation is considered as one of the major constraint which has been responsible for slow rate of industrial growth in India.

2. Political Factors:

During the pre-independence period, industrial policy followed by the British rulers was not at all favorable for the interest of the country. Thus, India remained a primary producing country during 200 years of British rule which ultimately retarded the industrial development of the country in its early period.

3. Lack of Infrastructural Facilities:

India is still backward in respect of its infrastructural facilities and it is an important impediment towards the industrialization of the country. Thus in the absence of proper transportation (rail and road) and communication facilities in many parts of the country, industrial development could not be attained in those regions in spite of having huge development potentialities in those areas.

4. Poor Performance of the Agricultural Sector:

Industrial development in India is very dependent on the performance of the agricultural sector. Thus, the poor performance of the agricultural sector resulting from natural factors is also another important factor responsible for industrial stagnation in the country.

Agriculture provides not only raw materials and foodstuffs but also generates demand for the goods produced by the industrial sector. Thus, this poor performance of the agriculture retards the development of industries in India.

5. Gaps between Targets and Achievements:

In the entire period of planning excepting 1980s, industrial sector could not achieve its overall targets. During the first Three Plans, against the target of 7, 10.5 and 10.7 per cent industrial growth rate, the actual achievements were 6, 7.2, 9 per cent respectively. Since the Third Plan onwards, the gap between the targets and achievements widened.

It is only during the Sixth and Seventh Plan, the industrial sector could achieve its targets. Again in first part of 1990s the industrial sector again failed miserably to achieve its target. This trend is all along against the smooth industrial development of the country.

6. Dearth of Skilled and Efficient Personnel:

The country has been facing the problem of dearth of technical and efficient personnel required for the industrial development of the country. In the absence of properly trained and skilled personnel, it has become very difficult to handle such highly sophisticated computerized machineries necessary for industrial development of the country.

Moreover, inefficiency and insincerity of those personnel engaged in industrial sector has been resulting in huge wastage of resources of the industrial sector. Moreover, social factors like immobility of labour and capital and lack of proper initiative and enterprises on the part

of people of India are also highly responsible for this slow pace of industrialization in the country.

7. Elite Oriented Consumption:

In recent years, a strong tendency to produce rich men's goods has been established among the large industrial houses. Accordingly, the production of "white goods" like refrigerators, washing machines, air conditioners etc. expanded substantially along with the other luxury products.

But the production of commodities for mass consumption has recorded a slow growth rate. This clearly reveals a 'distortion of output structure' of Indian industries, resulting in a recessionary tendency in the market of these luxury products in recent years.

8. Concentration of Wealth:

The pattern of industrialization in the country has been resulting in concentration of economic power in the hands of few large industrial houses and thus failed to achieve the objective of planning in reducing concentration of wealth and economic power. As for example, Tatas with 38 companies substantially increased their assets from Rs. 375 crore in 1963- 64 to Rs. 14,676 crore in 1991-92.

The assets of Birlas also increased from Rs. 283 crore in 1963- 64 to Rs. 6,775 crore in 1990- 91. Similarly other large business houses are also multiplying their assets at a very faster rate and are tightening their stronghold on the economy.

9. Poor Performance of the Public Sector:

In-spite of attaining a substantial expansion during the planning period, the performance of public sector enterprises remained all along very poor. A good number of such enterprises are incurring huge losses regularly due to its faulty pricing policy and lack of proper management necessitating huge budgetary provision every year. Thus, the public sector investment failed to generate required surpluses necessary for further investment in industrial sector of the country.

10. Regional Imbalances:

Concentration of industrial development into some few states has raised another problem of imbalances in industrial development of the country. Western region comprising Maharashtra and Gujarat attained maximum industrial development whereas the plight, of the poor states are continuously being neglected in the process of industrialization of the country in-spite of having a huge development potential of their own.

Although a huge investment in the public sector has been made in the backward states like Bihar, Orissa and Madhya Pradesh, but the 'trickling down effects' of such investment were

not also visible. Various fiscal incentives, capital subsidies and other facilities introduced for industrial development of backward area were mostly channelized to develop industries in backward areas of developed states leading to a gross neglect of the demand of backward states.

11. Industrial Sickness:

Another peculiar problem faced by the industrial sector of the country is its growing sickness due to bad and inefficient management. As per the RBI estimate, a total number of sick industrial units in India were 1,71,316 as on 31st March, 2003 and these sick industrial units had involved an outstanding bank credit to the extent of Rs. 34,815 crore.

The RBI estimate further disclosed that every seventh small scale unit in India was sick at the end of December 1983. Thus, this growing sickness of industrial units has resulted in a huge problem in the path of industrial development of the country.

12. Regime of State Controls:

Lastly, industrial inefficiencies resulting in perpetuation of regional state controls and regulatory mechanism are standing in the path of industrialization of the country. In recent years, the Government has undertaken some serious measures to make necessary economic reforms in the industrial structure of both the public as well as private sectors of the country. Although these measures are quite challenging in nature but these are expected to do much headway in removing various obstacles mentioned above and also in attaining industrial development of the country further in the years to come.

INDUSTRIAL POLICY 1991

In order to accelerate Industrial Development in India, and in accordance with the changing circumstances, various industrial policies were declared in the years 1948, 1956, 1977, 1980 and 1985, but in spite of all efforts, the pace and as well as the level of Industrial Development in India, could not reached according to its need. Therefore, in order to lift unnecessary restrictions on Industries, under the licensing policy, and to increase their efficiency, development and technological level, in order to make Indian goods usable in the competitive global market, on 24th July, 1991, in Lok-Sabha the Minister of States for industries, Mr. P. J. Kurian declared the Industrial Policy, 1991.

OBJECTIVES OF NEW INDUSTRIAL POLICY, 1991

- To liberalise the economy
- To increase employment opportunities
- To encourage foreign assistance and co-partnership

- To make the Public Sector more competitive
- To increase the production and productivity, give encouragement to industries
- To liberate the economy from various government restrictions
- Industrial development of backward areas
- To give liberty to private sector to work independently
- To make development for modern competitive economy
- To give encouragement for expansion of production capacity
- To increase exports and liberalize (facilitate) imports.

SALIENT FEATURES OF NEW INDUSTRIAL POLICY, 1991

• Liberalized Industrial Licensing Policy

Under this policy, with the exception of 18 industries, licensing system has been removed for all other industries. Some of those 18 industries, where the licensing system is still mandatory are; Army and Defence, Forest Conservation, Industries engaged in manufacturing goods which are harmful to the Environment and industries, which are manufacturing luxury goods, for the affluent (very rich) class, etc.

• Localisation Policy

Those industries which are situated in cities, where the population is less than 1 million, industrial permission from the government, to start any industry is not required. In cities having population of more than 1 million, with the exception of electronics and other pollution free industries, all industrial units may be 25 kilo meters away from the city's boundary.

• Foreign Investment

Provision has been made to invest up to 51 percent by foreign investors in the equity shares of Indian Companies. Earlier, this limit was limited up to 40% only. This will increase the flow of foreign capital into India and make possible technical exchange from developed countries.

• Workers' Participation in Management

Under this industrial policy, emphasis has been laid on safeguarding the workers' interest. Provision has been made for workers' participation in management, in order to manage sick units, provision has been made to form co-operative societies of workers, to run them.

• Role of Public Sector

Those public sector undertakings which are not doing well at present, but in which there are enough chances of improvement, shall be re-constituted. Public sector undertakings, which are facing constant financial crisis, shall be kept under observation by 'Board of Industrial and Financial Reconstruction' or by any other institution, which is fixed by the government.

• **Change in the MRTP Act**

In the industrial policy 1991, major changes have been made in the Monopolistic and Restrictive Trade Practice Act. Companies having investment of Rs. 100 crores, will not be required to take prior Government permission, for opening new subdivisions, or to expand the present industry or for amalgamation of companies. This industrial policy has also eliminated the investment limit, which was fixed by MRTP Act.

• **Creation of Productive Capacity**

In order to increase the productive capacity of new industries, all administrative controls have been removed. Industrialists will only have to inform the government of opening of new units or increasing their production capacity.

• **Foreign Technology**

No prior permission from government will be required in importing foreign technology, up to the limit of One Crore rupees. Indian companies, will be free to negotiate their terms and conditions, with their foreign collaborators, in matters of technology transfers (exchange of 'technical know-how').

• **Promotion of Industries in Rural Areas**

In order to remove the regional imbalances, under this industrial policy, various provisions have been made to encourage industries in rural areas. Reservation of Small Scale Industries This policy has stated that the government shall keep giving assistance to small scale industries. The limit for small scale industries has been reduced from Rs 3 Crores to Rs. 1 Crore, since 24 December, 1999.

Impact Of Industrial Policy, 1991

- The all-round changes introduced in the industrial policy framework have given a new direction to the future industrialization of the country.
- Industrial growth was 1.7 per cent in 1991-92 that has increased to 9.2 percent in 2007-08.
- The industrial structure is much more balanced.
- The impact of industrial reforms is reflected in multiple increases in investment envisaged, both domestic and foreign. This is due to encouraging response from the private sector.

- There has been dramatic increase in FDI since 1991. The foreign investment as a percentage of total GDP has increased from 0.5 percent in 1990-91 to 5.7 percent in 2006.
- Investments in infrastructure sector such as power generation have surged from players of various sizes in different states. The capital goods have grown at an accelerated pace, over a high base attained in the previous years, which augurs well for the required industrial capacity addition.

The Production Linked Incentive (PLI) scheme

The Production Linked Incentive (PLI) scheme is a flagship initiative of the Government of India aimed at boosting manufacturing in India across various sectors. The scheme was introduced to make India a global manufacturing hub by providing financial incentives to companies that produce goods in India. Below are detailed notes on the PLI scheme. The scheme was announced by the Government of India as part of its 'Make in India' initiative to encourage local manufacturing.

1. Introduction to the PLI Scheme:

Objective: The primary objective of the PLI scheme is to promote domestic manufacturing and reduce import dependency in various sectors, thereby boosting the country's economic growth and job creation.

2. Sectors Covered:

The PLI scheme covers a range of sectors, including but not limited to:

- Electronics and Technology Products
- Pharmaceuticals
- Automobiles and Auto Components
- Textiles and Apparel
- Food Products
- Solar Panels and Cells
- White Goods (Air Conditioners and LED Lights)
- Telecom and Networking Products

Objectives of PLI schemes

- **Boost Domestic Manufacturing:** The primary goal of the PLI scheme is to promote and boost domestic manufacturing across various sectors. By providing financial incentives to companies, the government aims to encourage the production of goods within the country.
- **Reduce Import Dependency:** Another key objective is to reduce India's dependency on imports for critical goods and products. By incentivizing domestic production, the government aims to decrease the reliance on foreign imports, which can help strengthen India's economic and strategic resilience.
- **Create Employment Opportunities:** The PLI scheme aims to generate employment opportunities by encouraging manufacturing activities. As companies expand their production capacities or set up new manufacturing units, they are likely to hire more workers, thereby contributing to job creation.
- **Promote Investment:** The scheme seeks to attract both domestic and foreign investments in various sectors. It does so by offering financial incentives, which can serve as a strong motivator for companies to invest in building manufacturing infrastructure and capacities in India.
- **Encourage Technological Upgradation:** The PLI scheme encourages companies to invest in advanced technologies and modern manufacturing processes. This focus on technological upgradation can lead to increased efficiency, improved product quality, and innovation within the manufacturing sector.
- **Export Promotion:** Many PLI schemes include provisions to promote exports. By manufacturing products for export markets, Indian companies can increase their international competitiveness and contribute to India's export growth.
- **Support the 'Make in India' Initiative:** The PLI scheme aligns with the broader government initiative known as 'Make in India,' which aims to transform India into a global manufacturing hub. It seeks to enhance the country's image as a destination for quality manufacturing.
- **Atmanirbhar Bharat (Self-Reliant India):** The PLI scheme is in line with the government's vision of making India self-reliant (Atmanirbhar) in various sectors. By promoting domestic production, the government aims to reduce dependency on foreign goods and strengthen India's economic sovereignty.
- **Economic Growth:** Ultimately, the PLI scheme aims to contribute to India's economic growth by increasing the contribution of the manufacturing sector to the country's

Overview:

Self-reliant India campaign is the vision of new India envisaged by the Hon'ble Prime Minister Shri Narendra Modi. On 12 May 2020, our PM raised a clarion call to the nation giving a kick start to the Atmanirbhar Bharat Abhiyaan (Self- reliant India campaign) and announced the Special economic and comprehensive package of INR 20 lakh crores - equivalent to 10% of India's GDP – to fight COVID-19 pandemic in India.

The aim is to make the country and its citizens independent and self- reliant in all senses. He further outlined five pillars of Aatma Nirbhar Bharat – Economy, Infrastructure, System, Vibrant Demography and Demand. Finance Minister further announces Government Reforms and Enablers across Seven Sectors under Aatmanirbhar Bharat Abhiyaan.

The government took several bold reforms such as Supply Chain Reforms for Agriculture, Rational Tax Systems, Simple & Clear Laws, Capable Human Resource and Strong Financial System.

The announcements made by the Finance Minister Nirmala Sitharaman concluded the relief measures undertaken in five tranches by the government as part of the economic package announced by PM Modi for 'Atmanirbhar Bharat'. The minister, also provided a tranche-wise complete break-up of the stimulus. Importantly, the overall package which stood at Rs 20,97,053 crore, included the Rs 1.92 lakh crore stimulus from measures announced by Modi recently such as the Pradhan Matri Garib Kalyan Package worth Rs 1.7 lakh crore. A big chunk, in fact the largest, worth Rs 8.01 lakh crore of the economic package belonged to the various measures by the Reserve Bank of India in February, March and April this year to inject liquidity.

As part of this initiative, the government has outlined five pillars to guide the strategy and implementation:

- **Economy:**

The first pillar focuses on building a robust and self-reliant economy. It includes measures to promote economic growth, investment, and job creation.

Initiatives under this pillar include economic reforms, infrastructure development, financial sector reforms, and support for various industries and sectors.

- **Infrastructure:**

The second pillar emphasizes the development of modern and resilient infrastructure. Adequate infrastructure is essential for economic growth and competitiveness. This

pillar includes investments in transportation (roads, railways, ports, airports), energy (power generation and distribution), digital infrastructure, and affordable housing.

- **Systematic Structural Reforms:**

The third pillar involves systematic structural reforms across various sectors to make the Indian economy more efficient, competitive, and self-reliant. Reforms may include changes in labor laws, agricultural reforms, simplification of regulations, and measures to improve ease of doing business.

- **Vibrant Demography:**

The fourth pillar recognizes India's demographic advantage, with a large and young population. It aims to empower and provide opportunities for this youth population. Initiatives in this pillar include skill development programs, education reforms, and measures to enhance employment opportunities, particularly in emerging industries.

- **Demand:**

The fifth pillar focuses on boosting domestic demand. A strong domestic demand base is crucial for reducing dependence on imports. This pillar includes measures to enhance the purchasing power of consumers, promote local products and services, and support small and medium-sized enterprises (SMEs).

Impact of this Stimulus Package

- **Primary Sector:** The measures (reforms to amend ECA, APMC, Contract farming, etc) announced for the agricultural and allied sectors are particularly transformative. These reforms are steps towards the One Nation One Market objective and help India become the food factory of the world. □ These would finally help in achieving the goal of a self-sustainable rural economy. □ Also, the MGNREGA infusion of Rs 40,000 crore may help in alleviating the distress of migrants when they return to their villages.
- **Secondary Sector:** Given the importance of MSMEs for Indian economy, the Rs 3 lakh crore collateral-free loan facility for MSMEs under the package will help this finance-starved sector and thereby provide a kickstart to the dismal state of the economy. Also, as the MSME sector is the second largest employment generating sector in India, this step will help to sustain the labour intensive industries and thereby help in leveraging India's comparative advantage. □ Additionally, limiting imports of weapons and increasing the limit of foreign direct investment in defence from 49% to 74% will give a much-needed boost to the production in the Ordnance Factory Board, while reducing India's huge defence import bill.

- **Tertiary Sector:** The government has adopted a balanced approach in addressing concerns across sectors. For example: The newly launched PM e-Vidya programme for multi- mode access to digital online education provides a uniform learning platform for the whole nation, which shall enable schools and universities to stream courses online without further loss of teaching hours. □ Public expenditure on health will be increased by investing in grass root health institutions and ramping up health and wellness centres in rural and urban areas.